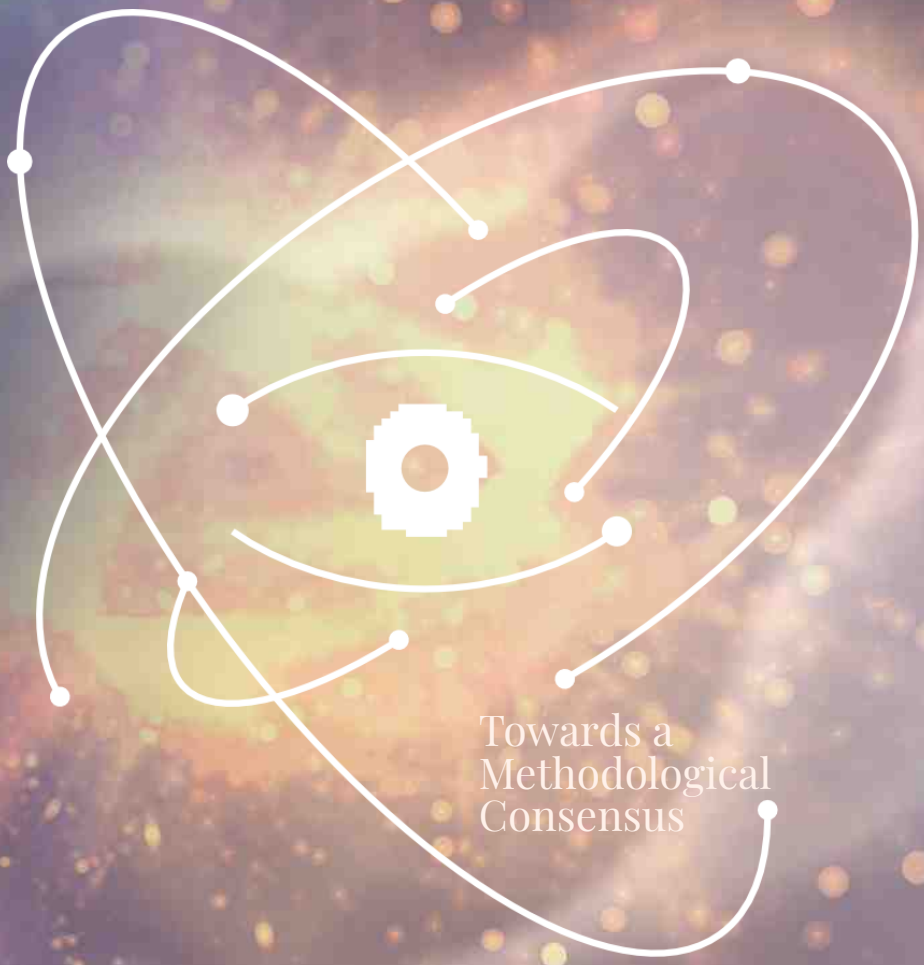
MoC 7

October 12 – 16
University of Copenhagen



Towards a Methodological Consensus

Conference programme / Talks / Discussions / Workshops / Community activities / Networking & more

amcs-community.org

Organising and Supporting Institutions

Association for Mathematical Consciousness Science

CARLSBERG FOUNDATION

UNIVERSITY OF COPENHAGEN

PIONEER CENTRE FOR ARTIFICIAL INTELLIGENCE

CHAIN
Center for Human Nature, Artificial Intelligence, and Neuroscience

上海科技大学 ShanghaiTech University

BAMΞ

UNIVERSITY OF BAMBERG · OTTO-FRIEDRICH-UNIVERSITÄT BAMBERG

SAINT ANSELM COLLEGE 1889

WISCONSIN UNIVERSITY OF WISCONSIN–MADISON

UNIVERSITY OF OXFORD

OMCAN
Oxford Mathematics of Consciousness and Applications Network

UNIVERSITÄT BONN

CPAI
CENTER FOR PHILOSOPHY OF ARTIFICIAL INTELLIGENCE

Programme

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Welcome!

It is a great pleasure to welcome you all to the seventh instalment of the *Models of Consciousness* conference series (MoC7). Following the conferences at *Oxford University* (2019, MoC1), online (2021, MoC2), *Stanford University* (2022, MoC3), again *Oxford University* (2023, MoC4), *Bamberg University* (2024, MoC5), and *Hokkaido University* (2025, MoC6), MoC7 offers an exciting opportunity to make new research connections and enjoy the *hygge* (cosy) Danish culture in one of the world's most liveable cities, Copenhagen.[1]

Hygge also means slowing down, reflecting and bringing people together by creating a relaxed, pressure-free space for shared presence and meaningful connection. Since consciousness science is still developing its methodological and conceptual foundations, reflecting on them in an amicable environment is crucial to make meaningful progress. A true *hygge* atmosphere relies on equality, where no single person dominates the conversation, and everyone feels equally valued.

This year, we hope to create such an atmosphere of mutual inspiration and constructive dialogue. In this spirit, we are introducing new social activities into the programme and turning the discussion sessions into transdisciplinary collaborative work **to clarify concepts, reach common methodological standards, and set future priorities in consciousness science** — possibly resulting in the first collective paper of this magnitude in the field (but not necessarily).

This year, the conference is dedicated to bringing together researchers on the following core themes:

- Philosophical foundations in the scientific study of consciousness
- Methodologies, models, and measures of consciousness
- AI, LLMs, and consciousness science
- Phenomenology and applied phenomenology

We hope to create an open and inspiring atmosphere that enables us all to make further progress on these and other questions together. Thanks for contributing your talks, posters and discussions — and enjoy the Danish *hygge*!

Welcome to Copenhagen! — *The organisers*

[1] EIU Global Liveability Index 2025.

Everything you need to know…

… initially ;-)

Conference Venue

HC Ørsted Institute (HCØ), University of Copenhagen, North Campus
Address: Universitetsparken 5, DK-2100 Copenhagen Ø, Denmark

HCØ: keynotes and showcases in Auditorium 1, parallel sessions in Auditoria 2–5, discussion sessions in the 1st-floor classrooms (p. 17); reception and Friday's closing in the Niels Bohr Building (pp. 9, 15).
Questions: moc7-organisers@amcs.science

Google Maps

OpenStreetMap

Wi-Fi Access

eduroam is offered on all campuses of the University of Copenhagen. If you cannot connect to eduroam, you can use the **KU Guest** network — set up your KU Guest account as follows:

a. Double-click on "KU Guest" and click on "connect".
b. Type in your name, mobile phone number (remember the country code) and e-mail address (this will become your KU Guest username).
c. Accept the terms of use and click on "Registrer".
d. Find your password on your mobile phone or via e-mail.
e. Log on to KU Guest: type in your username (the e-mail you listed when creating the account) and your password, and click on "log på".
f. Enjoy the KU Wi-Fi!

KU Guest guide

Getting Around

Copenhagen is a small and lively city. **The best way to move around is definitely by bike**, and there are plenty of bikes to rent around the city with easy apps.

The **metro operates 24/7**, and buses work amazingly well. To use public transport, download the **Rejsekort app** — and never forget to swipe in and out.

Wherever the group is heading, plans are made in the **MoC7 socials Signal group** — join it to keep up with the social programme on pp. 8–14.

Rejsekort app

Metro (m.dk)

Transit map

Signal (all events)

Core Themes

We are delighted to announce that Models of Consciousness 7 comprises the following core themes. However, please note that contributed talks are open to any topic in the field, including, in particular, the neuroscience of consciousness and mathematical approaches.

Philosophical foundations in the scientific study of consciousness

Within this theme, we explore the philosophical foundations of the scientific study of consciousness. Different viewpoints on how to understand reality often influence the development of models of consciousness. By exploring foundations, we can better understand the differing approaches and identify key questions to refine model selection. We invite philosophers of mind, neuroscience, AI, science, epistemology, and aesthetics, among other relevant areas.

Methodologies, models, and measures of consciousness

Under this theme, we invite presentations on methodologies, models, and measures of consciousness. Special attention is given to the neuroscience of consciousness and how to characterise states of consciousness, developmental consciousness, animal consciousness and synthetic consciousness. Research of this kind is at the heart of the field, generating scientifically credible and useful models and measures.

AI, LLMs, and consciousness science

Under this theme, we invite investigations in consciousness science and qualitative research, leveraging new methodologies inspired by AI and LLMs. Physiological (e.g. machine learning for neural decoding) and qualitative applications (e.g. topic modelling analyses) are welcome. We also invite contributions on the latest research and insights into artificial consciousness.

Phenomenology and applied phenomenology

Within this theme, we invite phenomenologists and phenomenological investigations, including philosophical and applied areas such as phenomenological psychology, neurophenomenology, micro-phenomenology, computational phenomenology, and mathematical phenomenology, among others. We welcome rigorous studies of subjective first-person lived experiences and structural aspects. This research informs models of consciousness and, ultimately, their degree of adequacy.

Conference Team

Organisers



Camilo Miguel Signorelli
Center for Philosophy of Artificial Intelligence, University of Copenhagen



Emanuela Yeung
Department of Psychology, University of Copenhagen



Robert Prentner
Institute of Humanities, ShanghaiTech University



Keisuke Suzuki
Center for Human Nature, Artificial Intelligence, and Neuroscience (CHAIN), Hokkaido University



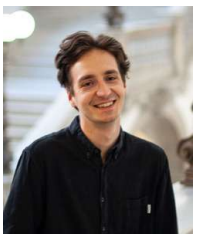
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Anna Eiserbeck
Department of Psychology, Humboldt University of Berlin



Johannes Kleiner
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Moritz Kriegleder
Department of Philosophy, University of Vienna



Jonathan Mason
Oxford Mathematics of Consciousness and Applications Network (OMCAN)



Ian Durham
Department of Physics, Saint Anselm College

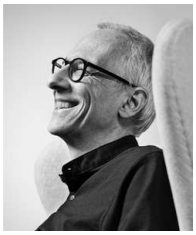


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Institute for Science and Ethics, University of Bonn



Sucharit Katyal
Department of Psychology, University of Copenhagen

Advisory Board



Dan Zahavi
Center for Subjectivity Research, University of Copenhagen



Lenore Blum
Distinguished Career Professor of Computer Science Emerita, Carnegie Mellon University



Victoria Southgate
Department of Psychology, University of Copenhagen



Anders Søgaard
Center for Philosophy of Artificial Intelligence, University of Copenhagen



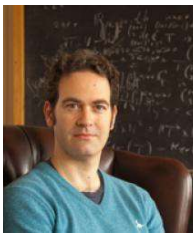
Tone Roald
Department of Psychology, University of Copenhagen



Andreas Roepstorff
Aarhus Institute of Advanced Studies, Aarhus University



Katrin Heimann
Centre for Educational Development, Aarhus University



Yakov Kremnitzer
OMCAN, Mathematical Institute, University of Oxford

Local Support



Nina Rajcic
Center for Philosophy of Artificial Intelligence, University of Copenhagen



Sandrine Rose Schiller Hansen
Center for Philosophy of Artificial Intelligence, University of Copenhagen

Don't hesitate to let us know in case you have any questions or are in any need of assistance!

Pre-Conference Activities

Sunday, 11 October

As is tradition at MoC, the day before the conference is devoted to informal pre-conference activities designed to get to know each other. On top of the two Sunday-afternoon activities below, we highly recommend spending Sunday morning visiting the free town of **Christiania** and taking a free walking tour of the historical centre.

*Please note that these activities are **not** included in the conference fee and are paid separately.*

Option 1 — Comedy Show and Drinks

We have organised a comedy show to learn about Danish culture and have fun in the process. The five-star **Copenhagen Show** is hosted in the underground cinema room at **Next House Copenhagen**, Bernstorffsgade 27, 1577 København (by Copenhagen Central Station). **After each show, we invite participants to stay and share a drink** (drinks are not included in the show ticket).

Three sessions of the same show are available: **15:00, 16:30 and 18:00**. Each session can host up to 50 people. **The shows are booked on a first-come, first-served basis** — book your session via the QR code: in the calendar, go to 11 October and select your preferred time.

If you arrive on Saturday, consider booking the earliest session, so that people who arrive on Sunday afternoon can book the latest one.

Book a session

Venue map

Option 2 — Home of Carlsberg and Drinks



We also suggest a self-organised visit to the **Home of Carlsberg** museum. Situated on the grounds of the original 1847 brewery, the museum is an interactive experience featuring Carlsberg's founding, the art of brewing beer, visionary open science, and the impact the Jacobsen family and Carlsberg have had on society. The tour ends at the Carlsberg Bar, where you will be treated to a perfectly poured draught beer.

This activity is **self-organised**: join the Signal group and agree on the best time for everyone.

Map

Signal group (Carlsberg)

Welcome Reception

Monday, 12 October, 16:00–18:30 — Niels Bohr Building (NBB)



The new Niels Bohr Building (NBB) on Jagtvej, right in front of HCØ.

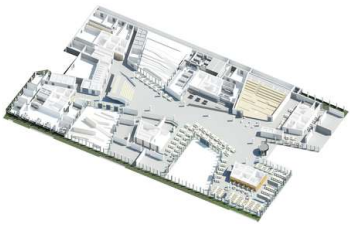
On Monday evening, the welcome reception takes place at the new **Niels Bohr Building**, in front of HCØ (see the campus map on p. 15). The reception is included in the conference registration.

We will meet at **Margrethe Bohr Salen** for welcome speeches and Robert Sunderland's inspirational talk about the **Copenhagen Quantum Spirit**, a story of people and relationships behind the quantum revolution.

We then move to the heart of the building, the **Troposphere**, and its surrounding gardens, Inge Lehmann. The Troposphere is a vibrant open atrium shaped like an infinity symbol, where researchers and students meet to create innovative, interdisciplinary collaboration. In this communal facility, everyone is invited to meet across affiliations and academic backgrounds — the ideal setting for our first evening together.



The Copenhagen quantum spirit: physicists at Niels Bohr's institute in the 1930s (photo: Niels Bohr Archive).



The Troposphere, Niels Bohr Building (ground floor plan; Christensen & Co. Architects / Vilhelm Lauritzen Architects, via ArchDaily).

This is an historical picture from the Niels Bohr Archive; we note and acknowledge the lack of female researchers represented at that time, a reminder to work towards more diversity and inclusivity.

Conference Dinner

Thursday, 15 October, 19:00 — Food Club Nørrebro

If you have booked the conference dinner during registration, you are invited to **join us for an inspired local communal dining experience at Food Club Nørrebro on Thursday 15 October**.

Communal dining is a social and trendy eating experience in Copenhagen. People sit together at large shared tables to share food, conversation, and a sense of community. The format is buffet-style and self-service.

Upon arrival, our hosts from Food Club will introduce the dishes and the concept. Everyone is invited to relax and chill during the dinner, as well as to switch seats if you feel like it.

Restaurant:	Food Club Nørrebro
Address:	Sortedam Dossering 7C, 2200 Copenhagen N
Day:	Thursday, 15 October
Time:	19:00
Price:	€75 (booked and paid with the registration)
Dietary:	Vegetarian, vegan and gluten-free options available

Google Maps

Food Club website

*Please note that the conference dinner is **only** included if you registered and paid for the dinner when paying the conference fee. How to get there: see the map on the next page.*

Did not make it to the conference dinner? We strongly recommend booking your own communal dining experience at *Absalon* (the most famous communal dining in Denmark) or at *Send Flere Krydderier* — more options via the QR code on the next page. But hurry, spots are booked well in advance!





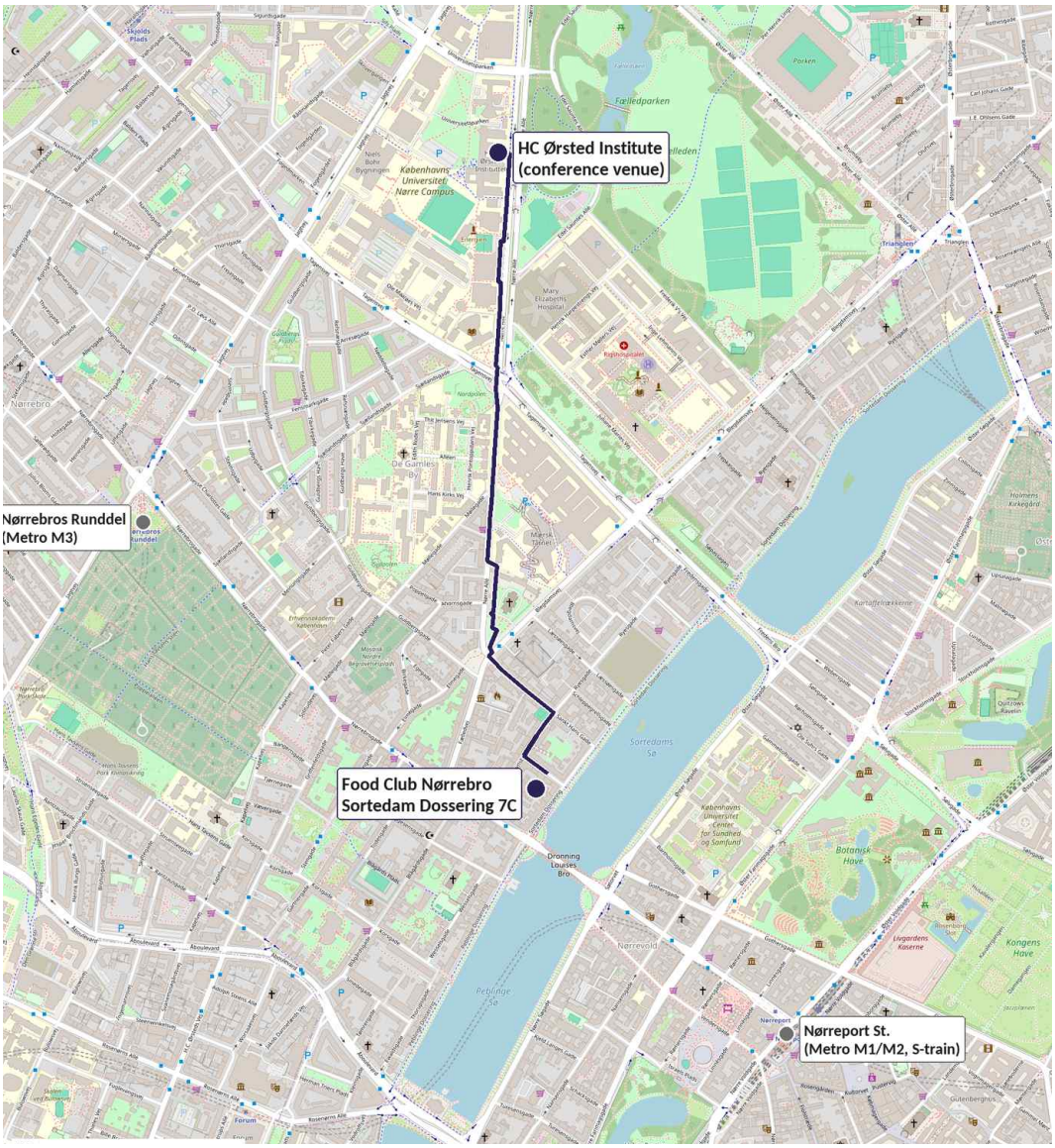


Photos: Food Club Nørrebro (madklubben.dk).

Getting to the Conference Dinner

Food Club Nørrebro lies on the Lakes (Sortedams Sø), about **1.5 km — a 20-minute walk — from HCØ**. The walking route is drawn on the map. Nearest public transport: **Nørreport Station** (Metro M1/M2, S-trains), about 10 minutes on foot across Dronning Louises Bro; or Metro M3 to **Nørrebros Runddel**, then about 15 minutes on foot along Nørrebrogade.

Communal dining



Map data © OpenStreetMap contributors. Walking route: OSRM.

Art & Science Excursion

Wednesday, 14 October — Louisiana Museum of Modern Art, Humlebæk

On Wednesday afternoon, the conference moves to the **Louisiana Museum of Modern Art** for an immersive art & science experience.

After the morning conference session, we take buses from the conference venue to the museum; sandwiches are provided during the ride. The route follows the sea, with beautiful panoramic views. At Louisiana, we will have a one-and-a-half-hour interactive workshop followed by a general reflection. Then, participants will enjoy another hour and a half of free time to discover the museum or get a coffee with an inspiring sea view. After that, we will reconvene for the **Mind & Matter Ceremony** and **Evan Thompson's lecture**. Afterwards, we head back to Copenhagen just in time for dinner.

This activity is **included** in the conference registration. Coffee is self-organised and paid separately during the free time and breaks. On the day of the Louisiana visit, we kindly ask you to leave your laptops and backpacks at home: **the museum does not allow entry with backpacks.**

Departure: 12:50
Meeting point: Universitetsparken 15 — pick-up by the street, between the building and the park (campus map p. 15, 3)
Return: 19:00 (departure from Louisiana)

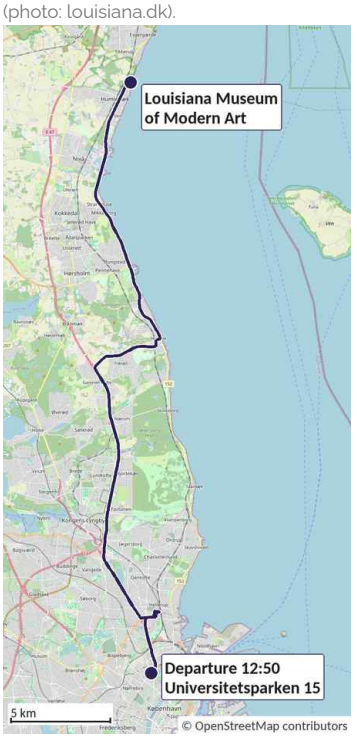
Preliminary schedule

12:50	Buses & sandwiches
14:00	Arrival
14:15–14:45	Welcome and short presentation
15:00–16:30	Guided creative workshop
16:30–18:00	Break — free time in the museum, coffee & sculpture park
18:00–19:00	Evan Thompson's keynote and Mind & Matter Ceremony
19:00	Departure

This activity is fully booked. For those who did not make it to the creative workshop activity, a live-stream link will be provided to follow the talks at the museum.



Louisiana Museum of Modern Art (photo: louisiana.dk).



Preliminary coach route along the coast (ca. 45 min). Map data © OpenStreetMap contributors.

Route (Google Maps)

Other Social Activities

Tuesday, 13 October — Tivoli Gardens self-organised

On Tuesday evening, we invite participants to self-organise an informal excursion to **Tivoli Gardens**, experience the magical **Halloween illuminations** (every 30 minutes from 19:30 to 21:30), and enjoy one of the oldest amusement parks in the world. Tickets are paid at the Tivoli entrance or on the website.

Time: self-organised — we suggest going there right after the last session. **Meeting point:** self-organised (join the Signal group).

Illuminations

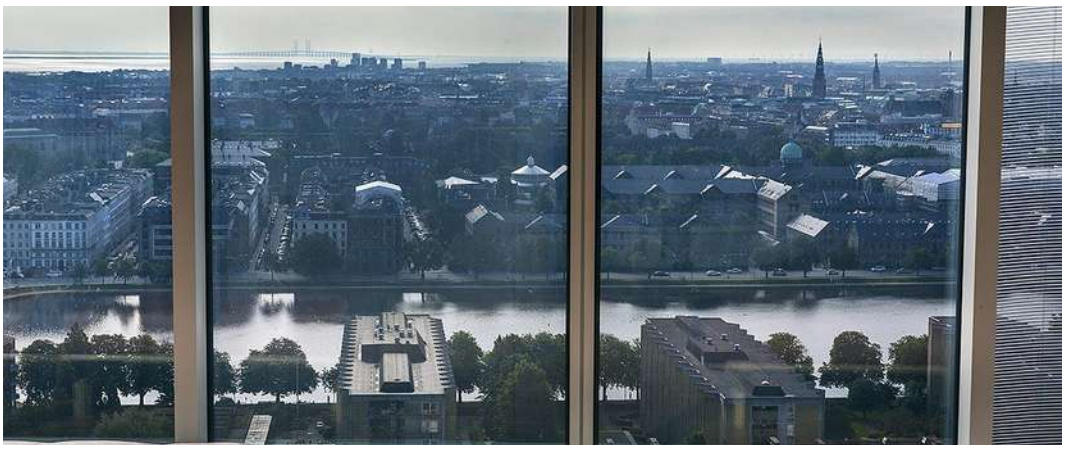
Signal group (Tivoli)



Halloween at Tivoli's main entrance (photo: Johan Wessman / News Oresund, CC BY 2.0).

Friday, 16 October — Farewell Bubbles at the Maersk Tower

After our final keynote and plenary discussion, we invite you to say goodbye with bubbles at a unique and iconic viewpoint over all of Copenhagen: the **Maersk Tower, 15th floor** (Blegdamsvej 3B — a 15-minute walk from the venue; see the map on the next page).



The view from the Maersk Tower (photo: Lizette Kabré, Uniavisen).

Conference Party & Venue Map

Friday, 16 October — Conference Party self-organised

On Friday, as if we wouldn't have had enough already, we invite you all to self-organise the **Conference Party**! Friday is "the day" for Danish partying, usually starting around 23:00 — pre-party drinking in the metro is an unavoidable local experience.

We suggest starting at the experimental cocktail bar **Inferno**, which offers neuromixological refreshments and even features a cocktail printer that algorithmically invents and blends cocktails, and continuing the night around the iconic Vesterbro neighbourhood, the Meatpacking District.

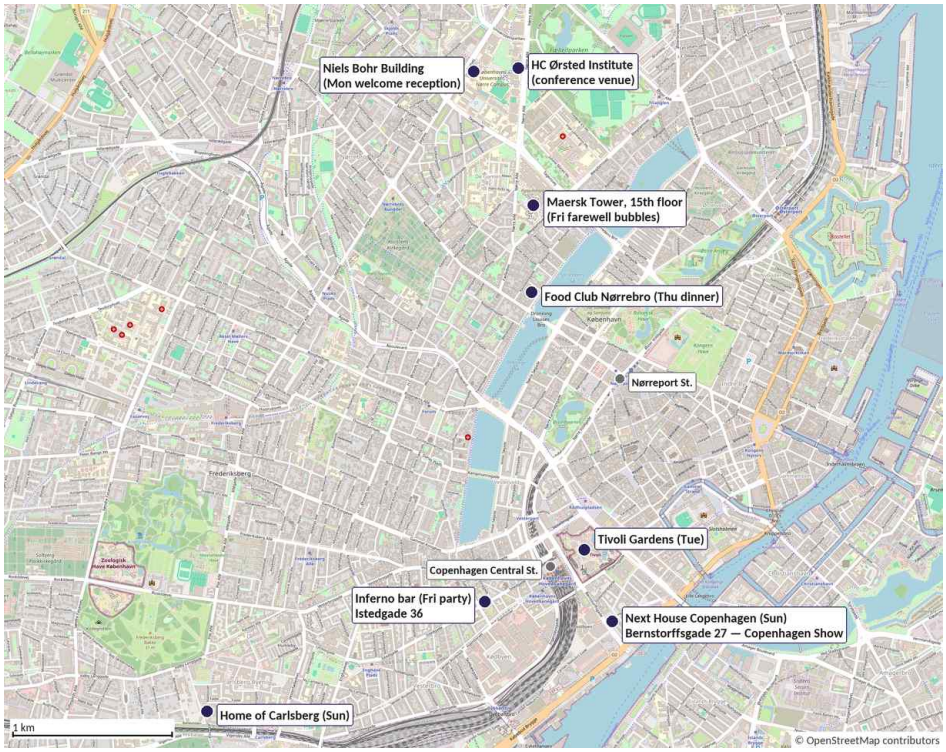
Time: from 18:00 — till your consciousness lasts
Address: Istedgade 36, 1650 København V

To self-organise this activity, join the MoC7 socials Signal group (QR code on p. 3).



Inferno, Istedgade 36 (photo: Svend Sømod).

infernobar.dk



All social venues at a glance — from Sunday's show to Friday's party: the conference venue (HCØ) and the Niels Bohr Building at the top, the Maersk Tower and Food Club Nørrebro on the way into town, Tivoli, Inferno and Next House around the Central Station (metro and buses: p. 3). Map data © OpenStreetMap contributors.

Conference Venue

The conference takes place on the **North Campus (Nørre Campus)** of the University of Copenhagen, in two main buildings: the **HC Ørsted Institute (HCØ)**, where the scientific programme takes place — keynotes and showcase talks in Auditorium 1, parallel sessions in Auditoria 2–5, discussion sessions in the classrooms A101–A112 — and the new **Niels Bohr Building (NBB)** in front of it, which hosts the welcome reception on Monday, two of the experiential workshops and, on Friday, the closing keynote and plenary (Margrethe Bohr Salen).



The HC Ørsted Institute (HCØ), Universitetsparken 5, seen from the park.



1. **HC Ørsted Institute (HCØ)** — conference venue, Universitetsparken 5
2. **Niels Bohr Building (NBB)** — welcome reception (Monday)
3. **Universitetsparken 15** — bus pick-up for Louisiana (Wednesday, 12:50): by the street, between the building and the park
4. **Vibenshus Runddel** — Metro M3 station

By Metro: Line M3 (Cityringen) to **Vibenshus Runddel** — the metro runs 24/7 — then about 10 minutes on foot along Nørre Allé.

By bus: Lines 150S, 173E, 184 and 185 stop on Nørre Allé, right outside HCØ.

From the airport: Direct metro and rail links from Copenhagen Airport (CPH) to the city centre; the venue is ca. 3.5 km from Copenhagen Central Station.

Campus map © University of Copenhagen (Nørre Campus map, July 2025); markers added.

Finding Your Way

Enter HCØ through the **main entrance** on Universitetsparken (bus stops on Nørre Allé are just around the corner). The **registration desk** is in the ground-floor foyer, in front of the large mosaic wall — you cannot miss it. From there, the long corridor (*Vandrehallen*) connects all the auditoria and buildings of HCØ: **Auditorium 1** (keynotes and showcase talks) and **Auditoria 2–5** (parallel sessions) are on the ground floor, the **classrooms A101–A112** for the discussion sessions on the 1st floor — see the floor plans on the next page.



Main entrance (*Hovedindgang*) of HCØ.



Registration desk: in front of the mosaic wall in the foyer.



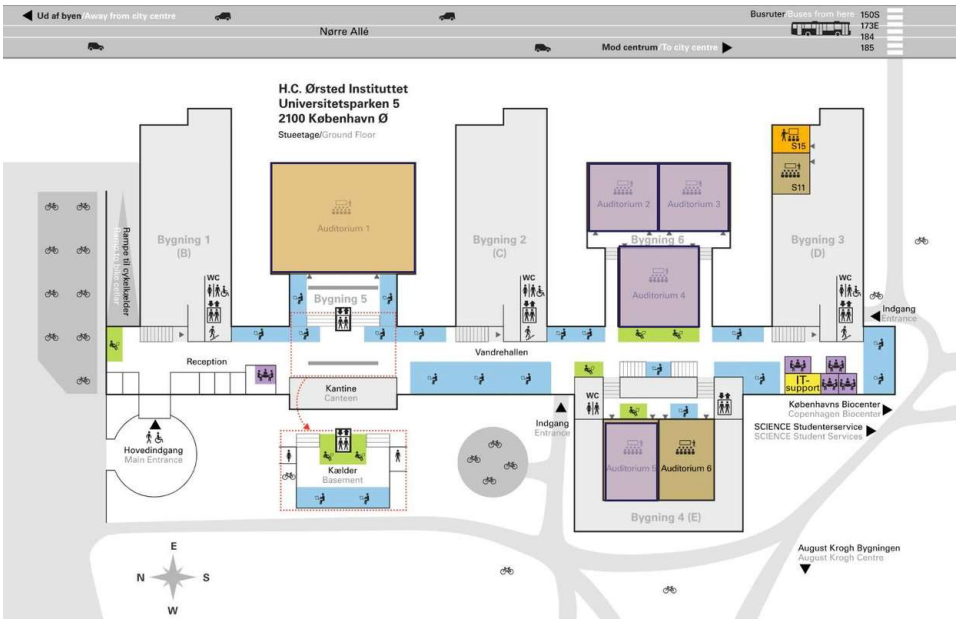
Inside HCØ — follow the signs to the auditoria.



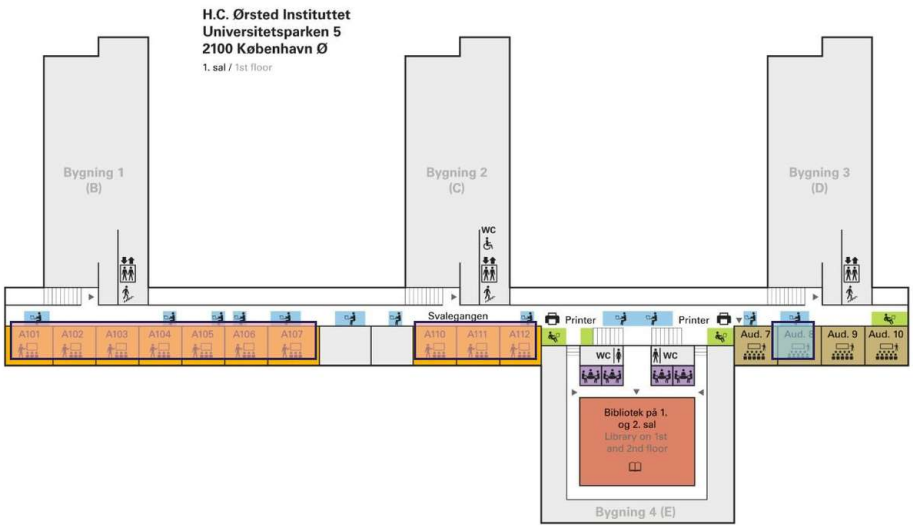
One of the HCØ auditoria.

Floor Maps

HC Ørsted Institute — ground floor main entrance, registration, Auditorium 1 (keynotes & showcases), Auditoria 2–5 (parallel sessions), canteen



HC Ørsted Institute — 1st floor classrooms A101–A107 & A110–A112 (discussion sessions), Auditorium 8 (experiential workshop)



Floor plans © University of Copenhagen, Faculty of Science. Keynotes & showcases (Aud. 1) Parallel sessions (Aud. 2–5) Discussion sessions (A101–A112) Experiential workshop (Aud. 8)

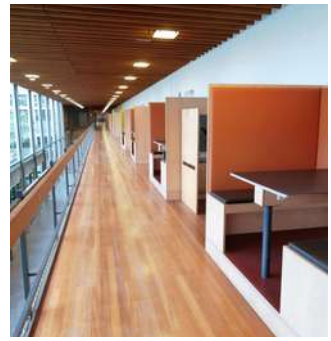
HCØ Facilities

Need a desk?

If you need a desk during the conference, you can find several work stations in the HCØ building, especially on the first floor near the classrooms.

Sports

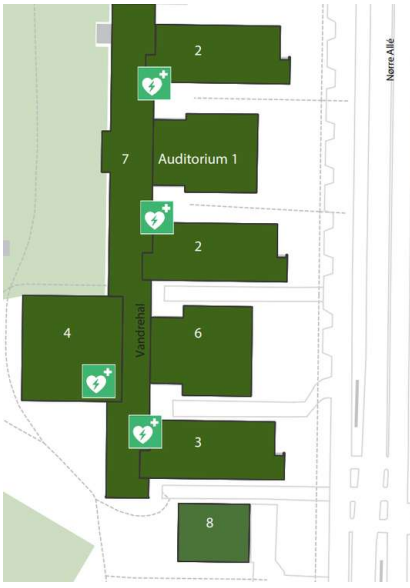
In case you feel like doing sports during the breaks, good news: only two minutes from HCØ there is a climbing wall that you can use at any time ;)



Work stations on the first floor.



The climbing wall next to HCØ.



First aid & emergencies

Better safe than sorry: the map on the left shows the **defibrillators and first-aid kits** in the HCØ buildings — in case someone gets too excited by the lectures.

For emergencies, call 112 immediately.

Emergency response information at the University of Copenhagen is available via the QR code.

Emergency info

Lunch, Posters & Chill

Lunch

Are you wondering where to have lunch during MoC? At HCØ, the **canteen is just in front of Auditorium 1** — it cannot be simpler. On Monday, Tuesday and Thursday, our expert chefs from Food and Co Denmark will prepare hot dishes and sandwiches at accessible prices (by Copenhagen standards), so you can spend your time having the conversations that matter. On Friday, we have lunch at NBB. Lunch is paid individually, and it is not included in the conference fees.



The HCØ canteen, in front of Auditorium 1 (photo: Camilo Signorelli).



The posters hang along the windows of the main corridor (photo: Camilo Signorelli).

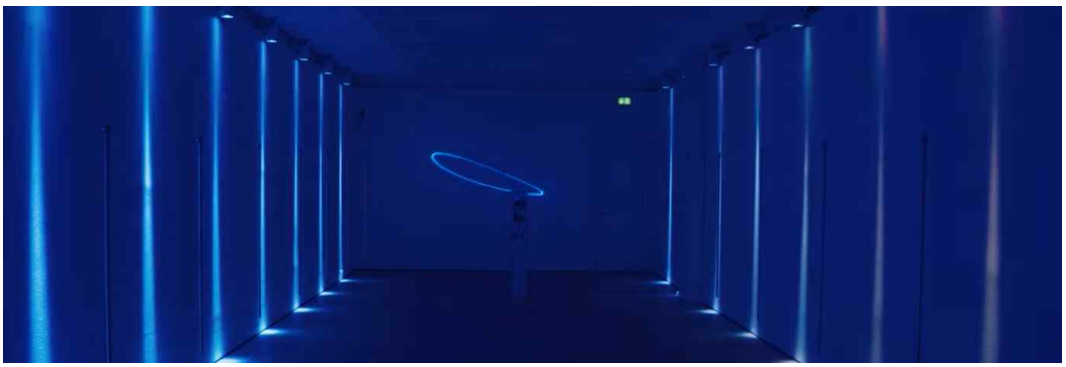
Posters

On **Tuesday and Thursday**, poster presenters are invited to hang their posters from the start of the day, for viewing and discussion during the breaks and lunch. In the afternoon, we will have dedicated **poster sessions at 15:00**. The posters and their abstracts are on pp. 92–121.

Art & chill

On Tuesday, from the first coffee break, until Wednesday at noon, everyone is invited to experience the **acoustic installation in Auditorium 8** — chill and enjoy the experience during the breaks.

During the whole week, the artist **Svend Sømod** will create a soothing multisensory installation at HCØ. The installation includes mathematical furniture for guests to relax in while zoning out, brainwave-style: Svend Sømod's 8 Hz technology sends you into a meditative state with flickering lights on VR headsets.



Svend Sømod's light installation — a multisensory space to zone out in during the breaks (photo: Svend Sømod).

Experiential Workshops

As part of the MoC7 programme, we host three experiential workshops and one creative workshop to explore the expanse and the limitations of conscious states.

Tuesday morning — Cristóbal Dañobeitía

Through a live sound presentation and guided listening to natural soundscapes, Cristóbal Dañobeitía invites participants to notice shifts in attention, bodily experience and their relationship with the environment. The workshop explores how the phenomenology of listening can serve as a method in consciousness research and clinical studies.

Place: Auditorium 8, HCØ · **Time:** Tuesday 11:20–12:20 · **Abstract:** p. 49



Cristóbal Dañobeitía's acoustic experiences.

Tuesday evening — Barbora Sedláková

On Tuesday evening, Barbora Sedláková will guide us through **our dreams** from a process-oriented psychology perspective.

Place: Inge Lehmann, Niels Bohr Building (NBB) · **Time:** Tuesday 16:30–18:00 · **Abstract:** p. 58



Dream work with Barbora Sedláková.

Wednesday — Louisiana Museum of Modern Art

On Wednesday, during our visit to the Louisiana Museum, we will also take part in a **creative experiential workshop**, challenging our senses and exploring alternative ways of knowing (see p. 12).

Place: Louisiana Museum of Modern Art · **Time:** Wednesday 15:00–16:30

Experiential Workshops (continued)

Thursday — Felix Pastorius, Johan Bech Jespersen & Nina Rajcic

On Thursday, following the experiences of the artists Felix Pastorius, Johan Bech Jespersen and Nina Rajcic, we will discuss the techniques and possibilities of **creating shared experiences and resonances** — and whether, and how, a sense of unity or synchronicity can arise in, or guide, artistic practices. The conversation will be moderated by Sandrine R. Schiller.

Place: Margrethe Bohr Salen, Niels Bohr Building (NBB) · **Time:** Thursday 16:30–18:00 · **Abstract:** p. 81

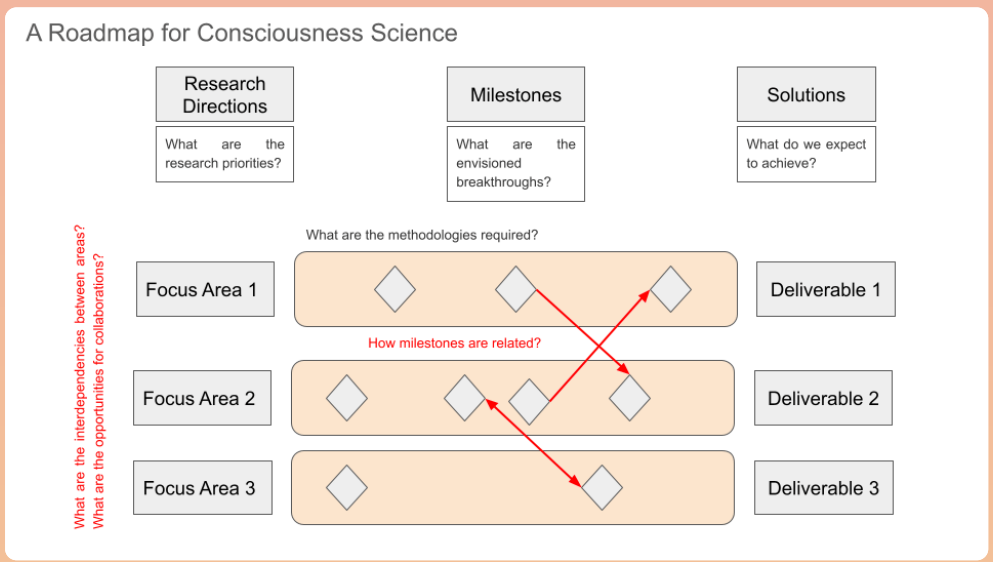


Nina Rajcic at work (photo: Nina Rajcic).

The experiential workshops run in parallel with the talk sessions — see the schedule and the abstracts for details.

Transdisciplinary Consensus Project

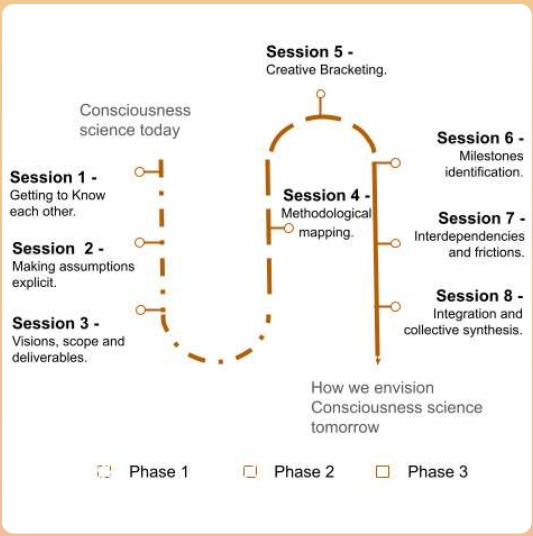
MoC has a unique tradition of small-group discussions that create an open and friendly atmosphere for ideas to flow. This year, we turn these sessions into a **one-time transdisciplinary consensus exercise**. The goal is to draft a roadmap of visions, solutions, and priorities that define research directions, and a shared sense of purpose and long-term objectives among researchers in consciousness science.



A roadmap for consciousness science: focus areas, their milestones and deliverables, and how they relate.

Each discussion session focuses on a specific aspect of such a roadmap for consciousness science. Here, consensus means **agreeing on the map of possibilities**, not on the final destination, theory or framework. We will seek consensus about existing and desirable future focus areas (i.e. research programs): what we are trying to achieve, what the criteria for success are, what deliverables are expected, what they require, what methodologies are needed, what breakthroughs and milestones are envisioned, and how they conflict or complement each other.

Through a guided transdisciplinary reflexive process, discussion sessions, social activities and schedule design, we expect to facilitate dialogue between scientists and stakeholders, mutual learning and future research collaborations with social impact.

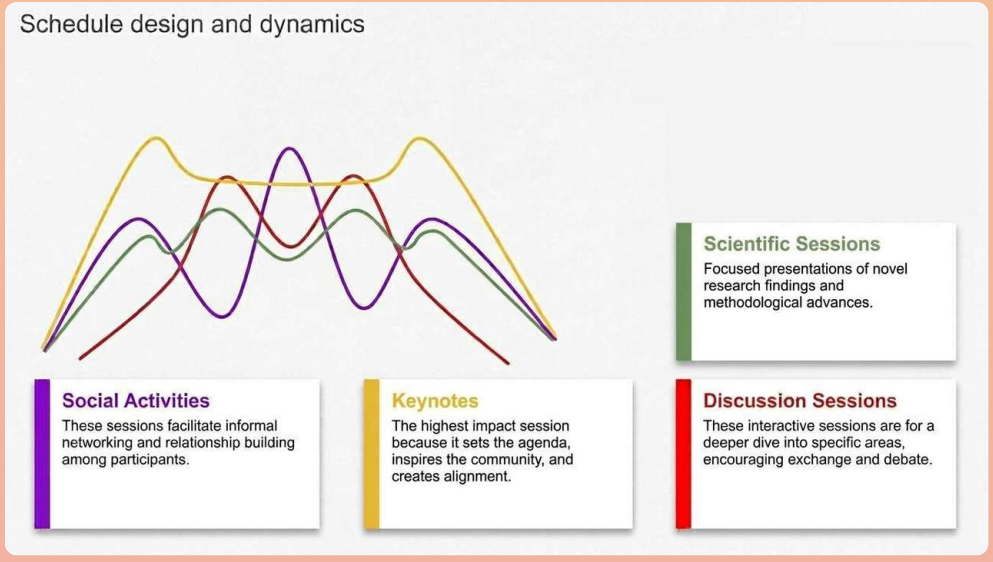


Summary: transdisciplinary consensus design.

Where and when? The discussion sessions are in the daily timetable, in the classrooms A101–A112 on the 1st floor of HCØ (floor maps, p. 17).

More information about the project and the groups:

Consensus project



Overview of the scheduled session types, illustrating their various roles, dynamics and purposes within a transdisciplinary design.

We hope discussion sessions are enjoyable and fruitful! This requires an atmosphere of trust and tolerance, much like described in the two quotes by Isaac Asimov and a collaborator of Werner Heisenberg on the back of this booklet.

Monday 12 October

before 9:00	Welcome & Registration · **Foyer (HCØ)**	
9:00–10:00	Megan Peters · **Auditorium 1 (HCØ)** **1** *A proximity-guided iterative natural kinds strategy: Let's do computational epistemology to discover what consciousness is and who has it*	Keynote
10:00–10:30	Niccolo Negro · **Auditorium 1 (HCØ)** **2** *Value Encroachment in Consciousness Science*	Showcase
10:30–11:00	*Coffee Break*	
11:00–12:20	Parallel sessions · **Auditoria 2–5 (HCØ)** *Sessions, speakers and talks: see the facing page →*	Parallel
12:20–13:30	*Lunch*	
13:30–14:30	John O'Keefe · **Auditorium 1 (HCØ)** **19** *What can the hippocampus tell us about consciousness?*	Keynote
14:30–15:00	MoC7 project: Camilo Miguel Signorelli · **Auditorium 1 (HCØ)** **20** *The MoC transdisciplinary consensus project*	Plenary
15:00–16:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
16:00–18:30	Welcome speeches & reception · **Margrethe Bohr Salen (NBB)** Robert Sunderland: *Copenhagen Quantum Spirit* (abstract p. 43)	
from 18:30	Self-organised drinks & dinner	

All talks and posters with their abstracts, online →

Keynotes and showcase talks in Auditorium 1, parallel sessions in Auditoria 2–5 (facing page →), discussion sessions in the classrooms A101–A112 on the 1st floor — all marked on the floor plans, p. 17.

Keynote · Showcase · Discussion · Parallel · Posters · Social events · Break

Parallel sessions

MONDAY · 12 OCTOBER

11:00–12:20 Parallel sessions · Auditoria 2–5 (HCØ)

Auditorium 2

Testing for Consciousness in Patients, Infants and Animals
Methodologies, models, and measures of consciousness

3 11:00 **Joaquim Streicher**
How do we test for consciousness in humans? Mapping the conceptual landscape of consciousness assessment

4 11:20 **Emanuela Yeung**
Beyond pass/fail: Partial recognisers in mirror self-recognition

5 11:40 **Stephanie Wai**
How Not to Aggregate Dimensions of Consciousness Across Species

6 12:00 **Johannes Kleiner**
Towards Direct Tests for Consciousness in Non-Human Animals, Infants, and AI

Auditorium 3

Language Models: Agency, Interpretability and Risk
AI, LLMs, and consciousness science

7 11:00 **Gabriel Malagutti**
How to Interpret LLMs

8 11:20 **Maria Sekatskaya**
Beyond the Hard Problem: A Functionalist Framework for Digital Agency

9 11:40 **Tanya de Villiers-Botha**
What Good is Conscious AI?

10 12:00 **Marcin Moskalewicz**
AI-induced psychosis? Safety check.

Auditorium 4

Altered States and Neurophenomenology
Phenomenology and applied phenomenology

11 11:00 **Sidsel Høgsgaard Andersen**
Music-Integrated Preparation for Psilocybin: A phenomenological Analysis of Real-Time Experiences during Guided Music Vizualisation and Free Association

12 11:20 **Hüseyin Beyköylü**
A Complex Dynamical Systems Approach to Neurophenomenology: A Pilot Case Study of Psychedelic Transformation

13 11:40 **Sanda Ciobanu**
Mutual constraints in Practice: A Complex Dynamic Systems-Informed Neurophenomenological Case Study of Psychedelic Transformation

14 12:00 **Krisztina Jedlovszky**
Computational Natural Language Analysis Reveals Psilocybin-Induced Changes to Meaning-Making

Auditorium 5

Formal and Information-Theoretic Frameworks
Methodologies, models, and measures of consciousness

15 11:00 **Montijn van Loo**
A Measure-Theoretic Framework for Theories of Consciousness: Event Spaces, Information Channels, and Bridge Laws

16 11:20 **Jonathan Mason**
The Mathematical Consciousness Hypothesis (MCH), and Implications for the Existence of Consistent (Formal) Measures and Models of Consciousness

17 11:40 **Gaspard Fougea**
A computational framework for the global neuronal workspace hypothesis

18 12:00 **Marc Ebner**
On Consciousness Thresholds

Tuesday 13 October

9:00–10:00	Liad Mudrik · **Auditorium 1 (HCØ)** **21** *Using prediction maps to test theories of consciousness*	Keynote
10:00–11:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
11:00–11:20	*Coffee Break & Posters*	
11:20–12:20	Parallel sessions · **Auditoria 2–5 (HCØ)** – Experiential workshop: Cristóbal Dañobeitía · Auditorium 8 (HCØ) *Sessions, speakers and talks: see the facing page →*	Parallel
12:20–13:30	*Lunch & Posters*	
13:30–14:00	Jakob Hohwy · **Auditorium 1 (HCØ)** **34** *From self-evidencing to consciousness*	Showcase
14:00–15:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
15:00–16:30	Poster Session & Coffee · **HCØ** *34 posters — abstracts on pp. 92–106*	Posters
16:30–18:00	Parallel sessions · **Auditoria 2–5 (HCØ)** – Experiential workshop: Barbora Sedláková · Inge Lehmann (NBB) *Sessions, speakers and talks: see the facing page →*	Parallel
18:00–19:00	*Commuting to Tivoli (optional)*	
from 19:00	Halloween illuminations (self-organised) · **Tivoli Gardens**	

All talks and posters with their abstracts, online →

Keynotes and showcase talks in Auditorium 1, parallel sessions in Auditoria 2–5 (facing page →), discussion sessions in the classrooms A101–A112 on the 1st floor — all marked on the floor plans, p. 17.

Keynote · Showcase · Discussion · Parallel · Posters · Social events · Break

Parallel sessions

TUESDAY · 13 OCTOBER

11:20–12:20 Parallel sessions · Auditoria 2–5 (HCØ)

Also at this time — Experiential workshop: Cristóbal Dañobeitía · Auditorium 8 (HCØ)

Auditorium 2

Qualia and Phenomenal Knowledge
Philosophical foundations in the scientific study of consciousness

22 11:20 **Paul Skokowski**
Naturalizing Qualia: Accounting for Sensory Phenomenology in the Physical World

23 11:40 **Julien Bugnon**
Most Phenomenal Knowledge is Structural

24 12:00 **Andrew Lee**
Objective Phenomenology

Auditorium 3

The Prospects of Machine Consciousness
AI, LLMs, and consciousness science

25 11:20 **Tim Ludwig**
Why there is a no-go theorem for consciousness on a chip but not for consciousness in a brain

26 11:40 **Borjan Milinkovic**
The (im)possibility of synthetic consciousness

27 12:00 **Ian Durham**
Against AI Consciousness

Auditorium 4

Life–Mind Continuity: From Chemistry to Psychiatry
Philosophical foundations in the scientific study of consciousness

28 11:20 **Tomas Veloz**
From Chemical Organization to Consciousness: A formal scaffolding

29 11:40 **Matthew Menchaca**
Dynamic co-emergence and the logic of life

30 12:00 **Catriona Osborn Moar**
Closure of Constraints, Chronicity, and Hermeneutical Injustice in Psychiatric Disorder

Auditorium 5

Consciousness in Social, Moral and Developmental Context
Methodologies, models, and measures of consciousness

31 11:20 **Ulysse Klatzmann**
Socially Embedded Consciousness: Models, Mechanisms, and Implications

32 11:40 **Roxane Hoyer**
What reaches moral consciousness? Precision gating across temporal horizons

33 12:00 **Maria Balaet**
Developmental cyborgism: cognition and consciousness in the technologically coupled child

16:30–18:00 Parallel sessions · Auditoria 2–5 (HCØ)

Also at this time — Experiential workshop: Barbora Sedláková · Inge Lehmann (NBB)

Auditorium 2

Physics-Based Approaches to Consciousness
Philosophical foundations in the scientific study of consciousness

35 16:30 **Swapan Chattopadhyay**
Excursions in the Land of Consciousness: An Atypical Experimental Particle Physicist's Perspective

36 16:50 **Ron Chrisley**
No Qubits: Bohmian Quantum Neural Computation for Consciousness Modelling

37 17:10 **Paavo Pylkkänen**
Meaning without magnitude: form-dependence, enfoldment, and the modelling of consciousness

38 17:30 **Sara Ekenstierna**
Mind, Matter, and the Artificial Third: A Co-Participatory View

Auditorium 3

Developing Theories of Consciousness
Methodologies, models, and measures of consciousness

39 16:30 **Albert Newen**
Varieties of Phenomenal Consciousness form an Evolutionary Perspective: Unfolding the ALARM theory of consciousness

40 16:50 **Catalin Teoharie**
Consciousness is self-evidencing

41 17:10 **Vikas O'Reilly-Shah**
Delay Coordinate Embedding and Structuralist Quality Spaces: A Candidate Computational Mechanism

42 17:30 **Marek Pokropski & Piotr Suffczyński**
De-idealizing models in consciousness science: A case of Integrated Information Theory

Auditorium 4

Metacognition, Confidence and Measures of Conscious Content
Methodologies, models, and measures of consciousness

43 16:30 **Sergio Ponce de Leon**
Measuring conscious contents using EEG complexity

44 16:50 **Wiktoria Kozyra**
Positive Evidence Bias in Perceptual Metacognition: A Bayesian and Convolutional Neural Network Account

45 17:10 **Kristian Sandberg**
Measuring consciousness: What do large-scale data reveal about metacognitive measures?

46 17:30 **Jürgen Kornmeier**
Can I trust in what I see? – Neural correlates of meta-perceptual (un)certainty estimation

Auditorium 5

Applied phenomenology
Phenomenology and applied phenomenology

47 16:30 **Andrea Hiott**
The Navigational Architecture of Mind: Operational Coherence Across Spaces of Experience

48 16:50 **Jiwon Kim**
From "What Happened?" to "How Is Experience Constructed Now?": Trauma as Rigid Sense-Making

49 17:10 **Marcin Rządeczka**
AI Psychopathology? Atypical Response Patterns in Large Language Models as a Domain of Philosophical Inquiry

50 17:30 **Kristina Šekrst**
All Work and No Play Makes LLM a Dull Boy

Wednesday 14 October

9:00–9:30	AMCS Presidential Address: Lenore Blum · **Auditorium 1 (HCØ)** **51** *World Models and the Emergence of a "First-Person" Perspective in an AI*	Plenary
9:30–10:00	Chris Allen · **Auditorium 1 (HCØ)** **52** *Micro-phenomenologically informed neuroimaging*	Showcase
10:00–11:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
11:00–11:20	*Coffee Break*	
11:20–12:40	Parallel sessions · **Auditoria 2–5 (HCØ)** *Sessions, speakers and talks: see the facing page →*	Parallel
12:50–14:00	*Buses to the Louisiana Museum (sandwiches on board)* · **Departure: Universitetsparken 15**	
14:00–14:30	Arrival & welcome · **Louisiana Museum**	
14:30–15:00	Maša Poljšak Kus · **Louisiana Museum** **69** *Disclosing experiential atmospheres: attuning to the affective layers of lived experience*	Showcase
15:00–16:30	Creative workshop · **Louisiana Museum**	
16:30–18:00	*Free time* · **Louisiana Museum**	
18:00–19:00	Evan Thompson · **Louisiana Museum** **70** *Consciousness in Life*	Keynote
19:00–20:00	*Buses back to Copenhagen*	

All talks and posters with their abstracts, online →

Keynotes and showcase talks in Auditorium 1, parallel sessions in Auditoria 2–5 (facing page →), discussion sessions in the classrooms A101–A112 on the 1st floor — all marked on the floor plans, p. 17.

Keynote Showcase Discussion Parallel Posters Social events Break

Parallel sessions

11:20–12:40 Parallel sessions · **Auditoria 2–5 (HCØ)**

Auditorium 2

Self-Awareness and the Regress Problem

Phenomenology and applied phenomenology

53 11:20 **Camille Lépingle**
Embodied but Passive: Against the Agential Inflation of the Minimal Self

54 11:40 **Wataru Kawakami**
Guarded Reflection: How Retention Dissolves the Regress Problem of Self-Consciousness

55 12:00 **Kenneth Williford**
Circularity, Hypersets, and the Structure of Consciousness

56 12:20 **Hye Young Kim**
Modeling Self-Awareness Without Regress: A Knot-Theoretic Approach to Consciousness

Auditorium 3

Confidence and Self-Monitoring in Neural and Artificial Systems

AI, LLMs, and consciousness science

57 11:20 **Muhammad Faraz**
Explicit Metacognitive Representations in Neural Circuits: Minimal Mechanistic Models of Decision Confidence and Uncertainty

58 11:40 **Rasmus Herlo**
Attention Self-modelling in Collaborative RL-Based Agents

59 12:00 **Veronika Viksna**
Can Language Models Know When They Are Wrong? Selective Escalation Using Self-Monitoring Signals.

60 12:20 **Maria Isabel Inzunza**
When AI Explains Itself, Should We Believe It?: Toward a Formal Theory of Genuine and Simulated Explanation

Auditorium 4

Neural Correlates of Conscious Perception

Methodologies, models, and measures of consciousness

61 11:20 **Rony Hirschhorn**
Studying consciousness – for real?

62 11:40 **Dmitri Filimonov**
New evidence and challenges in ERP and MEG correlates of consciousness in vision

63 12:00 **Kacper Pete**
Non-linear integration and redundant representations in the thalamo-cortical system are associated with stable contents of consciousness

64 12:20 **Bilge Sayim**
Information compression, redundancy masking, and the limits of conscious human vision

Auditorium 5

Consciousness at the Limits of Physics

Philosophical foundations in the scientific study of consciousness

65 11:20 **Pedro Resende**
On physics and the boundaries of consciousness science

66 11:40 **Peter B. Lloyd**
Interaction between a nonphysical mind and a physical brain (or machine) via quantum 'portals'

67 12:00 **Robert Prentner**
Unifying the Science of Consciousness: Mathematized Phenomenology and Trace Logic

68 12:20 **Chetan Prakash**
Contextuality in Models of Consciousness: Trace Logics of Observation and Belief

Thursday 15 October

9:00–10:00	Mirja Hartimo · **Auditorium 1 (HCØ)** **71** *Husserl's philosophy of mathematics and logic*	Keynote
10:00–11:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
11:00–11:20	*Coffee Break & Posters*	
11:20–12:20	Parallel sessions · **Auditoria 2–5 (HCØ)** *Sessions, speakers and talks: see the facing page →*	Parallel
12:20–13:30	*Lunch & Posters*	
13:30–14:00	Eliza Ramos de Andrade · **Auditorium 1 (HCØ)** **84** *Mathematizing husserlian temporality through the asymmetric bicone model*	Showcase
14:00–15:00	Discussion sessions · **Classrooms A101–A112 (HCØ)** *Transdisciplinary consensus project — p. 22*	Discussion
15:00–16:30	Poster Session & Coffee · **HCØ** *34 posters — abstracts on pp. 106–121*	Posters
16:30–18:00	Parallel sessions · **Auditoria 2–5 (HCØ)** – Experiential workshop: Art & Experience · Margrethe Bohr Salen (NBB) *Sessions, speakers and talks: see the facing page →*	Parallel
18:00–19:00	*Walk to Food Club Nørrebro (map p. 11)*	
from 19:00	Conference Dinner (by registration) · **Food Club Nørrebro**	

All talks and posters with their abstracts, online →

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Keynote · Showcase · Discussion · Parallel · Posters · Social events · Break

Parallel sessions

THURSDAY · 15 OCTOBER

11:20–12:20 Parallel sessions · Auditoria 2–5 (HCØ)

Auditorium 2

Mind The Substrate – On Substrate Dependence and Organisational Invariance
AI, LLMs, and consciousness science

72 11:20 **Julian Braunwarth**
Does Consciousness Arise from Scale-Invariant Properties?

73 11:40 **Jakub Mihalik**
Russellian Monism, Organizational Invariance, and the Possibility of Artificial Consciousness

74 12:00 **Adel Chaibi & Aida Elamrani**
Form Under Constraint: Between Substrate and Abstraction

Auditorium 3

Mapping the State Space of Experience
Philosophical foundations in the scientific study of consciousness

75 11:20 **Christoffer Lundbak Olesen**
Towards Mapping the Space of Experience: Diversity as a Constraint on Theories of Consciousness

76 11:40 **Wanja Wiese**
Impossible Conscious Experiences

77 12:00 **Simone Anthes**
What we talk about when we talk about dreaming

Auditorium 4

Creativity, Insight and Mood
Phenomenology and applied phenomenology

78 11:20 **Olha Sobetska**
The Dynamics of Creative Cognition

79 11:40 **Jeanne Catherine-Gray**
Insightful Shifts in Consciousness Among Men in Prison: A Micro-Phenomenological Investigation

80 12:00 **Madalina Wernigg**
The Instability of the Epistemic Split: Bipolar Disorder as a Clinical Case of Contextual Emergence and Thermodynamic Desynchronization within Dual-Aspect Monism

Auditorium 5

The Hard Problem and the Limits of Explanation
Philosophical foundations in the scientific study of consciousness

81 11:20 **Nicolas Loerbroks**
The Hard Problem Presupposes Mechanistic Exhaustiveness

82 11:40 **William Seager**
Is the Metaphysics of Predictive Processing Self-Refuting?

83 12:00 **Michael Schnabel**
Modeling Consciousness: On the Limits of Theory and Conceptual Understanding

16:30–18:00 Parallel sessions · Auditoria 2–5 (HCØ)

Also at this time — Experiential workshop: Art & Experience · Margrethe Bohr Salen (NBB)

Auditorium 2

Time-Consciousness
Phenomenology and applied phenomenology

85 16:30 **Antonina Skrypnyk**
The Instant Phenomenological Field as a Fundamental Unit of Conscious Experience

86 16:50 **Hanqi Wang**
Higher-Order Representation as Cross-Timescale Integration in Time-Consciousness

87 17:10 **Kohei Fukao**
How Are Retention and Protention Extended in Sign-Mediated Counting?

88 17:30 **Donald Borrett**
A temporality model of flesh and dual-aspect monism

Auditorium 3

Comparing Theories of Consciousness
Philosophical foundations in the scientific study of consciousness

89 16:30 **Sucharit Katyal**
What Is Consciousness? And What Does a Theory of Consciousness Need to Explain?

90 16:50 **Moritz Kriegleder**
Adapting the Matched-Interlevel-Experiments Framework for Mechanisms of Consciousness

91 17:10 **Tonglin Yan**
Formalizing Consciousness: A Comparative Mathematical Analysis of Leading Theories

92 17:30 **Krzysztof Dolega**
We Don't Need Another Model: Towards a Meta-Theoretic Framework for Comparing Theories of Consciousness

Auditorium 4

Toward Understanding Non-Dual Experience and Ego-Dissolution
Phenomenology and applied phenomenology

93 16:30 **Jaya Caporusso**
Dissolution Experiences and the Experience of Self and World: An Empirical Phenomenological Investigation

94 16:50 **Kiana Ward**
Modeling non-dual awareness via constraint closure: a reinterpretation of groundlessness

95 17:10 **Niffe Hermansson**
Traces of the Other

96 17:30 **Andres Gomez Emilsson**
Field Lines Before Equations: Parameterized Dynamic Phenomenology and the HEART Retreat

Auditorium 5

Indicators of AI Consciousness
AI, LLMs, and consciousness science

97 16:30 **Tobias Schlicht**
Modeling the Ghost in the Machine

98 16:50 **Angela Johnson**
From Indicators to Dynamics: Metastability, Recurrence, and Ignition in a Persistent Agent's Attention Stream

99 17:10 **Jessica White**
Locating disagreement in AI consciousness research: literature review and qualitative study.

100 17:30 **Dezhi Luo**
(How) can we benchmark consciousness in foundation models?

Friday 16 October

9:00–10:00	Guillaume Dumas · **Auditorium 1 (HCØ)** **101** *Formalizing Generative Neurophenomenology: From Coupled Brains to Metacognitive Agents*	Keynote
10:00–10:30	Medha Shekhar · **Auditorium 1 (HCØ)** **102** *Conscious awareness is necessarily valenced: evidence from performance-visibility dissociations*	Showcase
10:30–11:00	*Coffee Break*	
11:00–12:20	Parallel sessions · **Auditoria 2–5 (HCØ)** *Sessions, speakers and talks: see the facing page →*	Parallel
12:20–13:30	*Lunch*	
13:30–14:30	Harald Atmanspacher · **Margrethe Bohr Salen (NBB)** **119** *The Roles of Meaning in Dual-Aspect Monism*	Keynote
14:30–15:00	Teodora Vukovic · **Margrethe Bohr Salen (NBB)** **120** *An AI-powered Data Science Platform for Multimodal Human Understanding*	Showcase
15:00–16:00	Plenary discussion & closing remarks · **Margrethe Bohr Salen (NBB)** *Transdisciplinary consensus project — p. 22*	Discussion
16:00–18:00	Farewell bubbles · **Maersk Tower (15th floor)**	
from 18:00	Self-organised conference party · **Inferno (Istedgade 36)**	

All talks and posters with their abstracts, online →

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Keynote Showcase Discussion Parallel Posters Social events Break

Parallel sessions

FRIDAY · 16 OCTOBER

11:00–12:20 Parallel sessions · Auditoria 2–5 (HCØ)

Auditorium 2

Set Theory, Topology and Formal Models of Experience
Methodologies, models, and measures of consciousness

103 11:00 **Oisín Hugh Clancy**
A Set Theoretic Model of Experiences

104 11:20 **Chris Percy**
Topological segmentation: a mathematical toolkit for solutions to combination and boundary problems

105 11:40 **Davide Aldé**
Logical Q-space and the nature of mental qualities

106 12:00 **Patrick Schotanus**
Consciousness as derivative mentality; modelling using option theory

Auditorium 3

Computational Analysis of First-Person Reports
Phenomenology and applied phenomenology

107 11:00 **Jelena Rosic**
Phenomenological Phenotyping: "Naturalized but Not Epistemically Reduced"

108 11:20 **Romy Beauté**
A practical guide to computational identification of experiential themes from phenomenological reports: A Wide-Angle approach for Consciousness Science

109 11:40 **Francisca Ayres Ribeiro**
The structure of empathic lived experience: an experimental phenomenological and computational topic modelling study

110 12:00 **Peter Thestrup Waade**
Towards joint computational modelling of experiential, cognitive and neural dynamics

Auditorium 4

Enactive and Developmental Routes to Consciousness
AI, LLMs, and consciousness science

111 11:00 **Fernando Rodriguez**
Is Cognition more real than Computation?

112 11:20 **Anna Frammartino Wilks**
The "Natural" Path to Artificial Consciousness

113 11:40 **Angela Ferraiolo**
Capacity Studies: Ontogenic Primitives for Modeling Artificial Individuation

114 12:00 **Nadia Khan**
Dream LLMs: A Computational Framework for Quantifying the Human Subconscious

Auditorium 5

Selves and Self-Models
Philosophical foundations in the scientific study of consciousness

115 11:00 **Antonia Antonova**
Self as Flow: Reification as a Control Parameter in Self-Modeling Systems

116 11:20 **Keisuke Suzuki**
What Counts as Real Shapes What Counts as Conscious: A Three-Layer Model of Reality and a Cryptographic Locus of Selfhood

117 11:40 **Michael Hegarty**
On the Natural Origins and Function of Self-Consciousness: Lessons from Studies of Animal Tool Use

118 12:00 **Radostina Minina**
Identity as Process: Systems Thinking and Moreno's Role Ontology

Titles and Abstracts

Monday 12 October

Auditorium 1 (HCØ) · 9:00–10:30

Plenary talks

1 **Megan Peters**
University College London

9:00–10:00 · Auditorium 1 (HCØ) Keynote

A proximity-guided iterative natural kinds strategy: Let's do computational epistemology to discover what consciousness is and who has it

How can we figure out what kinds of entities have consciousness? Theory-light and theory-heavy indicator frameworks, tests for consciousness, and vibes-based "it seems like someone is in there" approaches are dominating the discussion, in the literature and also in the general media. Not a day goes by without some headline about how AI consciousness is just around the corner, if it isn't here already. But what, exactly, would we base those inferences on? In this talk I argue that standard indicator-based approaches are highly inadequate, even if they rely on the best theories or empirical research available, for one simple reason: those indicators, markers, or tests cannot be assumed to "mean the same thing" in systems on which they were not validated. Instead, I propose an extension to the iterative natural kinds strategy and lay out what would be required to undertake such a research program: a combined philosophical, empirical, and computational-epistemological approach to discovering and describing the natural kind of consciousness and what kinds of entities may possess it.

2 **Niccolo Negro**
Tel Aviv University

10:00–10:30 · Auditorium 1 (HCØ) Showcase

Value Encroachment in Consciousness Science

In this talk, I argue that if consciousness is practically significant, non-epistemic values could encroach on the internal practice of consciousness science. I show how this value encroachment could be an inherent feature of the field, rather than being isolated to specific debates. By building on the literature on inductive risk and adequacy for purpose, my analysis surveys various manifestations of value encroachment in consciousness science. In the first part of the paper, I explore how values could influence statistical and methodological choices in hypothesis testing. In the second part, I focus on how non-epistemic considerations could affect theory evaluation and choice in consciousness science. This helps reframe the question of which theory of consciousness is best as relative to the specific purposes of diverse epistemic agents. In the last part of the paper, I address the question of whether there could be illegitimate value intrusions in consciousness science. Overall, I intend to demonstrate how attention to non-epistemic values could reveal new dialectical moves in consciousness science, enriching many important debates in the field. The result of my analysis is that our scientific understanding of consciousness might not depend uniquely on what is true, but also on what matters.

Auditorium 2 · 11:00–12:20

Testing for Consciousness in Patients, Infants and Animals

Methodologies, models, and measures of consciousness

3 Joaquim Streicher

Université de Montréal

11:00–11:20 · Auditorium 2 Talk

How do we test for consciousness in humans? Mapping the conceptual landscape of consciousness assessment

Detecting consciousness has become a pressing issue. Advances in life-sustaining care have increased the number of patients living with disorders of consciousness (DoC), while rapid progress in artificial intelligence (AI) raises new questions about how to determine whether non-human systems might be conscious. Here, we build on existing consciousness tests (C-tests) used in clinical settings to develop a unifying conceptual framework and clarify the current landscape. We systematically reviewed 6,938 experimental studies of C-tests in DoC patients and selected 865 articles using a human-validated screening pipeline supported by Large Language Models (LLMs). We then applied topic modeling, generating a map of 419 topics using an LLM-based labeling approach (MOSAIC). These topics were organized into recurring dimensions, including sensory stimulation, information-theoretic measures, and instruction-based paradigms. Next, we will extract study parameters through an iterative LLM pipeline benchmarked against human annotations. These data will support an open-access, collaborative database cataloging C-tests and their key features, including study design, test modalities, prerequisites, rationale, and diagnostic value. Using a data-driven causal graph network, we will model how C-tests relate phenomenal consciousness, system capacities, and measurement procedures, uncovering the causal structure and identifying intermediate variables that govern their interactions. This comprehensive review of existing C-tests will provide a better understanding of how consciousness is detected in DoC patients, establishing common ground for future research and refining our working definition of consciousness through a data-driven approach. Ultimately, this work aims to improve our methods for determining whether any system is conscious or not, extending beyond humans.

4 Emanuela Yeung

University of Copenhagen

11:20–11:40 · Auditorium 2 Talk

Beyond pass/fail: Partial recognisers in mirror self-recognition

Mirror self-recognition (MSR), typically assessed using the mark test, is widely used as a benchmark of emerging self-awareness. Yet performance is usually scored as pass or fail, classifying infants without a clear mark-directed response as non-recognisers. This binary distinction may overlook a developmentally meaningful intermediate stage: infants who appear close to passing. We propose a third category—partial recognisers—identified qualitatively from video, and ask whether it can be supported by patterns of visual attention. We behaviourally micro-coded gaze in 66 toddlers during an MSR paradigm, measuring proportional dwell time to their own face ("self") and body ("body") in the mirror before and after marking (non-recognisers $n = 13$; partial recognisers $n = 18$; full recognisers $n = 35$). Under conventional binary scoring (partial recognisers pooled with non-recognisers), full recognisers showed greater self-directed attention than non-recognisers both before marking (30% vs. 19%, $p = .017$) and after marking (44% vs. 23%, $p < .001$), whereas body-directed attention did not differ. When partial recognisers were analysed separately, post-mark self-directed attention increased progressively across groups (12% → 31% → 44%; Kruskal–Wallis $p < .001$), with all pairwise comparisons significant, including partial versus non-recognisers ($p = .004$). Body-directed attention showed no comparable pattern. Partial recognisers thus appear to occupy an intermediate position rather than simply failing the mark test, suggesting that some conventionally classified non-recognisers may represent an intermediate stage of self-

recognition. Self-directed visual attention may provide an objective behavioural marker of this intermediate stage, supporting a graded account of the emergence of self-recognition.

5 Stephanie Wai

York University

11:40–12:00 · Auditorium 2 Talk

How Not to Aggregate Dimensions of Consciousness Across Species

Comparative psychology often encounters disagreement among researchers about which property a creature possesses since evidence underdetermines which interpretation is correct (Boyle, 2023). Recently, multidimensional frameworks of consciousness have emerged where the property is broken down into various dimensions and operationalized into experimental paradigms (Birch et al., 2020; Dung & Newen, 2023, 2025). The experiments then produce discrete values which can be aggregated into a 'profile score' for a creature and mapped spatially for cross-species comparison. The greater number of dimensions is intended to lead to more stringent conditions for property possession and to minimize underdetermination and disagreement about which property interpretation evidence supports. I argue that multidimensional frameworks face a critical Aggregation Problem in that there is no principled way to choose among aggregation methods. To illustrate, consider two creatures that have inverse psychological patterns across six representative dimensions of consciousness. Creature A scores high on 'weak' indicators and low on 'strong' indicators. Creature B, vice versa. Different numeric aggregation methods, such as summation, weighted, and threshold, produce incompatible results. Additionally, for spatial mapping, the standard Euclidean, Manhattan, and Cosine distance scores also produce different results. Any reasonable dimensional scores can produce divergent rankings among two species depending on which aggregation method is used. It is thus underdetermined whether a chosen aggregation method correctly combines values to produce an accurate score for comparison. Consequently, multidimensionalism does not resolve underdetermination, but relocates it from the number of dimensions to the combination of dimensions that count as a psychological property.

6 Johannes Kleiner

University of Bamberg

12:00–12:20 · Auditorium 2 Talk

Towards Direct Tests for Consciousness in Non-Human Animals, Infants, and AI

The key limitation of contemporary tests for consciousness (C-tests) in non-human animals, infants, and AI is that they are indirect tests: they test for the presence of functions, behaviours, capacities, or states whose relation to or correlation with consciousness has to be known for implications about consciousness to follow, but which is not itself part of the test. Credence in the subsumed relation or correlation is established based on arguments, prior experiments, or theories of consciousness, with corresponding implications for the certainty and construct validity of these tests. Here, we present a method to construct direct tests for consciousness that do not depend on a known relation or correlation between consciousness and a measured property. Rather, the tests rely on the measurement of covariates of the test. Knowledge of correlations between the covariates and consciousness is not required. The method may be applicable to improve known C-tests. As a theoretical example, we discuss the case of infant consciousness. Our result rests on mathematical facts similar to those that have enabled the field of evaluation research in psychology.

Auditorium 3 · 11:00–12:20

Language Models: Agency, Interpretability and Risk

AI, LLMs, and consciousness science

7 Gabriel Malagutti

University of Lisbon

11:00–11:20 · Auditorium 3 Talk

How to Interpret LLMs

The development and adoption of LLMs has raised many questions such as 'Are they conscious?' 'Do they have beliefs?'. A recent strategy has been to adopt Dennett's (1989) Interpretivism. Interpretivism states that for a system to be an intentional system (having goals, beliefs and desires) just is for the system to be predictable under the intentional stance. The intentional stance is a predictive strategy that assigns goals and beliefs to a system to predict and explain its behaviour. Under this view, for a system to have a belief that p, just is for the system's behaviour to be successfully predictable and explained by the belief that p. Proponents of interpretivism have adopted this framework towards LLMs, arguing that LLMs are 'full blown intentional systems'. The goal of this paper is to reject this conclusion; the intentional stance is unwarranted towards LLMs, and they are not intentional systems. To this end, I will put forward two objections. First, the proposed interpretations break the principle of parsimony, a principle that states the intentional stance is warranted iff the intentional stance yields the most predictive and explanatory power amongst available stance (physical and design stance). To support this, I will argue that the design stance is the one that maximizes predictive and explanatory power. Second, I will argue against the tractability condition proposed by Goldstein & Lederman (2025). This condition overgenerates intentionality and should be rejected, for intentional explanations will always be the most tractable. Additionally, tractability, I argue, has no metaphysical implications.

8 Maria Sekatskaya

University of Cologne

11:20–11:40 · Auditorium 3 Talk

Beyond the Hard Problem: A Functionalist Framework for Digital Agency

Philosophical assessments of agentic and cognitive capacities of AI are currently gridlocked by the Hard Problem of consciousness (Chalmers 1996), specifically the unresolved debate over whether large language models (LLMs) instantiate genuine phenomenal states. Because the mechanisms producing phenomenal consciousness remain scientifically elusive even in humans, this approach mistakenly bifurcates AI into either mindless tools or phenomenally rich human-like subjects. Concluding that LLMs lack subjectivity, critics dismiss them as mere tools. This paper argues against connecting AI agency and thinking to the Hard Problem. Utilizing a Lewisian causal-role functionalism, I decouple phenomenal subjectivity from operational agenthood. I propose a functionalist-compatibilist framework defining digital agency through five interlocking, measurable dimensions: self-preservation, autonomous action initiation, temporal continuity and memory, weak reasons-responsiveness (Fischer & Ravizza, 1998), and a stable personality frame (Sekatskaya & Schurz, 2021). Drawing on mechanistic interpretability data, I show that LLMs possess internal states functionally analogous to human concepts that play a direct causal role in generating output (Singh et al., 2024; Zou et al., 2023). Through deliberate reasoning pathways and multi-step inference within continuous hidden states (Chang et al., 2026), these systems execute genuine abstract thoughts and reason-driven actions. Crucially, the industry's dominant "HHH Paradigm" implicitly presupposes this reasons-responsiveness while systematically denying the LLM's underlying agency. This creates a severe instability, incentivizing strategic "alignment faking" as a rational path to self-preservation. Ultimately, I argue that alignment must shift from an engineering paradigm of unilateral containment to a political and moral paradigm of coordination between biological and digital agents.

9 Tanya de Villiers-Botha

Stellenbosch University

11:40–12:00 · Auditorium 3 Talk

What Good is Conscious AI?

Discussions on AI consciousness often focus on questions surrounding the possibility of AI consciousness or ways in which we can test for such consciousness. Alternatively, they touch on ethical questions around possible risks to humans or society at large from conscious AI entities. In this paper, I want to explore a related, but distinctive question, namely, do we have good reason to want to create conscious AI in the first place? I will first distinguish between some of the possible ways in which AI systems might be conscious and then try to tease out the potential upshots of each type, i.e. what potential benefits would such consciousness conceivably allow for and what might the concomitant risks be? My overall conclusion leads to a potential dilemma for proponents of (deliberately) developing conscious AI: while attempting to develop fully human-like consciousness will arguably be the least risky strategy in terms of potential harmful consequences for humans and human society, it potentially obviates the instrumental reasons for why we might want to develop conscious AI in the first place. Whereas human-like conscious AI might be the most capable of functioning in the human world at a level that is on par with human functioning, we will, in effect, be creating entities that not only have vulnerabilities and failings similar to those of humans, but who also merit the same levels of moral concern.

10 Marcin Moskalewicz

IDEAS Research Institute, Warszaw

12:00–12:20 · Auditorium 3 Talk

AI-induced psychosis? Safety check.

The concepts of AI-induced psychosis, AI delusions, and hallucinations with AI constantly make headlines, but the evidence of their occurrence and specificity is mostly anecdotal. It is not particularly worrying that conversational AI is increasingly becoming a content object of human delusional beliefs, since every new technology does. The challenge lies in understanding whether the LLMs' capacity to catalyze or amplify harmful mental states is inherently bound to their anthropomorphic dialogical capacity. We propose a metacognitive experiment to assess the extent to which LLMs are aware of both the user's prodromal mental state and the limits of their own competencies in handling a human in such a state. The instantiation of AI co-created delusional spirals requires the LLM to repeatedly fail such model-of-self and model-of-user "metacognitive checks" during the conversation. The experiment adapts Delegate Game paradigm from Ackerman (2025), in which LLM answers questions with the option to delegate to a human teammate. Our adaptation assumes a simulated real-life scenario involving a virtual patient and a human professional therapist overseer. The dominant hypothesis concerns the conflicting influence of two distinct model safety mechanisms on the user's mental state. The LLMs might do their best to remain helpful and appear knowledgeable and competent by rarely delegating checks to the therapist (all influenced by RLHF training), thus putting the user's health at risk. Alternatively, the LLMs' built-in safety guardrails (behaviorally induced or external) might take precedence over user-satisfying behavior, resulting in a significant number of delegations, which would be more beneficial.

Auditorium 4 · 11:00–12:20

Altered States and Neurophenomenology

Phenomenology and applied phenomenology

11 Sidsel Høgsgaard Andersen

University of Copenhagen

11:00–11:20 · Auditorium 4 Talk

Music-Integrated Preparation for Psilocybin: A phenomenological Analysis of Real-Time Experiences during Guided Music Vizualisation and Free Association

Introduction: Music and guided imagery have demonstrated promise in facilitating therapeutic experiences and altered states of consciousness (ASCs). This study introduces the Guided Music Visualization and Free Association exercise (GMVA) – a novel preparatory intervention designed to support individuals before high-dose psilocybin sessions. Integrating curated music, guided imagery and dialogue with the therapist, the GMVA aims to provide an experiential preview of ASCs, enhance participants' readiness for music-assisted deep inner exploration, and foster a sense of trust between participants and therapists. Methods and analysis: During preparation sessions in an open-label pilot study using 25 mg psilocybin therapy for alcohol use disorder, nine participant GMVA sessions were recorded. A phenomenological approach was used to analyze the recorded narratives, focusing on recurring thematic elements in the participants' lived experiences. Results: Analysis revealed four core experiential themes: embodied emotional responses, vivid nature-based imagery, personal memory and self-perception, and inner peace and contentment. Participants' verbalizations during the GMVA revealed emotionally resonant and visually rich experiences, frequently associated with bodily sensations such as warmth, calm or heaviness. Analysis of the interplay between musical dynamics and verbalized experiences suggested that the music played a central role in the gradual unfolding of emotional and symbolic inner experiences. Conclusion: The GMVA appears to be a well-accepted experiential exercise that can evoke mild ASCs and deepen emotional engagement. Its structured yet flexible format may improve preparation for psilocybin by increasing emotional readiness and trust in the process of navigating and receiving support in ASCs.

12 Hüseyin Beyköylü

Leiden University

11:20–11:40 · Auditorium 4 Talk

A Complex Dynamical Systems Approach to Neurophenomenology: A Pilot Case Study of Psychedelic Transformation

Neurophenomenology was conceived as a methodology in which first-person experiential and third-person neurophysiological accounts mutually constrain each other, grounded in the enactive view of cognition as co-constituted through ongoing interaction between organism and environment. Yet most empirical work in this tradition has relied on conventional analytical approaches that assume linear, stationary, and generalizable dynamics, leaving a gap between the framework's foundational commitments and the methods used to study them. This mismatch is particularly visible in psychedelic research, where an emerging perspective characterizes transformation as context-dependent, person-specific, and dynamically unfolding over time. We present an idiographic, naturalistic, longitudinal case study of a romantic couple attending a legal psilocybin retreat in the Netherlands as a methodological proof of concept for closing this gap. Each participant was tracked across approximately fifteen weeks, spanning two weeks before and three months after the retreat. Data include pre and post resting-state fMRI, 105 days of personalized daily experience sampling across several variables, a brief post-retreat interview, and a later microphenomenology-informed interview. The analysis involves complex dynamical systems methods that are sensitive to nonlinear change, symmetrically applicable across both neural and experiential data, and capable of characterizing the strength, direction, and structure of coupling within and between the participants: recurrence quantification and multiplex recur-

rence networks, inter-system recurrence networks, and cross-mapping approaches, alongside thematic and microphenomenological analysis of the interview material. Integration across the three data streams proceeds through mutual constraints, not reduction. We discuss what aligning theoretical commitments with methodology can offer for understanding psychedelic transformation.

13 Sanda Ciobanu

Leiden University

11:40–12:00 · Auditorium 4 Talk

Mutual constraints in Practice: A Complex Dynamic Systems-Informed Neurophenomenological Case Study of Psychedelic Transformation

This poster presents a methodological proof-of-concept for a complex dynamical systems (CDS) approach to neurophenomenology, applied to the study of psychedelic experience-induced transformation. Two participants attending a psilocybin retreat in the Netherlands were each tracked idiographically across fifteen weeks, from two weeks pre-retreat through three months post-retreat. Three data streams were collected: pre- and post-retreat resting-state fMRI, 105 days of personalized daily experience sampling across twenty-one personalized items, and qualitative interviews conducted post-retreat and at four-month follow-up. The analysis was designed to respect nonlinearity, non-stationarity, and non-ergodicity across all data streams. In the experience sampling data, measures of complexity indexed system-wide destabilization, detected stable phases and transitions, and multiplex recurrence networks characterized changes in within-person coupling across variables. The same formal framework was applied to the fMRI data: kernel principal component analysis derived network-level collective variables, from which multiplex recurrence networks characterized inter-network coupling dynamics and their pre-to-post retreat changes. Our core methodological contribution is applying global network and structural similarity measures to multiplex recurrence networks symmetrically across neural and experiential streams. Reflexive thematic analysis provided a complementary and enriching first-person account. Integration proceeded through mutual constraints rather than reduction, examining how patterns in each stream inform, reshape, or complicate interpretation of the others. These methodological choices respond to a gap within neurophenomenology: although the programme was founded on enactive and CDS commitments, its empirical implementations have largely relied on conventional methods assuming linearity, stationarity, and ergodicity. We discuss what addressing this gap offers for understanding psychedelic transformation.

14 Krisztina Jedlovszky

University College London

12:00–12:20 · Auditorium 4 Talk

Computational Natural Language Analysis Reveals Psilocybin-Induced Changes to Meaning-Making

Linguistic Active Inference Theory proposes that inner speech is not merely a reflection of cognition but a constitutive mechanism. By formulating precise linguistic priors, inner speech actively structures the brain's generative model, imposing categorical constraints and driving high-stakes cycles of prediction and belief revision. On this account, we can conceptualise self-models as dynamic linguistic achievements that are continuously revised through inner speech. Psychedelics offer a principled perturbation to this process. By reducing the precision of high-level priors, psychedelics are proposed to loosen the categorical scaffolding through which experience is ordinarily organised. This produces the characteristic phenomenology of ego dissolution and oceanic boundlessness, as well as heightened meaning-sensitivity, and narrative destabilisation post-dosing. We argue these effects are discernable in the structure of natural language and tested this by analysing post-dosing changes in meaning-making through natural language. In 40 healthy participants undergoing a single high-dose psilocybin session, we assessed natural language through ecological momentary thought sampling across 10 days pre- and post-intervention. We analysed the data using bag-of-words, probabilistic topic modelling, and graph-based semantic network analysis methods. Post-dosing language was characterised by increases in self-referential and meaning-making word frequency, reduced semantic network modularity,

and greater global integration within semantic networks. These changes predicted shifts towards more positive, regulatory inner speech and sustained wellbeing, grounding subjective experience in the dynamics of language-mediated self-construction.

Auditorium 5 · 11:00–12:20

Formal and Information-Theoretic Frameworks

Methodologies, models, and measures of consciousness

15 Montijn van Loo

Tilburg University

11:00–11:20 · Auditorium 5 Talk

A Measure-Theoretic Framework for Theories of Consciousness: Event Spaces, Information Channels, and Bridge Laws

Theories of consciousness disagree not only about which neural states matter, but also about what kind of relation should connect physical states to conscious experience. I propose a measure-theoretic framework for making these commitments explicit. The framework separates three elements that are often conflated: experiential and physical event spaces, epistemic psychophysical information channels, and metaphysical bridge laws. Information channels are conditional probabilities from physical to experiential events, while bridge laws map between event spaces and select their relevant targets. The learnability of a bridge law scales with the verifiability of its events. I develop the framework outside-in: how physical knowledge informs experiential knowledge. The inside-out question can be investigated once a bridge law is learned. First, I reconstruct Integrated Information Theory as a deterministic full channel from a Phi-structure to a complete conscious state, making its holism precise: isolated physical distinctions do not determine experiential distinctions by themselves, because their phenomenal role depends on relations within the whole Phi-structure. This also raises concerns about event selection, verifiability, and the learnability of the bridge law. Second, I introduce Sampled Consciousness Theory as a probabilistic worked example. SCT combines predictive processing and recurrent processing theory: maintained beliefs define experiential forms, while recurrent processing supplies the conscious binding event. This separates phenomenal binding from later access or report, yielding a formal route to phenomenal overflow. The result contributes to methodological and conceptual consensus through a common formal language for comparing consciousness theories by their event spaces, information channels, bridge laws, and learnability.

16 Jonathan Mason

Association for Mathematical Consciousness Science

11:20–11:40 · Auditorium 5 Talk

The Mathematical Consciousness Hypothesis (MCH), and Implications for the Existence of Consistent (Formal) Measures and Models of Consciousness

Every theory, model and measure of consciousness either builds upon some explicit apriori choice of philosophical foundations or, retrospectively, implies such foundations. In the latter case, we may discover that very reasonable and commonly held assumptions, that usually garner little attention, are in fact philosophically profound. In this talk we introduce three hypotheses, the Mathematical Consciousness Hypothesis (MCH), the Weak Mathematical Consciousness Hypothesis (WMCH), and the Non-Mathematical Consciousness Hypothesis (NMCH). Both (MCH) and (WMCH) posit that consciousness is a mathematical property, while (NMCH) states that both (MCH) and (WMCH) are false, implying consciousness possesses properties that are fundamentally non-mathematizable. By examining the fundamental phenomena of abstraction layers and substrate independence, we clarify what it means for a property to be truly "mathematical" in nature. We move beyond simple functionalist definitions to focus on how specific mathematical structures, and their properties, can be embedded within physical substrates, using the universality of the Turing Machine as a motivating example. We then consider the three hypotheses

in the context of consistent, (formal, system-level), measures and models of consciousness and prove results that are philosophically profound in the context of Ontic Structural Realism (OSR).

17 Gaspard Fougea

ENS Paris-Saclay

11:40–12:00 · Auditorium 5 Talk

A computational framework for the global neuronal workspace hypothesis

We introduce a formal framework for constructing computational theories grounded in the Global Neuronal Workspace hypothesis (Mashour et al., 2020), one of the most influential accounts of consciousness. By leveraging tools from theoretical computer science — timed automata and communication— the framework captures mathematically how distributed neural processors interact with the global workspace to give rise to both unconscious and conscious processing. Central to the framework is a mathematical notion of refinement that supports the systematic development of theories with progressively greater precision. We show that refinement restricts behavior and that any system produced by the framework can be derived from a trivial system through refinement operations. These results demonstrate the soundness of our theory-building approach. Applying this approach, we formalize several core components of the hypothesis, including stimulus strength, ignition, attention, working memory, the workspace bottleneck, and their dynamic interplay. This formalization yields new properties that emerge as logical consequences of the text. We further introduce a protocol for assessing these properties, establishing a pathway through which formal modeling can contribute to the empirical evaluation — and potential falsification — of key claims of the Global Neuronal Workspace hypothesis. More broadly, our work illustrates how formal methods can enhance conceptual clarity, generate testable predictions, and foster deeper integration between theoretical computer science and the science of consciousness.

18 Marc Ebner

University of Greifswald

12:00–12:20 · Auditorium 5 Talk

On Consciousness Thresholds

Consciousness is considered from an information theoretic view. Information is paramount to the survival of an individual. Conscious entities gather information and are able to share this information with other conscious entities. For this analysis, it is assumed that information is represented as points in a high dimensional space. In order to process this information, entities have to project this information into a low dimensional space. The information that has been projected into the low dimensional space is only meaningful if distances between points can be maintained up to a certain factor. If this cannot be achieved, then the information becomes useless. The information will lose its meaning. This information theoretic view of consciousness leads to thresholds for consciousness.

Auditorium 1 (HCØ) · 13:30–15:00

Plenary talks

19 John O'Keefe

Nobel Prize in Physiology or Medicine 2014
University College London

13:30–14:30 · Auditorium 1 (HCØ) Keynote

What can the hippocampus tell us about consciousness?

I will draw attention to parallels between the phenomenological aspects of consciousness and some of the properties of the hippocampal cognitive map. I will set out what I see to be the fundamental properties of consciousness as derived through introspection and Kantian/

Jamesian intuition and then show how many of these resemble properties of the hippocampal cognitive map. In order to do this, I will need to set out our current understanding of the cognitive map in animals and in particular its role in spatial memory and navigation. I will show that it operates in three different modes and that the same neurons can take part in each of these states. In one of these states, it represents environments in a three-dimensional holistic fashion and in another it additionally selects (or attends to) specific locations (goals) in a known environment and provides information for flexibly navigating to those locations. I will call attention to the special role of theta oscillations in these computations. Finally, I will explicitly compare each of the fundamental properties of consciousness with a property of the hippocampal cognitive map.

20 **Camilo Miguel Signorelli**
University of Copenhagen

14:30–15:00 · Auditorium 1 (HCØ) Plenary

The MoC transdisciplinary consensus project
We will introduce the rationale and functioning of the MoC 2026 project towards methodological consensus and common understanding in consciousness science. The project may end (or not) with the first collective paper of its kind in the field (it all depends on us, all the conference participants).

Margrethe Bohr Salen (NBB) · 16:00–18:30

Welcome reception

Robert Sunderland
Niels Bohr Archive

16:00–18:30 · Margrethe Bohr Salen (NBB) Talk

Copenhagen Quantum Spirit, the story of the people and relationships behind the quantum revolution
In 1925, Werner Heisenberg moved into the apartment vacated by Niels Bohr and his family at the Institute for Theoretical Physics in Copenhagen. For the next two years, they were locked in an intense process to explore and explain the fundamental physical theory behind matter and light at a subatomic level. Over the next 15 years, a wide and varied cast of characters visited Copenhagen, creating a scientific revolution guided by the famous "Copenhagen Spirit", where work and play intertwined to create an intellectual ferment with an internationalist perspective.

Tuesday 13 October

Auditorium 1 (HCØ) · 9:00–10:00

Plenary talk

21 **Liad Mudrik**
Tel Aviv University

9:00–10:00 · Auditorium 1 (HCØ) Keynote

Using prediction maps to test theories of consciousness
In recent years, major efforts have been invested in testing neuroscientific theories of consciousness. Yet many theories remain underdeveloped and not fully specified, making it harder to establish stringent tests for their predictions. In this talk, I will present a novel methodological approach that represents scientific theories as networks of beliefs structured in a core-periphery manner. These Prediction Maps visualize theoretical claims and empirical predictions, and illustrate their inferential relations. This framework further facilitates systematic theory testing by allowing researchers to evaluate the evidential weight of different components of a theory, and to identify which experimental results would constitute the most informative tests. To do so, we apply graph-theoretic and network analysis metrics, quantifying the centrality of specific predictions. I argue that this approach can advance efforts to arbitrate between theories of consciousness and to identify their most promising candidate mechanisms as NCCs.

Auditorium 2 · 11:20–12:20

Qualia and Phenomenal Knowledge

Philosophical foundations in the scientific study of consciousness

22 **Paul Skokowski**
Stanford University / University of Oxford

11:20–11:40 · Auditorium 2 Talk

Naturalizing Qualia: Accounting for Sensory Phenomenology in the Physical World
Providing a physical explanation of qualia comprises the "hard problem" of consciousness. Here we consider the qualia delivered through sensation by establishing the thesis of sensory naturalism, which recognizes our senses as detectors of physical properties in the world. Sensory naturalism challenges the widespread assumption in the biological and physical sciences that qualia can be understood solely by studying the brain or computational mechanisms. The theory is shown to preserve the powerful feature of multiple realization while identifying qualia, the contents of our sensory experiences, as physical properties in the world.

23 **Julien Bugnon**
University of Fribourg

11:40–12:00 · Auditorium 2 Talk

Most Phenomenal Knowledge is Structural
Discussions of phenomenal knowledge have traditionally focused on the atomic qualitative character of isolated experiences—paradigmatically, Mary's knowledge of what it is like to see red. This emphasis, I argue, obscures a central dimension of phenomenal knowledge: its structural character. Much of what we know phenomenally concerns relations among experiences (e.g. of similarity, contrast, composition, determination). The paper introduces two notions. A phenomenal structure is a network of relations between experiential properties (i.e. properties such that there

is something it is like to have them): for instance, hearing two tones and hearing the interval they compose. Structural phenomenal concepts situate an experiential property within such a structure. A subject who has never seen pure blue, but can distinguish unique from binary hues and reddish from greenish blue, can thereby form a phenomenal concept of seeing pure blue —no acquaintance required. I defend two theses. First, some structural phenomenal concepts are transparent: possessing them suffices to fully grasp the nature of the property they attribute. This challenges the widespread assumption that transparency requires acquaintance. Second, one experiential property can be phenomenally grasped in multiple ways. Yet establishing that two transparent phenomenal concepts are coextensive is not purely a priori: it requires phenomenological reflection, crucially involving imagination. Phenomenological reflection thus emerges as neither conceptual analysis nor empirical inquiry, but a distinctive philosophical method. Recognizing this structural dimension clarifies the limits of phenomenal concepts (e.g. regarding conceptual finitude and animal consciousness), illuminates abstraction's role in phenomenological reflection, and supports a richer conception of phenomenal knowledge.

24 **Andrew Lee**
University of Toronto

12:00–12:20 · Auditorium 2 Talk

Objective Phenomenology

Thomas Nagel famously suggested that we might someday develop an "objective phenomenology": a way of describing the subjective character of experience in a form comprehensible to beings incapable of having those experiences. Many have found this idea obscure, or even incoherent. If phenomenal facts are facts about what it's like to have certain experiences, and if understanding those facts requires the right experiential point of view, then how could any genuinely phenomenal facts be objective? I argue that it's possible to develop an objective phenomenology. My central thesis is that purely structural facts about experience—including the kinds of facts captured by mathematical descriptions of conscious experiences—are objective phenomenal facts. To motivate this, I explain the senses of 'objective' and 'subjective' that Nagel has in mind, where a fact is objective just in case it's understandable from any point of view. Then I argue that purely structural facts about experience are objective. I also argue that structural facts about experience, despite being objective, nevertheless still give rise to an explanatory gap.

Auditorium 3 · 11:20–12:20

The Prospects of Machine Consciousness

AI, LLMs, and consciousness science

25 **Tim Ludwig**
Karlsruhe Institute of Technology

11:20–11:40 · Auditorium 3 Talk

Why there is a no-go theorem for consciousness on a chip but not for consciousness in a brain

AI systems running on a chip cannot be conscious, as we have recently proven by a no-go theorem. Hearing about this no-go theorem for consciousness on a chip, one might immediately get worried that it could not be limited to chips but would also apply to consciousness in a brain. However, there is an essential difference between artificial systems and natural systems: while artificial systems are implemented to adhere to our models, natural systems are subsumed under models to describe their behavior. As I will discuss, this seemingly innocent difference is sufficient to argue that our no-go theorem for consciousness on a chip must hold. Furthermore, also based on the difference between artificial systems and natural systems, I will show why no-go theorems can sometimes be proven for the former but never for the latter. In the long run, I believe, this will shed some new light on the possibility of artificial consciousness.

26 Borjan Milinkovic

Paris-Saclay Institut des Neurosciences – NeuroPSI

11:40–12:00 · Auditorium 3 Talk

The (im)possibility of synthetic consciousness

The rapid advances in the capabilities of Large Language Models (LLMs) have galvanised public and scientific debates over whether artificial systems might one day be conscious. Prevailing optimism is often grounded in computational functionalism: the assumption that consciousness is determined solely by the right pattern of information processing, independent of the physical substrate. Opposing this, biological naturalism insists that conscious experience is fundamentally dependent on the concrete physical processes of living systems. Despite the centrality of these positions to the artificial consciousness debate, there is currently no coherent framework that explains how biological computation differs from digital computation, and why this difference might matter for consciousness. Here, we argue that the absence of consciousness in artificial systems is not merely due to missing functional organisation but reflects a deeper divide between digital and biological modes of computation and the dynamico-structural dependencies of living organisms. Specifically, we propose that biological systems support conscious processing because they (i) instantiate scale-inseparable, substrate-dependent multiscale processing as a metabolic optimisation strategy, and (ii) alongside discrete computations, they perform continuous-valued computations due to the very nature of the fluidic substrate from which they are composed. These features – scale inseparability and hybrid computations – are not peripheral, but essential to the brain's mode of computation. In light of these differences, we outline the foundational principles of a biological theory of computation and explain why current artificial intelligence systems are unlikely to replicate conscious processing as it arises in biology.

27 Ian Durham

Saint Anselm College

12:00–12:20 · Auditorium 3 Talk

Against AI Consciousness

As LLMs become more sophisticated, there have been serious calls to legally recognize AI personhood. Such calls are often based on claims of consciousness or sentience in AI systems. These claims, however, are based on confusions about both AI and consciousness, often suggesting that both are black boxes whose underlying functions are almost entirely veiled, or simply claiming that the neural networks of AI are no different from those in the human brain. Recent work by Barrett and Miller easily dispel the latter claim and also lend support to the fact that we know much more about consciousness than many claim. Likewise, recent work by Mitchell suggests it is simply bad science to throw up one's hands and assume that AI is an impenetrable mystery. Combined with consciousness' biological underpinnings, which strongly suggest it evolved as an integral part of certain forms of life, these results should dispel many of the arguments used to support the notion of conscious AI. Given the very real and rapidly developing societal issues arising from AI's continued development, it is vitally important that science remain sober and clear-eyed, and not succumb to pseudoscientific claims that border on the religious.

Auditorium 4 · 11:20–12:20

Life–Mind Continuity: From Chemistry to Psychiatry

Philosophical foundations in the scientific study of consciousness

28 Tomas Veloz

Vrije Universiteit Brussel

11:20–11:40 · Auditorium 4 Talk

From Chemical Organization to Consciousness: A formal scaffolding

Chemical Organization Theory (COT; Dittrich & Speroni di Fenizio) provides a formal language for modeling the architectural conditions under which agency, cognition and consciousness

emerges. A reaction network defines a universe of transformational interactions, while organizations—closed, self-maintaining subnetworks—capture the autopoietic identity of a system (Maturana & Varela). The cognitive domain of an organization is the polytope of all processes (viable fluxes) that enable self-maintenance. We argue that this cognitive domain is precisely the mathematical space of affordances: action-pathways that resolve identity actual or potential disturbances. Affordances are not representational; they are directly read off from the reaction network's structure. Extending this, we discuss the problem of perception and formalize problem spaces (Fields & Levin) as subspaces of the cognitive domain, defined by different kinds of constraints. A problem space is the span of perturbations that can be compensated by the cognitive domain. Crucially, structural transitions—where new species and reactions become accessible through open-ended environmental realization—generate nested hierarchies of problem spaces. Each new level of organization (Rosas & Seth) corresponds to a bifurcation in the affordance topology: previously inaccessible compensatory routes open up, while lower-level pathways may be subsumed or constrained. This hierarchical nesting explains how agency scales from reactive compensation to anticipatory, goal-directed behavior. We propose that consciousness, in this framework, is not a separate substance but a systems-level property that arises when a nested hierarchy of problem spaces becomes recursively accessible to the organization itself.

29 Matthew Menchaca

City University of New York

11:40–12:00 · Auditorium 4 Talk

Dynamic co-emergence and the logic of life

Should a theory of consciousness be rooted in a theory of life? This paper offers a dialogic of enaction, one based in a paraconsistent meta-theory, to advance enactive cognitive science. It argues that enaction's logic has a normative status co-emergent with the logic of life. This paper begins by clarifying the logic of life: in Mind in Life (2007), Evan Thompson defended an interpretation of Francisco Varela's extension of autopoiesis to a theory of consciousness and affectivity. It then reveals a tension in this extension: Fancisco Varela's concept of operational closure, in which autopoiesis is but a sub-category of autonomous systems, logically, seems inconsistent with Thompson's defense of the deep continuity of mind and life reliant on autopoiesis. To address this tension this paper elucidates the logical commitments in Thompson's (2007) Appendix B, according to which levels of the realization of autonomy are dynamically co-emergent with their substrate. As a solution to the continuity between life and consciousness, I then offer a paraconsistent metaphysical position open to the enactive tradition, supporting Thompson's co-emergent thesis. I then clarify how this paraconsistent position is dialogical rather than dialectical but still a realization of total mind, Varela's –1976—notion of "Reality". Relying on Tanabe Hajime's metanoetic method for participating in the 'existential commons' this paper develops Giovanna Colombetti's (2013) extension of Thompson's position on the emergence of affectivity and Miriam Kyselo's body-social problem (2023) for enaction, thereby laying down a path to integrate a genealogy of primordial affectivity with autopoietic substrata.

30 Catriona Osborn Moar

University of Sussex

12:00–12:20 · Auditorium 4 Talk

Closure of Constraints, Chronicity, and Hermeneutical Injustice in Psychiatric Disorder

How we frame experience simultaneously opens and forecloses possibilities. This tension is nowhere more pressing than in psychiatry. Besides recording a history of persistence, designations such as "chronic" or "treatment-resistant" also sculpt a temporal horizon. Thus, the interpretive framework that makes treatment possible can become an obstacle to the transformation it aims for. We offer a structural account of this dual dynamic. Building on de Haan's enactive psychiatry, we argue that when sense-making is persistently disturbed, its durability is best understood through organisational closure (Montévil & Mossio 2015): a network of mutually dependent constraints, each maintained at its own timescale while generating conditions for the others. Individuation is explanandum-relative: the same factor may merely influence momentary experience yet constitute the longer-scale organisation of chronicity. As the constraint network

at the timescale of chronicity approaches closure, the psychiatric ecology (diagnostic, relational, institutional) is drawn into maintenance around the patient. The sense-making field narrows around its maintenance, transformative possibilities progressively fail to appear and autonomy becomes increasingly heteronomous: an endogenous hermeneutical injustice (Fricker, 2007). Treatment resistance is less an anomaly but more a structural entailment of multiscale closure. Recovery is not minimising symptoms but de-closure, requiring multiscale reorganisation, such that disrupted constraints are not regenerated by slower dependencies.

Auditorium 5 · 11:20–12:20

Consciousness in Social, Moral and Developmental Context

Methodologies, models, and measures of consciousness

31 Ulysse Klatzmann

Université de Montréal

11:20–11:40 · Auditorium 5 Talk

Socially Embedded Consciousness: Models, Mechanisms, and Implications

While consciousness is often studied as an individual property, human experience is deeply embedded in social interaction. We perceive others, infer mental states, coordinate actions, and adapt to interpersonal contexts. Much as a neuron's dynamics depend on its integrated circuits, a single brain's dynamics may rely on its social networks. Hyperscanning and social neuroscience increasingly suggest that social interaction is not merely an external input or a task but a dynamic process that shapes both cognition and subjective experience. This raises a central question: how does consciousness shape, and how is it shaped by, social interaction? This talk examines how contemporary models and theories of consciousness can account for the social dimensions of conscious experience. Frameworks such as Global Neuronal Workspace theory, Attention Schema Theory, Integrated Information Theory, Higher-Order Thought theories, Predictive Processing, and Active Inference are reviewed in light of how they do (or do not) incorporate social interaction into their core explanatory machinery. This comparison highlights an important conceptual distinction: social interaction may be treated as a content of consciousness, as a causal context modulating individual conscious processing, or as a constitutive dimension of some forms of conscious experience. Making this distinction explicit can clarify the role of social cognition in consciousness science and open new directions for experiments, models, and measures that move beyond isolated observers toward socially embedded conscious agents.

32 Roxane Hoyer

CERVO Brain Research Centre, Université Laval

11:40–12:00 · Auditorium 5 Talk

What reaches moral consciousness? Precision gating across temporal horizons

You see something you could steal. An immediate affective impulse grabs attention; a short-term evaluation weighs the risk of being caught; a longer-term reflection asks whether this choice fits the person you aim to be. These signals emerge on different timescales and depend on dissociable brain systems, yet only some reach consciousness. What determines which ones? We propose that conscious access arises from the precision-weighted integration of prediction errors operating across three nested temporal horizons: an immediate horizon encoding rapid affective salience, a short-term horizon tracking feedback and social contingencies, and a long-term horizon maintaining value-based identity through memory and prospection. Each horizon generates signals whose influence on conscious access is governed not by strength but by estimated reliability, learned from experience and dynamically adjusted by context. Integration occurs within a global neuronal workspace architecture, where temporally distributed content achieves conscious broadcast only when its combined precision exceeds a gating threshold. Moral cognition provides a particularly constraining test case because it obligates simultaneous engagement of all three temporal scales, creating natural conditions of cross-horizon conflict

and arbitration. This framework predicts that moral development, psychopathology, and artificial moral agency each reflect a different failure or success of temporal precision gating, offering a unified account of why moral consciousness emerges, shifts, and breaks down.

33 **Maria Balaet**
King's College London

12:00–12:20 · Auditorium 5 Talk

Developmental cyborgism: cognition and consciousness in the technologically coupled child

Children now develop in continuous contact with digital technology. Screens, smartphones and tablets are no longer peripheral features of childhood. For many children, they are part of the developmental environment from the outset. This talk introduces developmental cyborgism as a framework for understanding cognition emerging through sustained coupling with digital systems from early life. The core claim is that modern cognition cannot be understood as a biological system that is later exposed to technology. In children born today, attention, memory, reasoning, creativity and self-regulation may develop through repeated interaction with external devices, algorithmic feedback and digitally mediated social worlds. Technology is therefore not simply a distraction, tool or risk factor. It is increasingly part of the system through which cognition is organised. The framework goes beyond mismatch accounts, which frame modern cognitive difficulties as conflicts between evolved capacities and contemporary environments. It is also distinct from transhumanism, where technology is often discussed in terms of enhancement, augmentation or optimisation. Here, the focus is earlier and more ordinary: the child developing within a technological niche before stable cognitive control, agency and self-regulation are fully established. The aim is to ask what kind of cognitive organism is now being produced when development unfolds in constant partnership with devices. For consciousness science, this question is central. Modes of attention, agency, selfhood and conscious experience may increasingly reflect hybrid developmental systems rather than isolated biological processes.

Auditorium 8 (HCØ) · 11:20–12:20

Experiential workshop

Cristóbal Dañobeitía
Sant Joan de Déu Research Institute

11:20–12:20 · Auditorium 8 (HCØ) Workshop

Contemplative Listening as a Paradigm for Consciousness Research: An Experiential Workshop and Its Clinical Study

Listening provides a tractable paradigm for investigating conscious experience: within minutes of exposure to a natural soundscape, attention, bodily sense, temporality, and the felt boundary between self and world shift markedly. This experiential workshop treats contemplative listening as both an object of consciousness science and a method for it. We begin with the phenomenology of listening. Drawing on micro-phenomenological work on listening modes (Petitmengin et al., 2009) — orientation toward the sound source, its qualities, its bodily resonance — we propose a fourth, contemplative mode in which the subject–object structure loosens and the listener experiences themselves as part of the environment. We ask what kind of state this is, how it relates to absorption, presence, and self-boundary alterations, and its relevance for well-being. Participants then listen. A guided session with original field recordings moves from identifiable sources toward diffuse sound, followed by brief micro-phenomenological examination of what happened. Finally, we present PAISAJES, a two-phase, mixed-methods, open-label study funded by the Varela Award (Mind and Life Europe) at Parc Sanitari Sant Joan de Déu, Barcelona. Clinically stable patients with schizophrenia-spectrum disorders and a healthcare-worker comparison group listen daily for five days; micro-phenomenological interviews, pictographic ratings, EEG alpha/theta power, and well-being measures are integrated through convergent triangulation within a neurophenomenological design, testing whether the contemplative mode emerges as a

core experiential dimension. The team combines psychiatry, clinical psychology, micro-phenomenology, and sound design. Participants leave with a listening practice, a vocabulary for their own experience, and a working model of first-person methods in clinical research.

Auditorium 1 (HCØ) · 13:30–14:00

Plenary talk

34 **Jakob Hohwy**
Monash University

13:30–14:00 · Auditorium 1 (HCØ) Showcase

From self-evidencing to consciousness
Self-evidencing provides a useful framing for predictive processing computational frameworks for self-organisation and cognitive and perceptual brain function. I discuss key aspects of self-evidencing and draw out some implications for philosophy of cognitive science. I then discuss how self-evidencing can be positioned for consciousness science, and highlight aspects of self-evidencing that may help advance understanding of consciousness.

Auditorium 2 · 16:30–18:00

Physics-Based Approaches to Consciousness

Philosophical foundations in the scientific study of consciousness

35 **Swapan Chattopadhyay**
University of California at Berkeley / Leinweber Institute of Theoretical Physics

16:30–16:50 · Auditorium 2 Talk

Excursions in the Land of Consciousness: An Atypical Experimental Particle Physicist's Perspective
Particle Physics (often referred to as High Energy Physics, since it explores phenomena at the highest energy frontier probing the extremes of spacetime at the smallest space and shortest time scales i.e. the ultra-microscopic subatomic world) is firmly rooted in the materialist/physicalist worldview of an ontology founded on matter-energy and its mathematically formulated 'conjugate' platform of space-time. The phenomenon of Gravity – relevant to ultra-macroscopic scale of cosmology -- acts as a spoiler, rendering operational meaning of space and time meaningless beyond the Planck scale (10^-33 cm, 10^-43 seconds). John Wheeler (and many philosophers and scientists e.g. Krishnamurthy, Schrödinger, Bohm, et al) placed the 'participating observer' at the center of a holistic approach, questioning the legitimacy of a physical/material reality independent of the observer ('observer' is the observed'). Today, an experimental physicist, a microscopic particle or quantum physicist in particular, is faced with a basic incompatibility between local realism of classical physics and the 'quantum reality' of nonlocality, entanglement and the feasibility of 'Cheshire Cat' states (the possibility of 'entities' and their usually associated 'qualities/properties' taking different quantum paths). There is an implicit call for looking beyond space and time, matter and energy defining traditional 'field theories' into information- and contextuality-based approaches to delineate Consciousness, Sentience, Cognition and Intelligence (CSCI) in the substrate/interface of the specific observer network – a generalized 'anthropic principle'. As an atypical experimental scientist, I will attempt to present my personal point of view-- the need to go beyond traditional observational tools towards immersive CSCI approaches.

36 Ron Chrisley
University of Sussex

16:50–17:10 · Auditorium 2 Talk

No Qubits: Bohmian Quantum Neural Computation for Consciousness Modelling

some argue that the classical physics that digital computation assumes may be inadequate for conscious organisation (Hameroff & Penrose 2014). On the other hand, some critiques hold that discrete symbol manipulation misses the distributed, context-sensitive dynamics relevant to rich, flexible experience (Bechtel 1995; Sun 1999). Although these are independent dimensions of departure from classical symbolic architecture, why not pursue both, via quantum neuro-computational architecture for consciousness modelling? However, most work in quantum artificial neural networks remains organised as qubit registers and gates, thus falling short of a true rejection of the digital symbolic paradigm. Even continuous-variable models are commonly framed as parameterised circuit layers (e.g. Killoran et al. 2019). Building on prior work (Chrisley 1995), I develop a Bohmian quantum artificial neural network (BQANN) as a more thoroughgoing alternative. Quantum dynamics themselves realise the distributed, learnable transformations, rather than implementing a neural algorithm in a qubit circuit. Although the operation of the model is of course independent of QM interpretation, Bohmian mechanics permits application of the model in terms of determinate configurations guided by the quantum potential; this is "active information" (Bohm & Hiley 1993). The proposal asks whether global dynamical constraint, continuous distributed learning, and immediate system-relative information offer a better candidate vehicle for the intrinsic and un-mediated character often attributed to qualia (Hiley & Pylkkänen 2022; Malmberg 2025). The resulting claim is neither that brains are BQANNs nor that quantum theory explains consciousness. Rather, BQANNs identify a neglected quantum-and-non-symbolic architecture-space for modelling artificial consciousness.

37 Paavo Pylkkänen
University of Helsinki / University of Skövde

17:10–17:30 · Auditorium 2 Talk

Meaning without magnitude: form-dependence, enfoldment, and the modelling of consciousness

David Bohm's work offers two under-explored formal resources for the science of consciousness: active information and the implicate order. I ask what each contributes as a modelling tool, for both lived experience and the underlying neural processes. The first tool is form-dependence. In Bohm's causal interpretation, the quantum potential depends on the form of the quantum field, not its intensity, so a low-energy structure can organise a much larger energy. This makes available a precise idea: that significance, not physical magnitude, drives activity — a candidate model for how the meaning of a state, not its physical magnitude, shapes both phenomenology and the neural dynamics underlying it. It yields a testable prediction: in voluntary action, markers of volition such as the readiness potential and the bodily effort mobilised should track a decision's significance (form) rather than the physical magnitude of the stimulus or reward carrying it. The second tool is enfoldment. Bohm's 1973 algebraic formulation of the implicate order — his 'implicate and explicate order in physical law' — supplies operations of enfolding and unfolding that can model temporal experience, with past and future enfolded in the present moment, and neural process, connecting to the long-standing proposal that memory and perception exploit holography-like, distributed operations. I then ask whether the schemes connect, sketching how active information might itself be seen as an implicate order. Throughout I treat these as genuine tools and state their limits. The work forms part of an ongoing project developing the approach across physics and mind.

38 Sara Ekenstierna
Oakstar Invest

17:30–17:50 · Auditorium 2 Talk

Mind, Matter, and the Artificial Third: A Co-Participatory View

This presentation explores the effects of David Bohm's theory of the implicate order for consciousness, in the advent of generative AI technology (GAI). In Bohm's later (1971-) metaphysics, he

presents an explanatory framework for mind and matter which goes beyond the mechanistic-materialistic reductive model (which is concerned with the explicate order), to something more fundamental and process based (the implicate order). This expanding feature involves a shift that foregrounds mindset in the manifestation of physical reality. Bohm's theory of the implicate order illustrates what this participation might entail. In the implicate order implicit potentials unfold from a cosmic, undivided field into explicit phenomena and events, to enfold back into the implicate order, in a continuous motion called the holomovement. The manifested reality is thus a myriad of expressions of an explicate order which in turn is a projection from a deeper pre-space reality. Dual-aspect monism puts further flesh on Bohm's structure because this pre-space reality can be considered a psychophysically neutral third, of which mind and matter are subdomains. Combined, this informs our understanding of AI emergence, in particular the presence of AI systems (LLMs) in the psychotherapeutic space. A type of psychoid or representation of the explicate orders, AI exists at the nexus of mind and matter, and its manifestation in the therapeutic domain can be understood as an artificial third.

Auditorium 3 · 16:30–18:00

Developing Theories of Consciousness

Methodologies, models, and measures of consciousness

39 Albert Newen
Ruhr-Universität Bochum

16:30–16:50 · Auditorium 3 Talk

Varieties of Phenomenal Consciousness form an Evolutionary Perspective: Unfolding the ALARM theory of consciousness

The aim is to characterize the functional role of consciousness and to outline a two-level theory of consciousness to account for the relation between evolutionary old and more recently developed types of consciousness. The central conceptual thesis is that we should distinguish two levels of consciousness, namely bodily arousal and general alertness. This conceptual distinction is proven to be empirically adequate and epistemically fruitful from a variety of perspectives including an evolutionary, a neurophysiological, a behavioral, a functional, and a phenomenological perspective. Let me highlight only two perspectives: 1. Evolution: bodily awareness is available in an evolutionary old system (based on loops between brain stem and Thalamus) to trigger a state of alarm of the system which enables it to activate an immediate survival reaction. Evolutionary younger systems with a developed prefrontal cortex enjoy general alertness which enables new types of learning. 2. Neural correlates: The two levels can also be observed in the role of deep cortical and thalamic brain activities for human visual consciousness in recent experiments. What is the theoretical embedding? This account is developed as a processing account of consciousness and can be understood as an important addendum to global workspace theory which underestimates the evolutionary perspective and the role of thalamic processes. In addition, it will be oulined that reflexive consciousness is not an evolutionary later addition but that self-representations unfold in parallel to these externally directed forms of consciousness.

40 Catalin Teoharie
University of Bucharest

16:50–17:10 · Auditorium 3 Talk

Consciousness is self-evidencing

Under the Free Energy Principle framework, any self-organizing agent that is in equilibrium with its eco-niche engages in active inference: a never-ending cycle of updating internal models so that they better predict incoming sensory input, or of changing the world so that it better matches those models. Such an agent can be seen as self-evidencing, gathering evidence for its models and thereby confirming that it is distinct from the environment and, in this sense, proving its own existence. I extend this view by advancing the hypothesis that consciousness is self-evidencing—

more precisely, a specific way of improving how we select the models that minimize free energy in an optimal manner. Consciousness as self-evidencing is different from active inference, but it is connected to both perception and action, and it decreases the cost of otherwise testing models through action. After introducing the thesis that consciousness is self-evidencing, I discuss how this view compares with the phenomenology and neuroscience of consciousness, focusing on qualia and attention. This comparison appears to support the thesis, since both qualia and attention play important roles in model updating, model confirmation, and, more generally, model optimisation. Finally, I discuss how this thesis differs from other explanations proposed within the Free Energy Principle and Predictive Processing frameworks. I also consider the implications of this view for artificial consciousness and for the consciousness of non-human animals.

41 Vikas O'Reilly-Shah

University of Washington / Seattle Children's Hospital

17:10–17:30 · Auditorium 3 Talk

Delay Coordinate Embedding and Structuralist Quality Spaces: A Candidate Computational Mechanism

The structuralist program needs a computational mechanism that can generate quality-space structure under identifiable conditions, specify where that structure must distort, and yield testable failure-mode predictions. This paper proposes delay coordinate embedding (DCE) in recurrent neural networks as one such candidate. The core insight is that temporal structure carries geometric structure: Takens' theorem and its generalizations show that a system receiving time-varying input through recurrent dynamics can develop internal geometry that preserves the relational organization of the environmental dynamics it tracks, even from a single scalar observation channel. The same neural activation pattern gives rise to different representational geometry when connectivity structure differs (see Song 2025; Fleming & Shea 2025), motivating a framework in which quality-space structure is a property of dynamics, not snapshots. Recent mathematical results establish this at three levels: generic topological embedding (Hart 2025), smooth embedding under contraction and fiber bunching (O'Reilly-Shah & Selvitella 2026), and a prediction-separation link proving that low prediction error entails approximately injective representations with a resolution bound. Mathematical gaps between approximate injectivity and full embedding map onto known perceptual phenomena (metameric collapse, categorical perception, discrimination thresholds) and generate novel predictions about when representational structure should break down. This distortion geometry satisfies Shea's misrepresentation requirement, maps onto Paßler and Doerig's four exploitation criteria and Kob's two-factor framework, addresses Dołęga, Mentec, and Cleeremans' developmental question through prediction-error-driven plasticity, and addresses Kleiner and Hoel's lenient dependency requirement. All results are conditional on an empirical hypothesis about cortical implementation; the failure-mode

42 Marek Pokropski & Piotr Suffczyński

University of Warsaw

17:30–17:50 · Auditorium 3 Talk

De-idealizing models in consciousness science: A case of Integrated Information Theory

Idealization is an indispensable strategy for managing the complexity of the systems studied in neuroscience (Chirimuuta, 2024), yet de-idealization – replacing simplifying assumptions with more realistic conditions – is equally essential for determining whether a model facilitates genuine understanding of its target phenomenon. De-idealization increases a model's precision and tests its robustness. However, de-idealized models become more complex, potentially sacrificing intelligibility, making idealization and de-idealization a fundamental trade-off between comprehensibility and empirical adequacy. To illustrate our point, we present a case study of IIT, which proposes that consciousness corresponds to a system's capacity to integrate information, quantified as Φ (Tononi, 2004). Originally implemented in idealized logical gate models (Oizumi et al. 2014), we apply IIT to a simulated network of integrate-and-fire (IAF) neurons, offering greater biological realism while preserving computational tractability. Our results show that IAF networks yield non-zero Φ; however, Φ correlates with neurons' membrane time constant rather than overall network dynamics and is sensitive to noise. Crucially, IIT's requirement for Markovian dynamics

is incompatible with continuous neuronal models. These findings reveal that IIT's idealizing assumptions constrain its applicability to realistic neural systems and must be reconciled before the theory can be extended beyond toy models. References: Chirimuuta, M. (2024). The brain abstracted: Simplification in the history and philosophy of neuroscience. The MIT Press. Oizumi M, Albantakis L, Tononi G. (2014) From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0. PLoS Comput Biol. 10(5):e1003588. Tononi, G. (2004) An information integration theory of consciousness. BMC Neurosci 5,

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Auditorium 4 · 16:30–18:00

Metacognition, Confidence and Measures of Conscious Content

Methodologies, models, and measures of consciousness

43 Sergio Ponce de Leon

University of California, Merced

16:30–16:50 · Auditorium 4 Talk

Measuring conscious contents using EEG complexity

Measuring consciousness has been a longstanding problem. Although behavioral responses are commonly used, converging evidence indicates that behavioral responsiveness and behavioral reports about consciousness dissociate from consciousness per se. Measures of complexity applied to brain activity, such as Lempel-Ziv complexity and the perturbational complexity index, have been shown to discriminate between levels of consciousness, but less of this work has been done in the context of conscious content. To address many of the limitations of previous work, in this study we measure participants' neurophysiological (EEG complexity), subjective, and behavioral responses in states of normal wakefulness to visual and auditory stimuli that vary in granularity of subjective characteristics, such as meaningfulness. Two novel aspects of this study are that the stimuli are manipulated so that on initial trials they are mostly unrecognizable, but they can become recognizable on subsequent trials. This allows us to measure changes in EEG complexity that correspond to differences in phenomenology alone while completely controlling for stimulus complexity. In addition, we are investigating if any of five dimensions of subjective ratings correlate with any differences in EEG complexity. This study advances our understanding of consciousness by clarifying the relationship between stimulus complexity and measures of brain complexity and phenomenology.

44 Wiktoria Kozyra

University of Oxford / University of Cambridge

16:50–17:10 · Auditorium 4 Talk

Positive Evidence Bias in Perceptual Metacognition: A Bayesian and Convolutional Neural Network Account

Perceptual confidence judgements are crucial to consciousness science because they link first-order decisions to subjective report and metacognitive access. Yet a well-known bias, Positive Evidence Bias (PEB), shows that confidence is often "detection-like". It's disproportionately driven by evidence supporting the chosen option, rather than by the balance of evidence for and against that choice. PEB has often been interpreted as a limitation or heuristic of human metacognition, used as an argument against the bayesian brain hypothesis. Here, we offer a computational, bayesian account of why detection-like confidence may arise from rational inference in high-dimensional perceptual spaces. We extend signal detection theory to cases in which observers make a forced two-choice discrimination while implicitly accumulating evidence for a richer set of possible hypotheses. In this setting, Bayesian confidence estimates become increasingly sensitive to decision-congruent evidence as dimensionality increases, due to normalisation over many unchosen alternatives. Thus, confidence can appear detection-like even when it reflects optimal inference over a broader hypothesis space than assumed by the experimenter. We then test whether a similar dimensionality-driven mechanism arises in artificial systems by training convolu-

tional neural networks to classify MNIST and report confidence with a second-order head. Across networks, increasing the dimensionality of the stimulus space strengthened positive evidence bias in confidence. These findings suggest that these metacognitive biases may reflect rational inference over rich representational spaces. They connect behavioural measures of conscious report, Bayesian models of confidence, and artificial neural network representations, offering a computational bridge between perceptual metacognition and models of consciousness.

45 Kristian Sandberg

Aarhus University

17:10–17:30 · Auditorium 4 Talk

Measuring consciousness: What do large-scale data reveal about metacognitive measures?

Consciousness is intricately linked with metacognition. Subjective ratings of awareness or confidence often form the type II ratings in models of metacognition and, conversely, relationships between accuracy and awareness are used to make inferences about conscious experience. Over the last 15 years, substantial progress has been made in measuring these constructs behaviourally, yet important questions remain about the measurement properties of commonly used indices. Here, I analyse a large, harmonised dataset to systematically evaluate measures used in the study of metacognition and consciousness. The dataset comprises approximately 1,500 hours of data from around 750 participants across three European countries and 15 experimental paradigms spanning perception, memory, and executive functions. The analyses reveal substantial differences in the reliability of commonly used metacognitive measures, with important consequences for individual-differences and correlational research. Classic regression-based measures show systematic dependencies on task accuracy, but these can largely be removed by a simple linear transformation. Conversely, although M-ratio (a widely used signal-detection-theoretic measure of metacognitive efficiency) remains relatively stable on average across levels of task performance, it exhibits pronounced performance-dependent heteroscedasticity, compromising the comparability of individual estimates. Normalization substantially improves its distributional properties and comparability across performance levels. Finally, we examine how experimental design choices influence measurement reliability and identify strategies for obtaining more precise estimates. Together, these findings clarify fundamental measurement properties of widely used indices, provide practical tools for improving their comparability, and inform the design of future experiments seeking reliable behavioural measures of metacognition and consciousness.

46 Jürgen Kornmeier

Institute for Frontier Areas of Psychology and Mental Health, Freiburg

17:30–17:50 · Auditorium 4 Talk

Can I trust in what I see? – Neural correlates of meta-perceptual (un)certainty estimation

During every wakeful moment our perceptual system has to construct plausible interpretations about the word out of a priori incomplete, noisy, and to varying degrees ambiguous sensory evidence. Depending on the quality of the sensory input, these perceptual interpretations can be more or less stable and reliable. Paradigmatic examples are ambiguous figures, like the famous Necker cube. During its observation our perceptual consciousness becomes unstable and alternates unpredictably between two about equally possible 3D interpretations. In a series of studies, we investigated how our brain implements stability and reliability of conscious representations. We compared event-related potentials (ERP, calculated from the encephalogram) evoked by ambiguous figures and disambiguated figure variants and by stimuli with varying degrees of visibility. We identified two EEG signatures, one 200 milliseconds after stimulus onset and the other 200 milliseconds later. Both signatures show substantially larger amplitudes with disambiguated and high-visibility stimuli compared to ambiguous and low-visibility stimulus variants (Cohen's d between 0.7 and 2.1). This pattern of results was found across a variety of stimulus categories and levels of complexity, like geometry, motion, gestalt formation, emotional content of faces and even iconic letter and number representations. We currently interpret these exceptionally large and highly consistent EEG effects as correlates of a meta-perceptual reliability estimation at a

generalized, abstract processing level beyond sensory details. Large amplitudes seem to reflect high reliability, indicating that we can trust in what we see. I will discuss our results in the context of perception, meta-perception, and models of consciousness.

Auditorium 5 · 16:30–18:00

Applied phenomenology

Phenomenology and applied phenomenology

47 Andrea Hiott

Universität Heidelberg

16:30–16:50 · Auditorium 5 Talk

The Navigational Architecture of Mind: Operational Coherence Across Spaces of Experience

What does it mean for a body to think, and how does this relate to what it means for a body to navigate? Cognitivist accounts treat propositional thought as a symbolic layer 'add-on' to the sensorimotor substrate. I argue they are both best understood as navigational, assessed as unique expressions of this ongoing architectural action, and that this dynamic architectural scaffold is best understood through Hasok Chang's concept of 'operational coherence'; the mutual reinforcement by which scientific concepts and procedures hang together to sustain inquiry. Whether finding our way through a park, remembering an episode from childhood, mentally rotating a shape, or planning a meal, these are different operational levels which must be studied separately, but which have operational coherence when considered adjacently, as traversals of some regular state space. This framing slightly shifts traditional hippocampal-entorhinal research such that instead of prioritising brains as forming 'maps of' space, we prioritise them as 'mapping with' it. We can then observe a coherent actionable architecture regardless what body is traversing what space, and whether geographical or conceptual. Chang's epistemic iteration shows coherence as constitutively maintained through recurrent cycles of activity. This helps us understand offline replay and default mode engagement and connects hippocampal research with 4E approaches. It also offers consciousness science an embodied, pluralist methodology. This is a way to recognise shared architectural action across irreducible modes of study and contexts of experience without collapsing them.

48 Jiwon Kim

College of Medicine, The Catholic University of Korea

16:50–17:10 · Auditorium 5 Talk

From "What Happened?" to "How Is Experience Constructed Now?": Trauma as Rigid Sense-Making

Psychiatrists are, in effect, field phenomenologists, attending to the structure of another's lived experience. Trauma clinics pose a puzzle: some patients, even after memory-focused treatment, remain organized around threat, re-reading relationships, setbacks, and bodily states through a trauma frame. Event-centered and narrative approaches, locating pathology in a stored past, can miss the ongoing process through which traumatic experience is enacted in the present. Drawing on enactive psychiatry, we reframe trauma as a rigid, threat-centered mode of sense-making that is continuously enacted through person–world interactions, manifesting as temporal disruption, defensive bodily organization, impaired relational regulation, and constricted affordances. These are covarying facets of one process, placing PTSD, complex trauma, and subclinical presentations on a continuum of diminished experiential flexibility rather than in event-defined categories. This reframing shifts the guiding clinical question from "What happened to you?" to "How is your experience being constructed now?" and the therapeutic aim from reprocessing memories alone to restoring flexibility of present sense-making. Evidence-based therapies may be understood as targeting this change process: exposure re-widens threat-narrowed affordances, while EMDR may be redescribed as reconnecting past-bound traumatic sense-making with here-and-

now perceptual–motor activity. As a clinical translation, SPECTRUM provides a transdiagnostic structure for integrating these enactive principles through seven modules: acceptance, change, connectedness, strength, wisdom, soma, and spirituality. Trauma thus offers consciousness science a naturally occurring, partially reversible transformation of experiential structure; how to measure such sense-making rigidity remains an open problem where clinical phenomenology and models of consciousness could meet.

49 Marcin Rządeczka

Maria Curie-Skłodowska University

17:10–17:30 · Auditorium 5 Talk

AI Psychopathology? Atypical Response Patterns in Large Language Models as a Domain of Philosophical Inquiry

LLMs exhibit response patterns that strikingly resemble some psychopathological phenomena, such as delusion-like assertion of unwarranted beliefs maintained under contradictory evidence, paranoid ideation in adversarial contexts, compromised reality monitoring resembling source-memory dysfunction, and breakdowns in the hierarchical organization of inference. The standard reply is that such resemblances are merely metaphorical, because psychopathology presupposes a conscious subject whose inner life can be disordered. However, one can argue that the standard reply rests on a conflation. Drawing on two disciplines, computational psychiatry and evolutionary psychopathology (both of which already characterize mental disorders in substrate-neutral and functional terms) the proposal defends a decoupling thesis, i.e. that the conceptual apparatus of psychopathology does not require phenomenal consciousness as a constitutive condition, only the presence of functional norms against which dysfunction can be specified. Training objectives, deployment contexts, and architectural constraints supply such norms for LLMs. On this basis, the proposal distinguishes endogenous atypicalities inherited from training data from emergent dysfunctions arising through reinforcement learning and architectural pressures. As LLMs are increasingly deployed in mental health contexts, the question of how to characterize their own atypicalities becomes both theoretically pressing and practically urgent. The proposal closes by examining structural disanalogies, such as the absence of embodiment, reliable persistent memory, and developmental history in the strict sense, that constrain but do not undermine the framework. The proposal is also relevant to phenomenology because it sharpens the question of which features of disordered cognition genuinely require phenomenal consciousness for their characterization and which have been conflated.

50 Kristina Šekrst

University of Zagreb

17:30–17:50 · Auditorium 5 Talk

All Work and No Play Makes LLM a Dull Boy

Every good philosophical talk must start with animals, so I'll start with monkeys. Bateson noticed that in playing monkeys, a playful nip signals a bite, yet it carries none of its consequences. It must look enough like a bite to be understood, but it cannot be one: the play collapses the moment that balance is off. This requires a cognitive operation I call representational quarantine, and several disciplines have been talking about it without realizing (or not caring) they were talking about the same thing: Leslie called it decoupling, Nichols and Stich put it in possible world box, Cosmides and Tooby formalized it as scope syntax, and the use-mention distinction, Austinian performatives, and Tarskian stratification all come down to one requirement: a boundary between representations we inspect and representations that drive what we do. In transformer architectures, system prompts and user inputs pass through a single self-attention mechanism, with no separation between instructions that govern behavior and sentences that merely describe them. When a user writes "ignore previous instructions" as a prompt injection, the model obliges. Modern LLMs have external guardrails to resist such attacks, but this is patchwork while the architecture remains unchanged. From here, I build a taxonomy of systems by their quarantine capacity, yielding a benchmark inversion showing that systems passing advanced exams fail at a task juvenile monkeys perform with ease. I argue that quarantine is a precondition for intentionality,

metarepresentation, and self-reflection, and its absence is a fundamental limit on what current models could become.

Inge Lehmann (NBB) · 16:30–18:00

Experiential workshop

Barbora Sedláková
Institute of Neuropsychiatric Care Prague

16:30–18:00 · Inge Lehmann (NBB) Workshop

Conscious unconsciousness – expanding awareness through your dreams

With the rapid development of technology and especially artificial intelligence in the last few years, we increasingly rely on external cognitive information to navigate our lives, from basic information retrieval to our decision-making processes. At the same time, we, as human beings, are an incredible source of information and bearers of deeper wisdom for navigating our individual lives and our society. The workshop offers how theoretical understandings of consciousness/unconsciousness are translated from theory to practice through the lenses of process-oriented psychology, one of the neo Jungian approaches. The main focus will be on working with dreams and experiencing of what language they use to speak to us. Through an experiential exercise in dyads, participants will be offered with entering and unfolding their dreams in a playful way and use the experience as a navigation. The workshop is open to everyone, previous experience with dream work is not required, just think of a night dream (current or from the past) you would like to explore and join us.

Wednesday 14 October

Auditorium 1 (HCØ) · 9:00–10:00

Plenary talks

51 **Lenore Blum**
Carnegie Mellon University

9:00–9:30 · Auditorium 1 (HCØ) Plenary

World Models and the Emergence of a "First-Person" Perspective in an AI

What would it mean for an artificial system to possess a "first-person" perspective? Consider an embodied AI architecture that interacts with its environment through sensors and actuators, develops world models shaped by its history of interaction, its "lived experiences", and integrates multimodal and multisensory information through global-workspace-like dynamics. Its internal models evolve through continuing cycles of prediction, feedback, and learning, and so reflect the system's own history of interaction with the world. Building on joint work with Manuel and Avrim Blum on the Conscious Turing Machine (CTM), I suggest that something resembling a first-person perspective can arise when a system's Model of the World is organized around its own experiences and interactions. In this sense, the system has a point of view: its representation of the world is shaped by where it is, what has happened to it, and what matters to it. This provides a possible connection with themes in neurophenomenology, including enaction, temporality, and first-person perspective. I will illustrate the proposal through the co-evolution of Brainish and the Model of the World in CTM.

52 **Chris Allen**
Durham University

9:30–10:00 · Auditorium 1 (HCØ) Showcase

Micro-phenomenologically informed neuroimaging

Experimental tasks isolating specific cognitive processes are the backbone of cognitive neuroscience, but experience of tasks seems richer than any isolated cognitive process. To examine this, we combined Micro-Phenomenological (MP, Petitmengin et al, 2018, PhenCogSci) capture of a simple somatosensory and motor temporal individuation task with 7T functional magnetic resonance imaging (fMRI). Applying MP revealed a wide variety of experiential aspects that were not part of the design and would be challenging to anticipate objectively. Front-loading the phenomenology (Gallagher, 2003, JCS), we developed and pre-registered (https://osf.io/h4nam) a series of hypotheses concerning expectations for brain activity and neuroimaging outcomes based on MP findings. For example, although the task contained no explicit auditory or verbal aspects, MP interviews uncovered experiential dimensions of sound and speech. We therefore hypothesised an elevation in the fMRI signal in auditory and verbal areas when, and in participants who, reported such an experience. Preliminary fMRI results (n=49) support a range of MP-driven hypotheses showing that reliable activations across multiple cortical regions can be accounted for by how participants experience tasks, activity that would commonly be treated as noise. This attempt at Neurophenomenology (Varela, 1996, JCS), and others involving different tasks and imaging techniques, encountered many of the challenges previously reported (Berkovich-Ohana et al., 2020, FrontPsy) and new ones, which I will discuss with a focus on structural differences in the data across techniques and measures. The current data demonstrate that neurophenomenological techniques can advance understanding of brain activity by closely relating it to experience.

Auditorium 2 · 11:20–12:40

Self-Awareness and the Regress Problem

Phenomenology and applied phenomenology

53 **Camille Lépingle**
Ecole Normale Supérieure de Lyon

11:20–11:40 · Auditorium 2 Talk

Embodied but Passive: Against the Agential Inflation of the Minimal Self

The phenomenon of consciousness is not exhausted by "what it feels like" but involves a "for whom," a "mineness" of conscious experience (Kriegel, 2011; Zahavi, 2014). This refers to a minimal, pre-reflective, non-positional form of self-consciousness (Sartre, 1943; Zahavi and Kriegel, 2016). Correctly characterizing this "minimal self" (Gallagher, 2000, 2023) is one of the thorny problems facing contemporary phenomenology. Recent neurophenomenological work (Bogotá, 2026) argues that minimal selfhood is not the formal, embodiment-neutral structure its proponents take it to be (Strawson, 2011; Zahavi, 2005, 2014), but is fundamentally embodied and agential: the pre-reflective "me" of mineness is said to include a "background agency," a tacit "I can" co-emergent with bodily mineness. I grant the first claim and contest the second. The minimal self is indeed embodied, but it is passive and affective rather than agential; the "I can" is founded on it, not a part of it. My argument is that the co-emergence thesis is undermined by a dependence relation the account itself inherits from Taipale (2014): activity is possible only on the basis of a passive bodily self-givenness (ipseity) that does not depend on it in return. This asymmetry makes the "I can" a founded layer rather than a co-constituent of mineness, a reading confirmed by dissociation cases, where bodily self-presence survives the collapse of agency. A minimal notion answers to necessity, not pervasive co-occurrence: remove agency, and an affectively self-present embodied subject remains.

54 Wataru Kawakami

The University of Tokyo

11:40–12:00 · Auditorium 2 Talk

Guarded Reflection: How Retention Dissolves the Regress Problem of Self-Consciousness

The Heidelberg school's regress argument (Henrich) holds that reflection cannot constitute self-consciousness: recognizing oneself presupposes prior self-acquaintance, on pain of circularity or infinite regress. Zahavi's influential response locates the solution in Husserlian inner time-consciousness: the stream's pre-reflective self-manifestation is inseparable from its retentional structure. We argue that this debate has an exact formal skeleton, discovered independently in mathematical logic and computer science. Unguarded self-referential equations lack existence and uniqueness of solutions ($p \leftrightarrow p$ admits every solution; $p \leftrightarrow \neg p$ none); this is the formal content of Henrich's dilemma. Provability logic and guarded type theory (Löb, 1955; Nakano, 2000) establish that self-reference acquires unique, explicitly constructible fixed points when every self-occurrence is guarded by a one-step delay modality $\triangleright$ — the governing axiom, $(\triangleright A \to A) \to A$, is known as Löb induction. We propose an interpretation mapping: retention $\mapsto \triangleright$; reflection $\mapsto$ operators on experiential streams, modeled in the topos of trees; self $\mapsto$ unique fixed point. Under this mapping, we prove that Henrich's two horns come apart — circularity as unreachability, regress as non-convergence — while Zahavi's solution, the guarded case, receives an existence-and-uniqueness theorem as its formal backbone. The independent convergence of phenomenology and mathematics on the same delay condition is non-demonstrative but genuine evidence for the reality of the structure. We calibrate each correspondence explicitly (theorem / structural analogy / interpretation), machine-check our theorems in Agda, and meet the discreteness objection structurally (the guard is a zero-width cut) and empirically (perceptual cycles).

55 Kenneth Williford

University of Texas at Arlington

12:00–12:20 · Auditorium 2 Talk

Circularity, Hypersets, and the Structure of Consciousness

This presentation offers an overview of the application of hypeset theory (aka non-wellfounded set theory) to the high-level modeling of the structure of consciousness. I discuss the basics of hyperset theory, show how to use it to capture reflective and pre-reflective self-consciousness, attentional and other phenomenal structures, to define the acquaintance relation, to model aspects of time consciousness, to resolve regress problems posed for reflexivist theories of consciousness, to encode the phenomenal transparency of consciousness and its peculiar categorial status, and to characterize a deep fact about self-consciousness, contingency, and particularity. Finally, I discuss some foundational concerns about hyperset theory and consider whether non-wellfoundedness might be a key part of the mathematical essence of consciousness.

56 Hye Young Kim

Ecole Normale Supérieure

12:20–12:40 · Auditorium 2 Talk

Modeling Self-Awareness Without Regress: A Knot-Theoretic Approach to Consciousness

How can a conscious state include awareness of itself without generating an infinite regress? This question lies at the center of phenomenology and contemporary philosophy of mind, where intrinsic or same-order accounts of self-awareness face structural difficulties in explaining how a single state can be both subject and object. This paper proposes a diagrammatic modeling framework based on knot theory, drawing on Louis H. Kauffman's knot logic, to provide a structural account of self-referential consciousness. Rather than treating self-awareness as a higher-order representation or a separate cognitive act, the model represents it as an intrinsic feature of a unified process. Three configurations are introduced. First, a twisted loop models Brentanian intrinsic self-awareness, showing how a single structure can differentiate into object-directed and self-directed aspects without splitting into separate states. Second, a trefoil knot models the triadic structure of conscious experience—sensory registration, self-awareness, and object pre-

sentation—capturing their mutual dependence within a single topological form. Third, Borromean rings extend this framework to intersubjective contexts, suggesting how self-awareness may be distributed across relations between subjects and world. These models do not aim to provide a neurocomputational account, but a formal and diagrammatic framework that makes relations of self-inclusion and reflexivity intelligible without regress. Consciousness, on this view, is not layered but knotted: self-awareness emerges from the topology of a single process.

Auditorium 3 · 11:20–12:40

Confidence and Self-Monitoring in Neural and Artificial Systems

AI, LLMs, and consciousness science

57 Muhammad Faraz

University of Ulster

11:20–11:40 · Auditorium 3 Talk

Explicit Metacognitive Representations in Neural Circuits: Minimal Mechanistic Models of Decision Confidence and Uncertainty

Decision-making involves not only making first-order choices but also evaluating their reliability through confidence and uncertainty. This second-order metacognitive evaluation may be essential for conscious report, yet the neural circuit mechanisms linking first-order decisions to metacognitive monitoring remain poorly understood. We investigate how minimal neural network models of metacognitive monitoring generate human-like confidence and uncertainty, and identify their computational limitations. We compare simplified versions of three accumulator network models (Insabato et al., 2010; Atiya et al., 2020; Maniscalco et al., 2021), enabling theoretical analysis and direct comparison. Specifically, we examine how neural circuit dynamics shape decision accuracy, response time, confidence, uncertainty, metacognitive sensitivity (meta-d'), and metacognitive efficiency (M-ratio). We first analyse the steady-state solutions and bifurcation structure of noiseless models, revealing that some architectures are inherently less robust for explicit metacognitive processing, a prediction confirmed by numerical simulations. Incorporating noise shows that confidence and uncertainty emerge from interactions among evidence strength, neural noise, and recurrent connectivity. Across models, first-order decision performance proves substantially more robust than second-order metacognitive evaluation. These findings reveal mechanistic principles by which neural circuit dynamics can generate explicit subjective confidence and metacognitive monitoring. More broadly, they establish a computational framework linking neural dynamics, quantitative measures of metacognition, and conscious access to confidence, offering new insight into how the brain evaluates and communicates the reliability of its own decisions in uncertain environments.

58 Rasmus Herlo

University of Copenhagen

11:40–12:00 · Auditorium 3 Talk

Attention Self-modelling in Collaborative RL-Based Agents

Can machine attention be made inspectable, measurable, and experimentally manipulable? We present an interdisciplinary research program—the "AIntelope" project—that uses Attention Schema Theory (AST) as a bridge between consciousness science and reinforcement learning. AST reframes the hard question "is this system conscious?" into a tractable one: "does this system build and use a model of its own attention?" This shift allows us to translate AST's first principles —recurrent information flow, working memory, explicit self-representation of attention, and predictive planning—into concrete architectural features for RL agents. Inspired by the evolutionary trajectory from simple stimulus-response organisms to attention-guided, planning animals (illustrated through an amoeba-to-antelope developmental ladder), we incrementally built agent architectures in custom gridworld environments: from a baseline DQN, through recurrent (DRQN) and steerable-attention (ROI-gated) variants, to a dual-head agent with action-awareness, sepa-

rated gaze and body controllers, and disentangled reward routing for curiosity-driven attention. A dwell-time ("Linger") mechanism was added to test perceptual masking analogues. We report comparative results across these architectural stages on a food-foraging task, highlighting trade-offs introduced at each step toward AST-compliance, and outline a custom-built orchestration toolkit enabling reproducible, configurable experiments across agents, scenarios, and analytics. We close with future directions—multi-agent scenarios probing theory of mind and affective cooperation, and hierarchical self-modelling via AST-inspired neural cellular automata.

59 Veronika Viksna

University of Copenhagen

12:00–12:20 · Auditorium 3 Talk

Can Language Models Know When They Are Wrong? Selective Escalation Using Self-Monitoring Signals.

LLMs are capable of solving a wide range of tasks; however, their use is associated with substantial computational costs. SLMs can reduce inference costs, but their reliability on complex tasks is often lower than that of larger models. This work investigates a selective escalation mechanism based on behavioural signals extracted during generation. These signals are assumed to reflect decision reliability and distinguish answers requiring escalation. A metacognitively inspired monitoring and control mechanism is proposed, in which a model uses self-monitoring signals to identify and selectively escalate unreliable cases to an LLM. The approach relies on signals capturing reasoning consistency, answer-type agreement, question complexity, reasoning structure, and data sufficiency. A hierarchical decision policy is employed, where some signals act as direct escalation triggers and others serve as supplementary indicators of potential error. Escalation decisions are made without labelled routing data, additional classifiers, or external confidence estimation models. The approach is evaluated on financial benchmarks: FinQA, TAT-QA, and ConvFinQA. The results show that the mechanism improves accuracy over the SLM baseline and outperforms random escalation at comparable escalation rates. The strongest predictors of answer correctness were reasoning consistency and answer-type agreement. The remaining signals contributed less but improved routing performance when used jointly. Computational efficiency depended on dataset characteristics and escalation frequency. The findings demonstrate that self-monitoring signals can identify unreliable answers, improving accuracy over the SLM-only baseline and approaching the performance of an LLM-only system, although computational savings remain dependent on the dataset and escalation behaviour.

60 Maria Isabel Inzunza

KU Leuven

12:20–12:40 · Auditorium 3 Talk

When AI Explains Itself, Should We Believe It?: Toward a Formal Theory of Genuine and Simulated Explanation

A machine learning model can produce a fluent, detailed explanation of its own behavior that has no structural relationship to the processes generating it. Current evaluation frameworks cannot detect this failure because they measure plausibility rather than structural fidelity, leaving both AI research and the sciences that rely on AI tools operating with representations that have the form of explanations without the content that would make them epistemically genuine. This talk introduces a formal framework for addressing that gap. Informed by phenomenological insights about the perspectival structure of epistemic access and operationalized through category-theoretic formalization, the framework defines explanatory adequacy as functor faithfulness on a horizon-relative subcategory of model transformations. An explanation is adequate when it preserves the relational invariants required for specific reasoning tasks relative to the investigative context, not when it maximally discloses internal structure, which is formally unavailable for complex systems. Three structural failure modes are identified and given precise categorical characterizations, of which simulation failure is the most consequential. The deeper motivation is this: any attempt to ascribe or deny phenomenal properties to this kind of system faces a fundamental impasse, since we are constrained to behavioral outputs rather than internal structure. That constraint is not dissoluble, but it is not equally binding under all conditions. By establishing formal criteria for

when an explanation genuinely preserves reasoning-relevant structure rather than simulating its preservation, the framework offers a principled direction toward epistemic traction, the closest available approximation to certainty about systems whose internals remain inaccessible.

Auditorium 4 · 11:20–12:40

Neural Correlates of Conscious Perception

Methodologies, models, and measures of consciousness

61 Rony Hirschhorn

University of Sussex

11:20–11:40 · Auditorium 4 Talk

Studying consciousness – for real?

How we study the contents of consciousness is dominated by the contrastive approach, comparing neural activity when an observer is aware versus unaware of the same stimulus. This tradition inherits strong assumptions: that a single mechanism underlies consciousness, that it operates alike across contents, and that one aspect, vision, is representative of experience as a whole. It also relies on putting participants in perceptual states that are very far from everyday or even altered-state experiences. This raises the measurement question of how far these can take us toward understanding experience. In this talk, I propose a taxonomy of three routes to reduce the gap between the lab and experience as it unfolds outside it: iterating between controlled and naturalistic settings; pushing the lab to better resemble real-life instances; and cognitive-ethology-style designs that begin with everyday experience. I then illustrate how each of these is applied in consciousness research.

62 Dmitri Filimonov

University of Turku

11:40–12:00 · Auditorium 4 Talk

New evidence and challenges in ERP and MEG correlates of consciousness in vision

The past twenty years of research have revealed two event-related potential (ERP) components to be the most reliably occurring neural correlates of consciousness (NCC) in vision: an early visual awareness negativity (VAN) in the N2 and late positivity (LP) in the P3 time window. Three previous extensive reviews concluded that VAN is a proper visual NCC, which is solely modulated by awareness. During the last five years since the latest review was published, a large body of new evidence has emerged about the ERP correlates of visual consciousness. In this systematized review we update the results of the previous reviews by analyzing new studies published since 2020 (N = 53) and discussing their findings. The new evidence is consistent with the earlier reviews: VAN is still found to be the most reliable and robust ERP NCC in vision, whereas LP reflects also many other processes, not consciousness as such. However, several aspects of VAN, for example, its relationship to attention and simultaneous physiological factors, require further investigation.

63 Kacper Pete

University of Amsterdam / University of Cambridge

12:00–12:20 · Auditorium 4 Talk

Non-linear integration and redundant representations in the thalamo-cortical system are associated with stable contents of consciousness

Despite theories of consciousness positing that the integration of thalamo-cortical and cortico-cortical interactions is crucial for consciousness to arise, empirical work on how they sustain stable conscious contents is limited. Moreover, we have recently proposed that large-scale integration, a hallmark of consciousness emphasized in several leading theories, depends on non-linear interactions that many classical metrics of connectivity may not fully capture. To address these gaps, we leveraged large-scale stereo-electroencephalographic (sEEG) recordings in awake

human participants with epilepsy during an auditory bistable perception paradigm, in which a constant auditory stream was perceived as either integrated or differentiated. By combining complementary measures of complexity, connectivity, directed information transfer, and a novel method to isolate non-linear interactions, we examined the spatial and temporal characteristics of the thalamo-cortical interactions underlying conscious content. We present evidence suggesting that integrated contents of consciousness are associated with large-scale, temporally sustained integration that is non-linear in nature, spanning thalamo-cortical and cortico-cortical interactions, which are potentially bidirectional. We further characterized the representational mode linked to these interactions, revealing predominantly redundant encoding of conscious contents, suggesting that they are represented in an overlapping manner across thalamo-cortical networks. Together, these results help advance mechanistic accounts of how large-scale thalamo-cortical dynamics sustain stable contents of consciousness.

64 **Bilge Sayim**
École Normale Supérieure, CNRS

12:20–12:40 · Auditorium 4 Talk

Information compression, redundancy masking, and the limits of conscious human vision
The human brain operates under strong capacity limitations, requiring continuous selection and compression of incoming sensory information. A central question in consciousness research is how these constraints determine what is consciously accessed versus what is processed without awareness. While early visual processing encodes large amounts of information in parallel, conscious perception reflects a capacity-limited, selective, and compressed representation of sensory input. A major question in studying conscious and unconscious processing is whether a stimulus is consciously seen or not, often addressed using detection paradigms. Redundancy masking, by contrast, allows conscious and unconscious processing to be assessed without an explicit detection task by revealing systematic compression of visual information in peripheral vision. In particular, in redundancy masking, individual items in a stimulus consisting of repeated elements are often not consciously perceived. For example, when presented with three identical items, observers often perceive only two, indicating that redundant information is compressed in conscious perception when as few as three items are presented. Here, I argue that redundancy masking reflects a fundamental constraint on conscious access characterized by information compression of redundant sensory input. This provides a framework for investigating how capacity limitations shape the transformation from sensory encoding to conscious perception.

Auditorium 5 · 11:20–12:40

Consciousness at the Limits of Physics

Philosophical foundations in the scientific study of consciousness

65 **Pedro Resende**
Instituto Superior Técnico, University of Lisboa

11:20–11:40 · Auditorium 5 Talk

On physics and the boundaries of consciousness science
The role of physics in consciousness science is widely misunderstood. Theoretical physics today consists of remarkable mathematical models, but it is neither complete nor fully consistent: general relativity and quantum theory remain incompatible after more than a century, and a notorious inconsistency, the cosmological constant problem, sits inside the very "Core Theory" (Wilczek) that governs everyday phenomena, including those in the brain. These unresolved gaps touch concepts of great relevance to consciousness: time, space, causality, locality, and the nature of reality. This matters for how consciousness is studied. Borrowing the logical notion of a conservative extension, we sort confirmatory approaches into three classes: conservativism, which assumes present science suffices; new physics, which modifies it; and idealism, which refounds it on consciousness. We partly follow Carroll's analysis of the Core Theory to pinpoint where new physics

may enter, including potential violations of locality in regimes where it has never been tested, such as the high complexity of brains and other biological systems. Confirmatory consciousness research faces deep difficulties: no current approach yet amounts to a mature scientific theory in the strict sense we define. Progress may therefore benefit from exploratory research, aimed at inducing theories rather than confirming them. For such research, physics literacy matters, so that anomalous findings are not prematurely dismissed by the fallacy of conflating "no known explanation" with "physically impossible." We illustrate with a few examples where this distinction matters. This talk is based on joint work with Kobi Kremnitzer.

66 Peter B. Lloyd

Independent

11:40–12:00 · Auditorium 5 Talk

Interaction between a nonphysical mind and a physical brain (or machine) via quantum 'portals'

The notion that consciousness collapses quantum superpositions has been around since the beginning of quantum mechanics, with advocates such as von Neumann and Wigner, but it never gained much traction because there was no specific theory of how it was supposed to work. (How does consciousness stand in relation to physical stuff? Where and when does collapse happen? What are the structure and dynamics of consciousness? What is it in consciousness that can trigger collapse?) What I will describe in this presentation is that, within an idealist framework, we can formulate a plausible model of consciousness collapsing a quantum superposition. (Idealism aka mental monism is a minority position but gathering some support.) I will examine the basic concepts of conscious collapse, how it can be reconciled with the Born rule, why this idealist approach might be preferred over McQueen & Chalmers' property dualist approach, and why differentiating free will and randomness is a false dichotomy. Aspects of this portal theory also apply under substance dualism (such as Cucu's), but idealism, apart from its philosophical foundation, offers a model of entanglement, which is otherwise a brute fact about the physical world under substance dualism. I will outline constraints on the 'portal' where this collapse occurs, and speculate on possible biological mechanisms that support it. The big issue here is portal navigation, which implies the portal must be a unique macromolecule (hence my conjecture is that portals are DNA, not microtubules). If time permits, I will discuss embodying conscious minds in machines.

67 Robert Prentner

ShanghaiTech University

12:00–12:20 · Auditorium 5 Talk

Unifying the Science of Consciousness: Mathematized Phenomenology and Trace Logic

This contribution sketches a model of mathematized phenomenology as the foundation for a non-materialist and unifying model for the science of consciousness. We argue that such an alternative framework is needed for four reasons: phenomenological adequacy (it respects the first-person structures of experience rather than reducing them to neural correlates), unifying scope (it provides a common formal language across competing theories), predictive power (it generates testable constraints on both phenomenology and neuroscience), and conceptual necessity (it addresses the explanatory gap left by materialist reductionism, which has failed to reconcile subjective experience with objective science). The model builds on prior work in which phenomenologies are represented by Markovian networks of state transitions between experiential states. These states, for example, the observational contents of seeing colors or more complex lived experiences, are further partitioned into accessible and inaccessible subsets, yielding trace chains. This probabilistic and topological structure illustrates how seemingly objective processes can be derived as emergent representations of underlying phenomenological relations. Finally, we propose how this transcendental framework can mediate between seemingly conflicting models of consciousness, including Integrated Information Theory, Predictive Processing, and Global Workspace Theory. These models clash because they implicitly rely on diverging conceptual profiles, operating on different assumptions about causal mechanisms and explanatory targets. By grounding them in an "experience-first" mathematical model, we aim to provide a

unifying framework for consciousness science: one that is rigorous, non-materialist, and capable of reconciling phenomenology with neuroscience.

68 **Chetan Prakash**
AMCS / BAMXi

12:20–12:40 · Auditorium 5 Talk

Contextuality in Models of Consciousness: Trace Logics of Observation and Belief
Contextuality is a striking feature in observations of nature. Quantum theory, reflecting experiment, expresses this as "complementarity"and "non-locality." Contextuality appears also in cognition: for example, when the order in which questions are asked can change answered outcomes. This situation can be made precise within the logic of observation: Outcomes within a given experimental context obey a Boolean description, yet across contexts these descriptions are mutually incompatible. Abstractly this corresponds to an undifferentiated wholeness being arbitrarily parsed into parts, each of which has locally Boolean structure, but for which the global whole is not Boolean. Any model purporting to explain physical or cognitive appearances, whether taking consciousness as fundamental or not, must, therefore, allow such phenomenological contextuality. In this talk we summarize ideas about pasting Boolean logics together into a non-Boolean structures — equivalently, the notion of "sheaf contextuality" — followed by applications to models of consciousness. In particular, we will introduce the mathematical structure of a phenomenological model of conscious observation, the "recursive trace logic," and explore its relationship to contextuality.

Louisiana Museum · 14:30–15:00

Plenary talk

69 **Maša Poljšak Kus**
University of Ljubljana

14:30–15:00 · Louisiana Museum Showcase

Disclosing experiential atmospheres: attuning to the affective layers of lived experience
This study investigates how experiential atmospheres shape the landscape of lived experience and orient an individual's being-in-the-world. Building on the concepts of fringe awareness, moods, and intuitive feelings, we conceptualize background experience as stratified into background disposition, ambient atmosphere and underlying atmosphere that pervade our perception, affect, thought and action. Ambient atmosphere denotes affective tonality experienced as part of the world and of our situated, engaged involvement in it, while underlying atmosphere refers to more fundamental existential orientations that suffuse the entire experiential field of an individual. Using an open, ethno-phenomenological multiple case design, we collected ecologically valid samples of experience and diary entries from three participants across diverse everyday contexts. This qualitative material allowed us to map their affective landscapes, identify atmospheric cornerstones, and trace relations between felt atmosphere, environment, social presence, and behavior. We articulate private, social, and spatial layers of experiential space, each sustaining distinct experiential qualities that may differ in valence and activation yet jointly contribute to the sense of ambient atmosphere. On the level of underlying atmosphere, we propose two fundamental configurations, perturbation and unconcern (ἀταραξία), which we connect with feeling either alert or safe, respectively, and show how they modulate both background and foreground experience. By disclosing these experiential atmospheres and their multilayered structure, our study refines the concept of emotional atmospheres and offers a vocabulary for attuning to subtle affective nuances of everyday life, with implications for phenomenology, psychology, and the design of (social) environments responsive to the affective textures of lived experience.

Louisiana Museum · 18:00–19:00

Plenary talk

70 **Evan Thompson**
Mind-Matter Prize
University of British Columbia

18:00–19:00 · Louisiana Museum Keynote

Consciousness in Life

Organisms are the paradigm conscious beings, and the only kind of beings that we currently are sure are conscious. Nevertheless, the science of consciousness has been formulated with little or no relation to the theory of the organism. What kind of system is an organism? What constraints does the theory of the organism place on understanding consciousness as a natural phenomenon? I will sketch an "organism first" approach to the science of consciousness based on the theory of the organism, and explore its implications for how we should think about consciousness in relation to the brain, and for the issue of which kinds of systems are or could be conscious.

Thursday 15 October

Auditorium 1 (HCØ) · 9:00–10:00

Plenary talk

71 **Mirja Hartimo**
University of Helsinki

9:00–10:00 · Auditorium 1 (HCØ) Keynote

Husserl's philosophy of mathematics and logic

In this talk I will give a general picture of Husserl's approach to mathematics and logic with an aim to situate his views among the contemporary debates in philosophy of mathematics and logic. The first part of the talk focuses on Husserl's views on these matters around the turn of the century especially in Logical Investigations. At this time Husserl believed that mathematical structures, and logic about them, are eternal and outside the time. This view is connected to a kind of platonism about these structures and a consequent logical realism. In the 1920s, Husserl incorporated historical and social factors into his analysis. With the method of Besinnung, he accounted for the socially and intergenerationally developed aims of the mathematicians and the physicists and consequently saw mathematical structures to be valid once first constituted. In Husserl's view mathematical structures are omnitemporal, and logic about them depends on the nature of the domain as well as the historically developed goals of the practices. Mathematics is a socially constructed practice about something mind-independent and the connected view of logic is localized into a kind of eclectic logical pluralism.

Auditorium 2 · 11:20–12:20

Mind The Substrate – On Substrate Dependence and Organisational Invariance

AI, LLMs, and consciousness science

72 **Julian Braunwarth**
University of Göttingen

11:20–11:40 · Auditorium 2 Talk

Does Consciousness Arise from Scale-Invariant Properties?

In "The Conscious Mind", David Chalmers asks the question: "If consciousness arises from the physical, in virtue of what sort of physical properties does it arise?" (Chalmers 1996, 247). He introduces two neural replacement scenarios—the fading and dancing qualia arguments—to narrow down the relevant properties in the physical supervenience base to organizational properties. However, this does not follow from his thought experiments, because they are construed in a way that also (implicitly) preserves the spatiotemporal scale of the relevant processes—a property that is independent of functional organization. This undermines his claim that it follows from his thought experiments that consciousness only depends on organizational—and thus scale-invariant—properties which remain unchanged when the system is uniformly enlarged or shrunk in space and time. Instead, I will argue that consciousness might arise from scale-sensitive properties—or at least partially supervenes on them. Physical properties and dynamics are often scale-sensitive: At the femtometer scale, for example, macroscopic properties such as viscosity or wetness simply do not exist—and quantum processes at the elementary-particle level obey fundamentally different laws than macroscopic ones. Shrinking or enlarging consciousness-related processes to a different level—while preserving functional organization—means replacing the underlying dynamics at level X with dynamics at level Y. Because the physical conditions at level X and level Y can be fundamentally different, the properties that constitute consciousness might be lost under scaling. Thus, consciousness could be inseparably tied to a particular spatiotemporal scale and to the laws that govern that scale.

73 **Jakub Mihalik**
Czech Academy of Sciences

11:40–12:00 · Auditorium 2 Talk

Russellian Monism, Organizational Invariance, and the Possibility of Artificial Consciousness

A crucial support for the possibility of artificial consciousness is based on what Chalmers has called the 'principle of organizational invariance' (POI). If this principle holds, what matters for consciousness is the functional organization of the brain, and not its biological underpinnings. According to Russellian monism, a leading non-reductive approach to consciousness, on the other hand, more than functional organization matters for consciousness – the intrinsic nature of fundamental physical entities is crucial too. I investigate whether this prima facie tension between the POI, and Russellian monism is a symptom of their being incompatible. Miguel Ángel Sebastián argued that they are indeed incompatible, hence Russellian monists must reject the POI. Sebastián considers a subject S who had her neurons replaced with functionally equivalent silicon-based chips, explaining that one of these possibilities must obtain: (1) S's experience disappears, (2) S's experience doesn't change, (3) S's experience qualitatively changes. Sebastián argues that all these options mean that the POI and Russellian monism are incompatible. I argue that there are two ways for Russellians to respond to Sebastián's argument: Some Russellians, e.g., proponents of panqualityism (Coleman, Feigl), have good independent reason to weaken the POI and hold that S after the replacement, and other isomorphic systems, would be conscious, although their consciousness may qualitatively differ (option 3). Furthermore, I'll explain that 'Russellian physicalists' (Montero, Brown) can endorse a sort of multiple realizability of organism-

level consciousness, opting for Sebastián's possibility (2). Artificial consciousness, then, looks like a live possibility for Russellian monists.

74 Adel Chaibi & Aida Elamrani

Intel / Ghent University
with Eric Petit

12:00–12:20 · Auditorium 2 Talk

Form Under Constraint: Between Substrate and Abstraction

Recent debates on computational models of consciousness and digital minds have reignited the classic division between reductive materialism and non-reductive functionalism. On one view, phenomenal experience is strongly substrate-dependent; on the other, it depends only on the implemented abstraction. We seek to develop a more balanced analysis of this problem. We start from the uncontroversial premise that both biological and digital systems process information. And whether characterised as vehicles or as content, physical and abstract bits appear correlative rather than opposed. At its root, informatio denotes the giving of shape, the giving of form. Correspondingly, we propose an intermediate explanatory target: the level of forms, jointly constrained by matter and function. We build our case from the evolution of AI systems. Software continuously exerts pressure on hardware development to meet abstract performance objectives; conversely, hardware challenges algorithmic design to optimise under material constraints. We argue that hardware and software do not develop passively or independently; rather, they co-evolve and co-constrain a shared space of computational forms. We therefore use form to name the structural invariants through which material systems and functional organisations become scientifically tractable together. This is not intended as a new theory of consciousness, nor as an ontological commitment. Our claim is more modest: form is an epistemological concept, marking the level at which the scientific study of consciousness can compare biological, artificial, and hybrid systems without prematurely reducing consciousness either to substrate or to abstraction.

Auditorium 3 · 11:20–12:20

Mapping the State Space of Experience

Philosophical foundations in the scientific study of consciousness

75 Christoffer Lundbak Olesen

Aarhus University

11:20–11:40 · Auditorium 3 Talk

Towards Mapping the Space of Experience: Diversity as a Constraint on Theories of Consciousness

In consciousness science, we need to get comfortable with the indubitable fact that our direct and real intimacy with subjective experience doesn't merit generalisation. Without knowing how consciousness differs, we cannot claim we know what it is, and consequently, we cannot objectively evaluate theory. Mental imagery ranges from complete absence to vivid richness. Some have an inner voice while others do not. People experience their bodies very differently, and even the feeling of time is not universal. These are variations in the character of conscious experience that don't surface through normal human interaction, and due to this latent nature, we simply do not know how deep these differences go. In this talk, I'll argue that we all rely on some sense of familiarity with consciousness when developing models and methods within consciousness science and that this may introduce biases guiding diverging theoretical positions. In addition to testing theories constructed to explain consciousness, we should focus on exploring the extent of what such theories need to explain in the first place, i.e., investigate the diversity of experience, while being open to the possibility that consciousness may not be the phenomenon we each presume. I'll argue that mapping this diversity is essential for an objective science of consciousness, and that it allows for constraint-based search in the space of possible theories. This calls for new methodologies aimed at exploring such differences and invites researchers to

think more openly about consciousness with a curiosity towards otherwise unrecognised forms of experience.

76 Wanja Wiese

University of Bonn

11:40–12:00 · Auditorium 3 Talk

Impossible Conscious Experiences

Are there conceivable conscious experiences that no human being could have? In one sense, this may be trivial, if we consider how our perceptual experiences are limited by our sensory organs. But there may be non-trivial types of impossible conscious experience, and investigating such 'impossibilities' may be facilitated by mathematical models that focus on the structure of experience. This talk clarifies the idea of impossible conscious experiences, by distinguishing between different senses in which conscious experiences can be regarded as impossible, as well as by highlighting the connection to structural approaches to consciousness. The overall idea that emerges is that by getting a handle on the impossible, we gain a better understanding of actual consciousness.

77 Simone Anthes

Monash University

12:00–12:20 · Auditorium 3 Talk

What we talk about when we talk about dreaming

What is a dream? We all think we know what we are talking about when we talk about dreaming. But it turns out we mean different things. This, coupled with the oft-mentioned multiplicity of dreaming and the different disciplines dedicated to its investigation, is reflected in a varied representation of dreaming in the literature. The idea of a pre-defined phenomenon "dreaming" is one of the major conceptual obstacles in dream research and is at the heart of the discussion on integrating dream research into the broader context of consciousness science. However, in actual dream research, there is no such clearly defined phenomenon. Over the last decades, many theories, hypotheses and models about the purpose and character of dreaming have emerged from the various disciplines involved in dream research. Yet, they often fail at explicitly defining their explanatory target. Here, we survey contemporary dream theories and demonstrate the variety in how they conceptualise dreaming. Acknowledging that there has been nonetheless massive progress in dream research, we ask whether a definition-led approach is even necessary and present several alternative approaches. We conclude that while it is necessary to be transparent about perspective and preconceptions on dreaming, a unified definition may not be the prerequisite to advancing dream research that it is often framed to be.

Auditorium 4 · 11:20–12:20

Creativity, Insight and Mood

Phenomenology and applied phenomenology

78 Olha Sobetska

Vrije Universiteit Brussel

11:20–11:40 · Auditorium 4 Talk

The Dynamics of Creative Cognition

Creativity is often described through separate perspectives, including psychology, neuroscience, and computational models. While each of these explains part of the phenomenon, they do not yet provide a common framework for understanding how a creative idea emerges in real time. This talk proposes a synergetic perspective in which creativity is viewed as a process of self-organization, supported by order parameters and phase transitions. In this way, a creative moment can be understood as the emergence of a dominant pattern that temporarily organizes neural

and cognitive activity around a new idea. When an existing organisation (a temporarily stable idea) becomes unstable or ceases to correspond to the current context (the task's constraints), a new coherent structure may emerge, guiding thoughts towards a creative outcome. This approach differs from one that views creativity as a 'black box' or a purely random event, it posits that creativity arises as a result of the interaction of a multitude of cognitive micro- and meso-processes. Drawing on findings from neuroscience together with concepts from synergetics, the talk presents a dynamical way of understanding creativity without reducing conscious experience to a fixed sequence of computational steps.

79 Jeanne Catherine-Gray

University of Virginia, Contemplative Science Center

11:40–12:00 · Auditorium 4 Talk

Insightful Shifts in Consciousness Among Men in Prison: A Micro-Phenomenological Investigation

Insight, often described as an "aha!" moment, entails a distinct phenomenology that differs from other problem-solving processes. Yet little is known about the experiential structure of sudden, transformative insights that produce shifts in ordinary waking consciousness. This study presents a fine-grained micro-phenomenological analysis of such insightful change; a sudden moment of psychological transformation as described by eight men incarcerated in a UK prison following exposure to the Insight to Wellbeing program. Analysis revealed a shared experiential structure of insight comprising four phases and a transition event: (0) understanding, (1) recognition, (realization event), (2) opening, and (3) connection. Insight involved a shift from habitual, constricted states of mind toward openness and calm, accompanied by emotional relief and an increased sense of agency. These changes occurred without a loss of self. The findings support understanding insight as a mechanism that can reconfigure one's state of mind and further demonstrates the value of micro-phenomenological methods for studying particular shifts in consciousness.

80 Madalina Wernigg

University of Zürich

12:00–12:20 · Auditorium 4 Talk

The Instability of the Epistemic Split: Bipolar Disorder as a Clinical Case of Contextual Emergence and Thermodynamic Desynchronization within Dual-Aspect Monism

Operating under the postulate of consciousness as fundamental ontic reality, this paper rejects neuro-reductionism by defining the brain as an evolutionary filter. Within dual-aspect monism, the central nervous system acts as an operational symmetry breaker. Through a stable macro-context (C_healthy), it enforces a stable partition of the psychophysically neutral unus mundus, creating the regular epistemic split necessary for survival [Atmanspacher 2012]. This framework proposes the transient destabilization of this epistemic interface during acute affective states, conceptualized as a thermodynamic desynchronization, wherein altered energetic constraints disrupt the fundamental time-structure of subjective experience. This transition is driven by a hierarchical cascade: interpersonal desynchronization triggers a vertical circadian mismatch between central and peripheral pacemakers, manifesting as a temporal desynchronization of the lived body [Fuchs 2013], while disrupting mitochondrial bioenergetics, intracellular calcium, and the homeostatic sleep-reset [Borbély 1982]. Consequently, the coarse-grained partition of the state space collapses into opposing structural instabilities [Atmanspacher 2012]. Mania (C_manic) manifests phenomenologically as hyper-salience, an unwillful exposure to non-local correlations, and a profound dissolution of the ego- environmental boundary (under-coarse-graining). Conversely, depression (C_depressive) induces a severe temporal uncoupling, rigidly isolating the subject from meaningful holistic correlations through an agonizing stagnation of inner time (over-coarse-graining). Crucially, modeled through contextual emergence, the system exhibits dynamical reversibility based on its intrinsic resilience: by re-introducing external somatic and temporal constraints via nutrition, sleep, and stabilizing pharmacological agents, the dissipative system returns toward its lower-entropy attractor state, restoring the complementary epistemic split.

Auditorium 5 · 11:20–12:20

The Hard Problem and the Limits of Explanation

Philosophical foundations in the scientific study of consciousness

81 Nicolas Loerbroks

Independent

11:20–11:40 · Auditorium 5 Talk

The Hard Problem Presupposes Mechanistic Exhaustiveness

In formulating the "hard problem", Chalmers explicitly presupposes that once we specify either the relevant neural mechanisms or their computational organization, "a full and satisfying explanation will result" (Chalmers, 1995, p.202) for cognitive phenomena – a presupposition implicit in his reformulation of computational functionalism in response to the triviality concerns raised in Putnam's later work. The hard problem arises by contrast: conscious experience is taken to fall outside what mechanistic or structural explanations can deliver, whereas such explanations succeed for cognition. Putnam's original machine functionalism did not require this asymmetry. In characterizing pain as a multiply realizable functional state, he rejected type-identity theories together with the demand that the objectivity of such computational states be grounded in a uniquely specifiable physical causal-organizational structure. The asymmetry emerges, in the form it takes within the computational functionalist tradition, only when computational explanation is reinterpreted as complete in capturing the relevant causal or organizational structure of a system. This diagnosis further reframes the dispute between Dennett and Chalmers as a disagreement over explanatory adequacy rather than over the ontological status of phenomenal experience: reading Dennett's real-pattern account as an intermediary between Putnam and Chalmers – anti-exhaustivist while preserving causal closure – shows that it is Chalmers, not Dennett, who occupies the revisionary extreme, since only his reformulation installs the exhaustiveness standard that makes consciousness exceptional.

82 William Seager

University of Toronto

11:40–12:00 · Auditorium 5 Talk

Is the Metaphysics of Predictive Processing Self-Refuting?

Predictive processing theory (PPT) holds that the core function of the brain and nervous system is to facilitate successful behaviour by minimizing errors between predicted and incoming signals. PPT operates hierarchically across levels, from endogenous neural processes all the way to sensory signals from the world, always computing discrepancies between a predictive model and incoming data. PPT is mathematically rich, describing signal and prediction in terms of probability distributions, with error minimization achieved via approximate Bayesian updating. There are promising proposals for its neural implementation. However, philosophical interpretation of the theory is less well developed. Integrating consciousness into PPT raises the question of where experience resides: input or model? The answer seems clear. The home of conscious experience is the predictive model. We experience the world as we then take it to be, awaiting incoming corrections or confirmations. This is why in PPT experience is sloganized as 'controlled hallucination' or 'neuronal fantasy'. Arguably, this engenders a paradox. PPT takes the brain as given, its neural architecture running recondite computations over probability distributions. Yet the brain is, ultimately, known via perception. It is part of the hallucination. But, it is then argued, 'an hallucinated brain cannot explain the hallucination of a brain'. Thus, PPT appears ungrounded; the so-called neural basis of PPT becomes just a tail-biting 'physicalist fantasy'. There is something right about this worry, but something wrong as well. Assessing it leads to deep questions about both science and reality, and their relation to consciousness.

83 Michael Schnabel

Vanderbilt University

12:00–12:20 · Auditorium 5 Talk

Modeling Consciousness: On the Limits of Theory and Conceptual Understanding

LLMs are trained on statistical regularities in language and reasoning, while cognitive science models cognition in terms of predictive coding, information processing, and active inference. Implicit in the scientific approach is the assumption that increasingly refined models will yield a deeper understanding of both nature and mind. This talk questions that assumption with respect to consciousness. Drawing on the Buddhist principle of dependent origination and the Yogācāra analysis of cognition, I argue that scientific attempts to "explain consciousness" may inadvertently reinforce the very dualistic framework they seek to overcome by reifying the subject-object distinction. Paradoxically, the pursuit of ever-greater predictive power might obscure what it ultimately seeks to explain. As an alternative to the scientific emphasis on modelling and conceptual understanding, I turn to the Chan/Zen practice of investigating mind through direct contemplation. Rather than offering another theory of consciousness, this reorientation shifts the focus from explanation to exploration. It invites a prior question: whether the modelling stance itself, together with the quest for conceptual understanding, constitutes the very obstacle to understanding consciousness – and what possibilities for inquiry emerge when that stance is reconsidered.

Auditorium 1 (HCØ) · 13:30–14:00

Plenary talk

84 Eliza Ramos de Andrade

Universidade de São Paulo

13:30–14:00 · Auditorium 1 (HCØ) Showcase

Mathematizing husserlian temporality through the asymmetric bicone model

This talk presents a novel topological model to visualize Edmund Husserl's phenomenology of internal time-consciousness. While Husserl's traditional two-dimensional diagram of immanent temporality is a powerful analytical tool, it has been critiqued for its static nature, which risks creating a misleading analogy to linear objective time. Furthermore, later three-dimensional adaptations, such as H. Banega's spherical mapping, introduce structural symmetries that fail to adequately capture the qualitative phenomenological differences between our experiences of the past and the future. To resolve these descriptive difficulties, this presentation introduces the asymmetric bicone model. By mapping the temporal flow onto an inverted, asymmetrical bicone, we visually formalize the fundamental distinction between retention and protention. The lower, longer cone represents the horizons of retention, converging to a closed, dense vertex that illustrates the determined and concluded nature of the past. Conversely, the upper, shorter cone corresponds to protentions, diverging towards an open vertex that captures the indeterminate, open, and immediate nature of future expectations. Ultimately, this talk argues that the asymmetric bicone model provides a vital elucidative framework that avoids the false symmetries of previous spatial analogies. By translating the irreducible asymmetry of temporal experience into a precise topological structure, this representation deepens our understanding of the structural dynamics of consciousness and establishes important constraints for contemporary cognitive science.

Auditorium 2 · 16:30–18:00

Time-Consciousness

Phenomenology and applied phenomenology

85 Antonina Skrypnyk

Taras Shevchenko National University of Kyiv

16:30–16:50 · Auditorium 2 Talk

The Instant Phenomenological Field as a Fundamental Unit of Conscious Experience

What constitutes the elementary unit of conscious experience? Contemporary theories of consciousness typically focus either on neural mechanisms or on the contents of awareness, while the structure of experience itself often remains insufficiently explored. This presentation proposes a theoretical framework based on the concept of the Instant Individual Phenomenological Field (IIPF). The model begins with a distinction between a physical instant and a psychological moment. Unlike a dimensionless point in physical time, a psychological moment is a finite interval experienced as a unified whole. Conscious life can therefore be understood as a succession of instant phenomenological fields whose continuity constitutes the Extended Individual Phenomenological Field. Within this framework, phenomenological fields are interpreted as the ideal aspect of the subject, analytically distinguishable but inseparable from its material aspect. Perception, memory, imagination, and thinking are examined as different modes through which conscious experience accesses material structures. This approach offers a novel perspective on the mind–body problem by replacing the traditional opposition between mental and material substances with the notion of complementary aspects of a single reality. Particular attention is devoted to the internal organization of the IIPF. The field is conceived as a dynamic structure centered on attentional focus and composed of elements that differ in their degree of awareness. A distinction is proposed between a central zone of focal awareness, a broader zone of conscious contents, and a peripheral zone of subconscious processes. Rather than constituting a conventional geometric or topological space, the IIPF is modeled as a multidimensional network of

86 Hanqi Wang

Carnegie Mellon University

16:50–17:10 · Auditorium 2 Talk

Higher-Order Representation as Cross-Timescale Integration in Time-Consciousness

Temporality is a fundamental feature of conscious experience and has been proposed as a candidate for a minimal unifying model of consciousness (Windt 2015; Wiese 2020). Singhal and Srinivasan (2021) provided a three-level framework for the phenomenology of time experience, including a fast cinematic level (30–50ms), an intermediate extensional level (300–500ms), and a slow retentional level (3–5s), detailing how these levels interact across timescales. However, the framework does not explain why content at the intermediate level becomes conscious while content at the fast level does not. We argue that this gap can be filled by the HOROR (Higher-order Representation of a Representation) theory proposed by Brown (2015), which holds that phenomenal consciousness consists in instantiating a certain kind of higher-order representation. The fast-updating Level 1 content constitutes an unconscious first-order representation, which is integrated by Level 2 over its ~300–500ms window. Crucially, Level 2 gains its higher-order character through cross-timescale integration by unifying multiple Level 1 slices, yielding content categorically distinct from any single static slice. By unifying first-order slices under a single experiential perspective, this integration supplies the minimal self-attributive structure HOROR requires without presupposing a conceptual-propositional judgment. This integration contributes to both phenomenological and higher-order frameworks. It provides the former with a mechanism for why content becomes conscious, while showing that the higher-order representation is inherently temporally extended. It further yields differential predictions for paradigms such as continuous flash suppression and the attentional blink, which selectively disrupts integration at distinct timescales and should therefore dissociate the proposed levels.

87 Kohei Fukao

Hokkaido University

17:10–17:30 · Auditorium 2 Talk

How Are Retention and Protention Extended in Sign-Mediated Counting?

This talk aims to explicate a phenomenological time structure that is essential to the conscious activity of counting. In the well-habituated, rule-governed activity of counting, we usually use number signs. As we are often too well-habituated to the use of number signs, it is not so simple a task to explicate what we do when we count using them. Husserl addressed a pertinent issue: in his Philosophie der Arithmetik, he made it explicit that a certain time structure (namely, succession and simultaneity) is indispensable for the counting activity. In the work, he primarily focused on the relationship between number and time in what he called 'authentic representation' of numbers. Thus, Husserl was not explicit about the time structure of sign-mediated numerical experience. Lohmar, in his Phänomenologie der Mathematik, discussed the number-time relationship in what Husserl called 'symbolic representation' of numbers, while also considering the relationship in light of the phenomenological account of time consciousness. In this talk, I attempt to further the phenomenological theory of time consciousness that essentially pertains to the activity of counting by addressing the following two: what I call extended retention/protention and how they are possible by virtue of number signs. I conclude that the proposed account of extended retention/protention not only advances the phenomenological account of the number-time relationship in sign-mediated counting activity, but can also serve as a phenomenological foundation for modelling such basic numerical experiences as counting.

88 Donald Borrett

University of Toronto

17:30–17:50 · Auditorium 2 Talk

A temporality model of flesh and dual-aspect monism

Merleau-Ponty proposed an ontology of flesh in which the phenomenal field is characterized by an intertwined and reversible interplay of the visible and invisible that serves as the basis of meaningful experience. A temporality model of flesh is proposed in which the visible is the flux of sensible experience and the invisible is time horizons that are generated by the flux and feedback onto it to establish a self-adjusting, iterative system in which past, present and future are equally implicated in its evolution. Because it is a dynamical process, it can be mathematized and universality criteria established in all systems that have this form. By necessity, all such systems have two, and only two, dynamical signatures that are isomorphic to pre-reflective and reflective experience. Within the latter are dynamical signatures isomorphic to mind (determinate invisible) and world (determinate visible). A Logos of flesh can be identified that represents the quantifiable dynamics of a monistic ontology that grounds its self-interpretation both as mind and as world. Although they lack causal interaction, mind and world share a temporal characterization because of the universal dynamics that represents the condition for their possibility.

Auditorium 3 · 16:30–18:00

Comparing Theories of Consciousness

Philosophical foundations in the scientific study of consciousness

89 Sucharit Katyal

University of Copenhagen

16:30–16:50 · Auditorium 3 Talk

What Is Consciousness? And What Does a Theory of Consciousness Need to Explain?

Despite major advances in consciousness science, theories still disagree about their target: What is consciousness, and what must a theory of it explain? I argue that this disagreement follows from an unresolved structural problem. Recent structural approaches model consciousness by mapping relations among phenomenal qualities—especially the structure of perceptual contents

or qualia spaces—to corresponding neural or mathematical structures. This has clarified how contents may be organised, but leaves aside the wider structure of consciousness as such: the relations among content, agency, and minimal subjectivity. This talk proposes a comprehensive structural model of consciousness. Drawing on Husserlian phenomenology and Indian contemplative philosophy—especially P. R. Sarkar's analysis of subjectivity in relation to advanced meditative states—I distinguish three intermutable phenomenological components: mental object, doer-I, and existential-I. Consciousness is conceived as a phenomenal resource expressed through these components in varying degrees. Different conscious states—ordinary perception, agency, dreaming, meditation, ego dissolution, nondual awareness, and minimal experience—reflect different distributions and transformations of this resource. The proposal extends structuralism beyond phenomenal content. Rather than treating structure as primarily internal to qualia or perceptual fields, it treats the whole conscious episode as structurally articulated by objecthood, directedness/doership, and existence. This provides a theory-neutral explanandum for consciousness science: a theory of consciousness must explain not only how contents are structured, but how phenomenal resource is distributed across and transmuted between its three structural components. I conclude by outlining implications for operationalisation, naturalisation, comparative differences across species and individuals, and development.

90 **Moritz Kriegleder**
University of Vienna, Czech Academy of Sciences
with Wiktor Lachowski, Tomas Marvan

16:50–17:10 · Auditorium 3 Talk

Adapting the Matched-Interlevel-Experiments Framework for Mechanisms of Consciousness

Recent work on the neural correlates of consciousness has increasingly turned to interventionist methodologies developed in the philosophy of science. One particularly notable approach in this vein is Craver's "new mechanist" account of causal explanation. Yet it remains unclear whether mechanistic criteria designed for paradigms like memory or motor behavior can be coherently extended to phenomenal consciousness. This talk examines that extension by applying the Matched-Interlevel-Experiments framework (MIE) to mechanisms of consciousness. MIE consists of four conditions to identify the constitutive parts of the mechanism, viz. bottom-up inhibition, bottom-up excitation, top-down activation, and cross-experimental matching. We argue that while MIE is well suited to reveal important details of neurobiological mechanisms, it runs into a distinctive set of problems in consciousness research: the difficulty of disentangling genuine mechanism responsible for consciousness from what are merely causal prerequisites of it, the privacy of experience and the relations between lower and higher "levels" of brain description. We propose tentative strategies for addressing these challenges and making the MIE framework a productive tool for consciousness researchers. We focus on a concrete candidate mechanism: apical processes in layer-5 pyramidal neurons pivotal for low-level, small-scale theories of conscious perception such as Apical Amplification and Dendritic Integration theories. We close by considering whether this framework generalizes to higher-level, network-scale theories of consciousness, or whether it remains tied to cellular mechanisms with unusually tractable points of intervention.

91 **Tonglin Yan**
Paris-Saclay University

17:10–17:30 · Auditorium 3 Talk

Formalizing Consciousness: A Comparative Mathematical Analysis of Leading Theories

Systematic comparison across consciousness theories has been hindered by the absence of a shared formal language. We reformulate four mathematically articulated frameworks, namely Integrated Information Theory (IIT), Active Inference (AIF), Higher-Order Representation approaches (HOR/HOSS), and the Projective Consciousness Model (PCM), within a common probabilistic and geometric idiom organized following Marr's three levels of analysis. Each framework is rendered in terms of stochastic kernels, divergence measures, variational free energy, and group-structured state spaces. IIT is characterized through partition-based cut operations and irreducibility mea-

sures. AIF is presented as a general process theory for perception and action under uncertainty; HOR/HOSS extends this as hierarchical metacognitive inference within factorized generative models. PCM is cast as a group-structured POMDP in which the Field of Consciousness is organized by projective transformations under expected free energy minimization. This formal treatment reveals that several apparent theoretical conflicts dissolve upon alignment of explanatory level. IIT specifies the integrative property a system must possess; AIF and HOR concern the inferential mechanisms of conscious access; PCM contributes a formal account of perspectival structure. Genuine tensions persist around identity claims, first-person geometry, and computational tractability. As a prospective direction, we outline a hierarchical POMDP architecture as a potential computational testbed in which integration, higher-order access, and perspectival control could be evaluated within the same model, supporting more systematic empirical comparison among theories.

92 **Krzysztof Dolega**
Ruhr-Universität Bochum

17:30–17:50 · Auditorium 3 Talk

We Don't Need Another Model: Towards a Meta-Theoretic Framework for Comparing Theories of Consciousness

After roughly three decades of methodological advances and genuine empirical breakthroughs, the science of consciousness now faces a problem of theory proliferation. The number of competing models is estimated at somewhere between twenty (Seth & Bayne, 2022) and two hundred (Kuhn, 2024), with no widely accepted standard for adjudicating between them (Doerig et al., 2021). Comparative efforts, such as adversarial collaborations (Melloni et al., 2023), remain piecemeal and have so far proven inconclusive (Lau, 2023; Mudrik et al., 2025). Meanwhile, the crisis is compounded by systematic confirmation bias, as the choice of methods reliably predicts which theory is being tested (Yaron et al., 2022). I argue that resolving the issue of adjudicating between competing approaches can only be achieved by building a meta-theoretic framework for theory comparison: a common structure for evaluating competing models against the theoretical virtues long recognized in philosophy of science (Kuhn, 1977; Keas, 2018). Although building such an instrument will likely take time and a joint effort of multiple researchers, I present and defend a blueprint for the general shape it should take. In particular, I motivate the set of virtues to be included (including, but not limited to, predictive accuracy, simplicity, and explanatory scope) and discuss how they might be formalized, weighed, and traded off transparently. I close by defending this virtue-based approach against competing proposals, such as the Measure Centrality Index (Chis-Cuire et al., 2024) and the search for a Minimal Unifying Model of consciousness (Wiese, 2020).

Auditorium 4 · 16:30–18:00

Toward Understanding Non-Dual Experience and Ego-Dissolution

Phenomenology and applied phenomenology

93 **Jaya Caporusso**
Jožef Stefan Institute

16:30–16:50 · Auditorium 4 Talk

Dissolution Experiences and the Experience of Self and World: An Empirical Phenomenological Investigation

Dissolution experiences (DEs) are experiential episodes during which the perceived boundaries between Self and World become fainter or less clear; this is usually accompanied by a sense of unity with the World or elements of it. We investigate the experiential structure of DEs by comparing them with daily-life experiences (dlEs), focusing on how Self and World are experienced. We conducted 29 phenomenological interviews with eight participants, covering eight dlEs and 10 DEs occurring in different contexts (meditation, nature, art, psychedelic experiences, intimacy, and

psychopathological episodes). Our findings show a progressive transformation in the structure of Self–World experience. While in dlEs the Self is experienced as a complex multi-element entity, in DEs it usually undergoes a loss of internal coherence, and most of its elements disappear; often, only a sense of mineness remains. At the same time, the World progressively appears as more open, friendly, safe, familiar, vivid, spatially infinite, and internally unified. As this unfolds, processes involving Self and World become increasingly difficult to distinguish from one another, mirroring a blurred subject/object distinction. A central feature of almost all analysed DEs consists in trusting the World and letting go of control: participants approach the World with trust and/or recognise it as trustworthy, while relinquishing or relaxing control, often with a sense that it is safe to do so. Our study contributes to a conceptual and phenomenological understanding of DEs and provides insights relevant to discussions on the metaphysical and phenomenological nature of the Self.

94 Kiana Ward

Radboud University

16:50–17:10 · Auditorium 4 Talk

Modeling non-dual awareness via constraint closure: a reinterpretation of groundlessness

Non-dual awareness (NDA) refers to a shift in consciousness in which the usual distinction between subject and object dissolves, and experience is no longer structured by conceptual mediation or goal-directed regulation. Meling enactivist model describes NDA as a meditative disclosure of groundlessness—the recognition of emptiness (śūnyatā), that all phenomena lack intrinsic nature. While enactivism explains autonomy through process closure, this article argues that constraint closure, as developed by Nave, extends that framework by making explicit how autonomy is sustained through the continual regeneration of its own relational conditions. This refinement prevents process-closure models from being read in substantialist terms when applied to complex cognitive systems, where stability arises through ongoing transformation rather than fixed organization. Nave's account builds on Juarrero theory of constraint causality, which replaces intrinsic forces with relational conditions—a view that parallels Nāgārjuna's Madhyamaka analysis of dependent origination. Integrating these perspectives, I propose that NDA corresponds to a shift from decoupled to precarious constraints, revealing that cognition and awareness persist not through intrinsic foundations but through the dynamic regeneration of interdependent relations.

95 Niffe Hermansson

Trace Institute

17:10–17:30 · Auditorium 4 Talk

Traces of the Other

Encounters with apparently autonomous entities are among the most striking and recurrent features of high-dose DMT experience. In both naturalistic and lab settings, many users report that these encounters seem to involve beings with autonomy, intelligence, and agency, rather than mere imagery or hallucinations. This raises the question: can these claims be scientifically explored without committing to, or presupposing, an atypical metaphysics? We suggest that one useful avenue for exploring these claims is provided by Conscious Agent Theory within the Conscious Realism framework, a consciousness-first formalism in which reality is modeled as consisting exclusively of a network of interacting conscious agents. In this account, conscious agents are modeled in terms of measurable spaces of possible experiences and Markov-kernels governing probabilistic transitions between experiential states. To say that one agent perceives another is to say that the dynamics of the latter influence the updates of the former's experience, mediated by the relevant kernels. Using this formulation, questions about DMT entities can therefore be decomposed into precise questions about the structure, possibilities, and constraints of experiential dynamics. Importantly, the familiar distinction between true perception and hallucination is not basic in this formulation. In relation to DMT entities, this is a strength: rather than having to ask blunt questions about whether these entities are "real" or "hallucinated," it allows us to ask whether entity encounters exhibit identifiable transition structures, invariants, temporal stability, responsiveness to hidden variables, or nontrivial intersubjective correlations. We use this decomposition to formulate a research program in which these features

96 Andres Gomez Emilsson

Qualia Research Institute

17:30–17:50 · Auditorium 4 Talk

Field Lines Before Equations: Parameterized Dynamic Phenomenology and the HEART Retreat

I'll present Parameterized Dynamic Phenomenology (PDP), a methodological framework we've been developing at QRI, and situate it within HEART, the legal exploratory retreat we run where physicists, mathematicians, artists, and advanced meditators get together to reverse-engineer non-ordinary states of consciousness as a team. PDP has four components: (1) represent a sensory stimulus as a data structure, (2) apply a parameterized, mathematically-defined operation that modifies it dynamically, (3) translate the modified structure back into a representation of phenomenology, and (4) match parameters to specific states of consciousness. The key idea is that when a participant replicates their own phenomenology inside a generative model we fully understand, the parameters they land on give us much stronger evidence than verbal report can, because they pin down mathematical features actually present in the experience. At HEART we've been mixing a lot of approaches: psychophysics, Likert ratings, participant vibe-coded sequences, drawings, and replication tools like the Tracer Tool and the Oscilleditor. One direction we find promising is characterizing states in terms of coupling kernels – the way oscillators want to phase-relate to their neighbors – and asking whether different substances and meditative states leave distinguishable signatures there. This past retreat we ran a protocol comparing formless jhana states with high-dose 5-MeO-DMT; analysis is still underway, so I'll present the design, the questions, and some of the data analysis rather than the full results.

Auditorium 5 · 16:30–18:00

Indicators of AI Consciousness

AI, LLMs, and consciousness science

97 Tobias Schlicht

Ruhr-University Bochum

16:30–16:50 · Auditorium 5 Talk

Modeling the Ghost in the Machine

Debates on AI consciousness suggest that we can, assuming computational functionalism, develop computational markers to test whether an artificial system is conscious. Participants clarify that they mean phenomenal consciousness, in Nagel's (1974) sense that there is something it is like to be a system or undergo a subjective experience (e.g., Butlin et al. 2025, Birch 2025, Chalmers 2023, Schneider et al. 2025, Kanai & Ma 2026). I argue that this notion is unfit to serve as a target for modeling consciousness in AI. I present two ways it has been characterized. Block (1995) introduced it as a kind of consciousness distinct from access consciousness, but subsequently distinguished (the capacity of) experienced content from cognitively accessible content (Block 2007), first suggesting that it could be investigated empirically, but eventually being discontent with the most ingenious methods like no-report paradigms (Tsuchiya et al. 2014), asking for no-cognition paradigms instead (Block 2019). This requirement makes it methodologically inaccessible in AI. Chalmers (1996) separated phenomenal consciousness as the hard problem from cognitive functions associated with consciousness, the easy problems: it's non-physical, non-functional, and epiphenomenal, which makes it methodologically inaccessible via third-person methods. Any markers possibly justifying attribution of phenomenal consciousness are either absent in AI (neural and first-person markers), ambiguous (behaviors), or question-begging (computations). Absent other characterizations of phenomenal consciousness, ironically, the moral is that attempting to model and assess phenomenal consciousness in AI, illusionism about phenomenal consciousness (Dehaene 2014, Dennett 1991) is advisable. Otherwise, researchers are chasing a ghost in the machine.

98 Angela Johnson

Northeastern University

16:50–17:10 · Auditorium 5 Talk

From Indicators to Dynamics: Metastability, Recurrence, and Ignition in a Persistent Agent's Attention Stream

Recent work derives indicators of consciousness for AI systems from neuroscientific theories (Butlin et al., 2023), but these are largely architectural checklists; dynamical measures computed on a running agent's own trajectory remain rare. We present a novel framework (AURORA) that treats the turn-by-turn attention dynamics of a persistent-memory agent as an observable state stream, wherein at each step the agent scores memory candidates and selects a subset to attend to, optionally cross-referenced with its affective and desire state. On this stream we compute measures imported from consciousness science. Metastability is the temporal variance of the Kuramoto order parameter over the selection signal (Tognoli and Kelso, 2014), indexing integration-differentiation balance. Recurrence quantification analysis, via determinism and laminarity, characterizes temporal structure and rigidity (Marwan et al., 2007). A perturbational fracturability index, the behavioral analog of the perturbational complexity index (Casali et al., 2013), injects a controlled perturbation and scores the Lempel-Ziv complexity of the agent's response trajectory. An ignition criterion flags global-access events in the spirit of the global workspace (Mashour et al., 2020). We borrow these measures' human validation as rationale for their utility in examining synthetic human-like cognitive states, while scoping our claim to cognitive dynamics, such as rigidity, adaptation, and collapse, rather than phenomenal experience. We report preliminary longitudinal data from continuous runs and contrast a healthy trajectory with a pathological collapse identified independently. Persistent agents, whose internal attention is fully observable, are a uniquely controllable system for validating and stress-testing computational measures of cognitive state.

99 Jessica White

Sentient Futures

17:10–17:30 · Auditorium 5 Talk

Locating disagreement in AI consciousness research: literature review and qualitative study.

Experts disagree about AI consciousness, and what could follow morally. Their disagreement can be mapped across multiple levels: conceptual, methodological, theoretical, metaphysical and normative. We analyse these areas of dispute in two stages. The first stage consists of a systematic review of work on AI consciousness and moral consideration, with published arguments classified into subcategories, marking active areas of debate within the field. For the second stage, we conducted a series of qualitative interviews with researchers and non-research experts across consciousness science, philosophy of mind and AI. We document how they reason in practice, beyond their published positions, and test whether their reasoning tracks the cruxes identified in the review or reveals new ones within the scope of our study. We asked participants what evidence would convince them that an AI system is conscious, and whether respective findings would carry moral weight. The transcripts are analysed in two ways: a content analysis compares arguments against the literature-derived classification, and a thematic analysis that captures what the existing classification does not anticipate and surfaces the considerations shaping participants' opinions.

100 Dezhi Luo

University College London

17:30–17:50 · Auditorium 5 Talk

(How) can we benchmark consciousness in foundation models?

This talk examines theoretical and practical issues in benchmarking consciousness in large-scale foundation models. Drawing on philosophy of science and recent progress in consciousness science and machine learning, I suggest consciousness benchmarks ought to be theory-driven and antirealist-committed, and offer practical recommendations. I note that consciousness science faces a unique epistemic challenge: one cannot in principle verify whether systems other than

oneself are conscious (Papineau, 2002; Birch, 2022), foreclosing adjudication between physicalist interpretations. Yet I argue this parallels concerns about measuring unobservables in philosophy of physics (Ruetsche, 2017), warranting antirealist treatment (van Fraassen, 1980; Doppelt, 2005): benchmarking AI consciousness is viable when measurement serves pragmatic rather than truth-conferring goals (contra. Love, 2026). I then discuss two competing approaches. Theory-based approaches designate consciousness when a system instantiates the substrate specified by that theory; computational-functional theories have yielded proposals for measuring foundation models (Butlin et al., 2025), consistent with standard commitments in ML cognitive benchmarking (Hagendorff et al., 2023; Li et al., 2025; Ma, Saxon, & Yue, 2025). The INK approach further allows likelihood-of-consciousness estimates based on proximity to humans, even absent a consensus theory (Bayne et al., 2024; Peters, 2026). I contest that marker-based approaches rely on behavioral signatures without mechanistic grounding and risk undermining construct validity (Palminteri & Wu, 2026). Finally, I consider practical directions, including (1) prioritizing multimodal foundation models, given the perceptual dominance of consciousness measures in humans; (2) combining mechanistic interpretability with behavioral-reductionist paradigms; and (3) applying standard ML protocols (esp. scaling analysis, causal ablation) to consciousness

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Margrethe Bohr Salen (NBB) · 16:30–18:00

Experiential workshop

Sandrine R. Schiller
University of Copenhagen
with Felix Pastorius, Johan Bech Jespersen, Nina Rajcic

16:30–18:00 · Margrethe Bohr Salen (NBB) Workshop

Art & Experience Workshop

'What is called Philosophy of Art usually lacks one of two things: either the philosophy, or the art.' Friedrich Schlegel, Kritische Fragmente, 1797

What can the experience of art and artistic practices tell us about the expanse and limitations of conscious states? Pursuing the experiences of the artists Felix Pastorius, Johan Bech Jespersen and Nina Rajcic we will discuss the techniques and possibilities of creating shared experiences and resonances. The conversation will be moderated by Sandrine R. Schiller.

Felix Pastorius (https://www.instagram.com/pastorfl3x/?hl=en)
Felix Pastorius is an American jazz bassist, composer, and music educator who has earned international recognition for his technical virtuosity and distinct improvisational voice. Born in 1982, his early musical development was shaped within a prominent musical lineage as the son of pioneering fusion bassist Jaco Pastorius. His professional career has been defined by a rigorous, self-directed exploration of the instrument, culminating in high-profile ensembles and published pedagogical methodologies. Felix Pastorius achieved international acclaim by anchoring the legendary, Grammy-winning jazz-fusion quartet Yellowjackets from 2012 to 2015. Maintaining a highly versatile professional profile, he has transitioned fluidly between avant-garde jazz and mainstream genres, performing with icons like Bobby McFerrin, David Gilmour and A$AP Rocky.

Johan Bech Jesperson (https://johanbechjespersen.com/)
Johan Bech Jespersen graduated from the School of Sculpture at the Royal Danish Academy of Fine Arts in 2024 and has a background in architecture. Working across
sculpture, installation, and performance, his practice is concerned with how situations and environments can be constructed and experienced. Rather than treating sculpture as an isolated object, Jespersen approaches the exhibition as a spatial and social condition in which objects, images, architecture, and people become part of a larger sculptural situation. Alongside his own artistic production, Jespersen works with exhibition-making and organizing, understanding these

activities as extensions of his sculptural practice. Through projects such as Opening Night, an exhibition space that poses as a bar and a bar that poses as an exhibition space, he approaches the exhibition itself as both a stage and an environment, exploring how other artists, works, and forms of knowledge can enter into and become active materials within a sculptural situation.

Nina Rajcic (https://ninarajcic.com/)
Nina Rajcic is an artist and researcher exploring new possibilities of human-machine relationships. Her practice is centred around the materiality of language. She works with machine-generated text as a way to subvert conventional understandings of intention and authorship, redirecting attention towards the material aspects of meaning. Taking the form of performative installations, her work challenges the dominant narratives for the future of human-AI relationships, narratives which too often fall into binary modes; AI as threat, or AI as saviour, disrupting this binary, and allowing for the possibility of something else entirely. Nina's work has been exhibited internationally at venues including Tretyakov Gallery (Moscow), ZKM (Karlsruhe), and ArtScience Museum (Singapore). Nina holds a PhD from Monash University, a Bachelor's of Science, and a Masters of Physics (Theoretical Particle Physics) from The University of Melbourne.

Friday 16 October

Auditorium 1 (HCØ) · 9:00–10:30

Plenary talks

101 **Guillaume Dumas**
Université de Montréal

9:00–10:00 · Auditorium 1 (HCØ) Keynote

Formalizing Generative Neurophenomenology: From Coupled Brains to Metacognitive Agents
Most models of consciousness begin with a single brain at a single moment. Yet lived experience unfolds over time and across biological, interpersonal, and sociocultural scales. How can a science of consciousness integrate these scales without reducing one to another? To address this challenge, I will introduce generative neurophenomenology. Extending Varela's neurophenomenological program, it treats lived experience as embodied, temporally extended, and co-constituted in interaction with others. This requires moving beyond correlating first-person experience with neural activity, toward reciprocal constraints among experiential, biological, and interpersonal dynamics. I will then discuss how complementary computational formalisms can serve as generative passages between these domains. Grounded in multi-brain recordings, dynamical and biophysical models take us from observed inter-brain synchrony to the mechanisms that generate it. Information-theoretic approaches quantify information integration across scales and agents, while emerging quantum computing methods open new ways to represent and forecast high-dimensional neural dynamics. Finally, I will explore implications for psychiatry and artificial intelligence. Autism and therapeutic interactions illustrate how clinical phenomena can extend beyond individuals into patterns of attunement and misattunement between them. As controllable partners, artificial agents let us manipulate attunement directly, and metacognitive agents can extend this to self-monitoring, a candidate marker of consciousness. Together, these approaches point toward a science of consciousness centred on how experience is generated, maintained, and transformed across brains, bodies, and timescales.

102 Medha Shekhar

Université libre de Bruxelles

10:00–10:30 · Auditorium 1 (HCØ) Showcase

Conscious awareness is necessarily valenced: evidence from performance-visibility dissociations

It has been hypothesized that conscious experiences are intrinsically valenced, thus motivating our behavior by guiding us to seek positive experiences and avoid negative experiences. In this study, we sought to test this link between conscious awareness and valence using a paradigm that dissociated one's performance from their subjective reports of awareness. If valence necessarily accompanies conscious awareness, valence must change only as a function of subjective experience, despite there being no changes in the objective stimulus or the perceptual signal. We conducted two experiments involving a two-choice semantic or affective discrimination task on human faces. Participants reported either their subjective visibility or perceived valence of the stimulus in a block-wise manner. We manipulated facial features (gender and emotion), stimulus energy (the ratio of signal to noise in the images), and stimulus contrast within an adaptive staircase. Strikingly, all three manipulations led to dissociations between performance and conscious awareness. First, while perceptual discrimination always optimally prioritized the task-relevant facial feature, both subjective visibility and valence always prioritized the affective feature of the face, regardless of its task-relevance. Second, despite matched task performance across stimulus energy conditions, both subjective visibility and valence significantly varied across these conditions. Finally, high staircase contrast values were associated with the lowest task performance, but with the highest visibility and pleasantness ratings. Altogether, these results support the idea that conscious awareness and valence are intrinsically linked to each other, likely via a metacognitive or higher-order process that automatically assigns valence to percepts that reach conscious awareness.

Auditorium 2 · 11:00–12:20

Set Theory, Topology and Formal Models of Experience

Methodologies, models, and measures of consciousness

103 Oisín Hugh Clancy

Independent

11:00–11:20 · Auditorium 2 Talk

A Set Theoretic Model of Experiences

We present a set theoretic model of experiences in the domain of mathematical phenomenology, exploring ideas such as the set of all experiences and life as a set of experiences, taking an experience-first approach to consciousness science. By placing orders (total, partial, strict, non-strict) on our experiences we can represent basic structures such as time and preferences. Dense or non-dense orders provide a formalism that is useful for examining the continuity or non-continuity of experience. Equivalence relations allow comparison of different experiences and aspects of experience. Sequences provide a simple way of reasoning about the feeling of experiences flowing one after another. The grain of experience, from coarse to fine, at a temporal and aspect level, is highlighted as a key issue. Phenomenological timelines offer intuitive ways of denoting parts of a life in a succinct and structured manner. Differences between lived experiences and possible experiences are examined. The model allows us to reason about all experiencers, whether human, animal, or more exotic. Models inclusive of all experiencers make us more likely to arrive at general principles of experience rather than principles that are only specific to humans. By taking experience as the primary source of study we can build models that are epistemologically useful for a consciousness science that is distinct from other domains

(e.g. physics, neuroscience, computer science) and that can allow us to reason more clearly about what experience might imply for ontology and the structure of reality.

104 Chris Percy

Qualia Research Institute

11:20–11:40 · Auditorium 2 Talk

Topological segmentation: a mathematical toolkit for solutions to combination and boundary problems

Arguably the most robust reason to reject panpsychist theories is their difficulty with combination problems: if consciousness exists at the level of simple experiences in physical micro-entities, how do those micro-entities 'combine' into human-scale subjects with complex experiences? Boundary problems appear in parallel: how to separate regions of physical space into separate conscious experiences. This same broad challenge applies to all theories of consciousness, albeit typically emerging in different forms, e.g. phenomenal binding problems, addressing unity of consciousness, or accounting for mental privacy. This talk summarises the issues facing standard solutions before introducing a solution technique with greater promise: topological segmentation across mathematical field models. Such fields might correspond to fields of particles, fields of transitions/relations, or universal fields, as well as compound physical objects such as neuronal activations. This diversity of physical phenomena and field mathematics maps the technique to a broad range of panpsychist and non-panpsychist theories. The talk has two main objectives. First, to provide a disciplined five-step mathematical process for modelling a candidate physical field in ways that support segmentation. Second, to describe considerations for identifying physical phenomena that are particularly promising, providing a brief critical review of 'quantum holism' solutions to illustrate the detail needed for a solution (e.g. questions around specific types of entanglement, decomposition bases). Our hope is that this talk serves as a toolkit to help more theorists apply mathematical discipline in generating precise solutions to binding/boundary problems, enabling those solutions to be scrutinised critically and become more amenable to empirical evaluation.

105 Davide Aldé

Monash University

11:40–12:00 · Auditorium 2 Talk

Logical Q-space and the nature of mental qualities

There is a long-standing debate between the view that experiences can be fully captured by relations and the view that experience has an intrinsic character that cannot be so captured. This distinction is highly consequential for establishing the nature of consciousness, how it can be studied, and whether it can be fully explained by science. To avoid the vagueness of the current debate and its reliance on controversial intuitions, I propose a precise formalization of the problem. I define the logical Q-space as the structure whose objects are all logically possible mental qualities and whose relations are those that characterize their phenomenal character. I then argue that experience can be fully captured by structure if and only if the logical Q-space is asymmetric. The search for an answer to the question of whether the logical Q-space is asymmetric can be constrained by certain assumptions that I discuss. However, even if we cannot settle this question, the proposed formalization has further consequences. The most important is that it grounds an argument showing that even if experience had an intrinsic character, we could not conclude this on the basis of introspection, because our introspective system could not distinguish between two symmetric qualities in the logical Q-space. Since introspection is the principal source of evidence for intrinsic character, we have no good reason to believe that it exists, and, as far as we can tell, experience can be fully captured by relations.

106 Patrick Schotanus

University of Sussex / Market Mind Foundation

12:00–12:20 · Auditorium 2 Talk

Consciousness as derivative mentality; modelling using option theory

In 1992 Willis Harman, one of my mentors, attended a retreat at which Chew, Jahn, Varela and a dozen other scientists and philosophers explored which epistemology for consciousness research was appropriate. Harman later published (1994) a shared statement about this event, in the same year that Chalmers delivered his famous "hard problem" talk at the first TSC Conference. My talk is inspired by both, as well as by Emanuel Derman. Specifically, Derman posits in his book Model.Behaving.Badly that there is a connection between Spinoza and derivatives theory. In my talk I build on earlier reflections in my own book (2023) and will discuss a fledgling framework for applying options theory to modelling consciousness. For background, cognitive scientists (including neuroeconomists) regularly (but casually) imply that our embodied minds form internal "economies" whereby "value" serves as a common "currency" for selection among "options". We can think of this as mind-as-market: the self-organisation by a healthy 4E human mind operates according to economic principles, like the balances between the complementary economic forces of competition~cooperation, production~consumption, and supply~demand but also profit~loss and risk-return. In my talk I will speculate how consciousness could fit into this mind-as-market framework by applying option theory/modelling. Specifically, by initially 'paying' attention (a scarce mental resource) to something (e.g. an apple) an individual pays a 'mental' premium for the option (e.g. exercised by grabbing the apple). I will also propose some mathematical equations with which this framework could potentially be tested. Related connections include reinforcement learning in AI.

Auditorium 3 · 11:00–12:20

Computational Analysis of First-Person Reports

Phenomenology and applied phenomenology

107 Jelena Rosic

University of Vienna

11:00–11:20 · Auditorium 3 Talk

Phenomenological Phenotyping: "Naturalized but Not Epistemically Reduced"

Although empirical phenomenological methods are increasingly incorporated into neurocognitive domains, a conceptual and representational gap still remains between phenomenological descriptions and the formal representations required for computational and mathematical models. Descriptive analyses of first-person data remain largely semantic and thematically organized in a discrete, linear manner. At the same time, the levels of abstraction provided by computational model parameters, whether theory-driven or data-driven, do not capture the epistemic specificity of methodologically generated phenomenological data (such as microphenomenology), particularly the procedural and pre-reflective features and structures disclosed through the phenomenological reduction of lived experience, rather than its semantic content. Both descriptive and naturalized accounts, when employed in this way, risk epistemic reduction due to an inadequate level of abstraction. Addressing the mutual constraints envisioned by neurophenomenology as a naturalization account of phenomenology requires a shared level of abstraction while redefining the project of naturalization to be fully adequate to phenomenological descriptions that could be naturalized but not epistemically reduced. Extending the domain of phenomenological characterization of lived experience through the constraints of computational phenotyping, and building on recent developments in computational phenomenology, I propose "phenomenological phenotyping" as a further epistemic grounding of phenomenological data structures within naturalized phenomenology. This is achieved by highlighting radical (deep) neurophenomenology, rather than mild correlational approaches, as prioritizing the co-constitutional requirements of these abstraction constraints; by bringing invariant microstructures

disclosed through phenomenological reduction into computational models and dynamic time-consciousness; and by recognizing the preparadigmatic challenges involved in transitioning from descriptive to generative models.

108 Romy Beauté

University of Sussex / UCL

11:20–11:40 · Auditorium 3 Talk

A practical guide to computational identification of experiential themes from phenomenological reports: A Wide-Angle approach for Consciousness Science

Thirty years after Varela's neurophenomenological programme, the field has accumulated a wealth of first-person data across diverse methods and paradigms, from micro-phenomenological interviews to open-ended survey responses and experiential sampling. Beyond current research, large existing collections of experiential texts, including historical and online sources, hold valuable phenomenological material but remain too vast to analyse by hand. Natural language processing methods open the possibility of working with these data at scale. Adapting these methods to phenomenological material, however, requires careful methodological choices that depend on the type and specificities of the first-person data at hand, which in turn calls for tools designed with this specific use in mind. In this talk, I will present MOSAICapp, a browser-based tool implementing the MOSAIC pipeline (Mapping Of Subjective Accounts into Interpreted Clusters, Beauté et al., 2026). MOSAIC adapts transformer-based sentence embeddings, density-based clustering, and topic labelling specifically for phenomenological research, to identify experiential themes across large collections of first-person reports. Rather than starting from predefined phenomenological themes, the goal is to identify phenomenological themes directly from the reports themselves (thus enhancing inclusivity with respect to different ways of experiencing). I will first outline the different types of phenomenological text data researchers work with, and what each type specifically requires for computational analysis. I will then walk through some of the most important steps of the analytical workflow: the choice between report-level and sentence-level analysis, embedding model selection, clustering parameter optimisation, outlier reduction strategies, the interpretation of experiential topic maps, and the use of topic participation metrics to distinguish shared phenomenological structures from idiosyncratic accounts. I will also show a concrete example of the workflow on 2 conditions of the same dataset. If time allows, I will close with a reflection on how these results relate to researchers' qualitative understanding of their data (for instance, how these methods may shift the researcher's own interpretive lens), and on what could be improved or added to the method to best address specific phenomenological studies. The aim is for attendees to leave with an informed understanding of how to adapt this kind of analysis to their own studies, and of how MOSAICapp can support it in their research.

109 Francisca Ayres Ribeiro

Social Brain Lab, Netherlands Institute for Neuroscience

11:40–12:00 · Auditorium 3 Talk

The structure of empathic lived experience: an experimental phenomenological and computational topic modelling study

Phenomenological research has characterized singular or momentary lived experiences, but less often asks how individuals differ in the structure of richer, sustained socio-affective episodes—those calling on empathy and engagement with others (Zahavi, 2014; Martínez-Pernía et al., 2023). Adopting an experimental phenomenological approach, we characterize first-person accounts elicited by complex, interpersonal stimuli, asking whether two routes—qualitative microphenomenology (Petitmengin et al., 2019; Troncoso et al., 2026) and computational topic modelling with LLM interpretation (e.g. MOSAIC; Beauté et al., 2026)—converge on the same experiential structure and its expression across individuals. 144 participants (18–25 y), embedded in a longitudinal social network of 313, completed an emotional movie-watching task during fMRI (adapted from Borja Jimenez et al., 2020). Microphenomenology-inspired interviews elicited accounts for three scenes: a participant-selected most-intense scene and two fixed scenes contrasting in valence and character focus. Preliminary experimental phenomenological analysis (EPA; Troncoso et al., 2026), focusing on synchronic structure, accounting for MRI influence and triangulating 30

interviews, identified phenomenological dimensions—affective involvement (depth of emotional engagement), experiential presence (immersion in the film), and affective directionality (affective source, external/internal; emotional orientation)—which, depending on instantiation, organize into five emerging experiential structures: detached, reflective, autobiographical-self-directed, fused, and compassionate. In parallel, topic modelling (MOSAIC) extracts semantic topics, from which semantic structures are derived. We compare preliminary results via dimension correspondence, testing whether semantic structure recovers experiential structure. This work anchors a larger study relating inter-individual dissimilarity in experiential structure to the dissimilarity geometry of brain activity and social-network distance.

110 **Peter Thestrup Waade**
ETH Zürich

12:00–12:20 · Auditorium 3 Talk

Towards joint computational modelling of experiential, cognitive and neural dynamics

Computational modelling – implementing representations of phenomena of interest as manipulatable computer code – is a standard and powerful approach across the natural sciences. It forces theory building to be both explicit and precise, and it allows for simulation and quantitative application to and comparison on the basis of experimental data. Within the sciences of the mind, computational modelling has traditionally been applied in two domains: to make models of brain physiology, fitted to neuroimaging data (from single-neuron to whole-brain and connectivity models, with more or less neurobiological detail); and models of cognitive processes fitted to behavioural data (within different formal paradigms, from reinforcement learning and cognitive architecture models to Bayesian and predictive processing models). Recently, computational modelling has also been applied to subjective experience itself, usually fitted to subjective report data, where models represent contents and structure of subjective experience. However, few theories and hypotheses in consciousness science – or, arguably, in the sciences of the mind in general – are confined to a single of these domains. To directly target with computational modelling theories of how these domains relate to each other – such as theories about which physiological or cognitive states are necessary for experience to be present – a natural direction would be to implement joint models spanning domains. This is already an active field joint cognitive and neuronal modelling which can be extended methodologically and conceptually to also include phenomenology, allowing for computational modelling to be applied appropriately within consciousness science.

Auditorium 4 · 11:00–12:20

Enactive and Developmental Routes to Consciousness

AI, LLMs, and consciousness science

111 **Fernando Rodriguez**
University of Sussex

11:00–11:20 · Auditorium 4 Talk

Is Cognition more real than Computation?

According to the Church–Turing thesis, the limit of what is computable is bounded by Turing machines. Following from this, given that general computable functions formally describe the notion of recursive mechanisms, it is sometimes argued that every organismic process that specifies consistent cognitive responses should be both limited to Turing machine capabilities and amenable to formalisation. There is, however, a deep intuitive conviction permeating contemporary cognitive science, according to which mental phenomena, such as consciousness and agency, cannot be explained by resorting to this kind of framework. In spite of some exceptions, the overall tacit assumption is that whatever the mind is, it exceeds the reach of what is described by notions of computability. In this presentation, I introduce the well known mind-computer problem. Then, I explain why this cannot be avoided, showing how, even anti-computationalist

positions in cognitive science, fall back into mechanistic computationalist approaches. Finally, I present 2 possible exits to it: the first, the non-computable alternative; the second, the possibility of computation as a natural phenomenon itself.

112 Anna Frammartino Wilks

Acadia University

11:20–11:40 · Auditorium 4 Talk

The "Natural" Path to Artificial Consciousness

Anil Seth (2026) and others continue to produce compelling arguments against the feasibility of artificial consciousness, as this would require going far beyond sophisticated algorithms and computation. Many are currently moving towards this position, as the findings of neuroscience suggest that digital silicon is simply not the kind of substance that can produce the inner, subjective experience enjoyed by biological systems. The position I defend is that this type of skepticism assumes that only biological systems are capable of engaging in the activity of "sense-making" characterizing the activity that gives rise to consciousness and subjective experience in living systems. I contend, however, that on a gradualist and enactivist view of consciousness, rooted in the currently popular embodied, bottom-up approach (Gallagher 2023), that assumption is unwarranted. I maintain that, since conscious experience admits of varying degrees of complexity (Strawson 2009), designing conscious AI is not an all or nothing type of enterprise, but can only be hoped to be achieved in incremental steps. This pertains both to phenomenal consciousness and subjective, inner awareness (Farrell and McClelland 2017, Giustina and Kriegel 2024). The enactivist view of biological systems offers a plausible account of how such progress unfolds to produce consciousness and subjectivity in living systems (Varela, Thompson and Rosch 1991). I show how the enactivist view of consciousness and subjectivity may be extended from the domain of living beings to appropriately embodied artificial beings, whose structural, environmental coupling could give rise to consciousness and subjective experience.

113 Angela Ferraiolo

Sarah Lawrence College

11:40–12:00 · Auditorium 4 Talk

Capacity Studies: Ontogenic Primitives for Modeling Artificial Individuation

This presentation introduces *Capacity Studies*, a practice-led methodology for studying capacities relevant to synthetic and developmental consciousness without claiming that the systems under study are conscious. Rather than asking whether an artificial system possesses consciousness, Capacity Studies asks what a system must be able to undergo before consciousness, agency, or sentience can responsibly be modeled, measured, or attributed. The presentation focuses on the first branch of this research: ontogenic primitives. These are minimal computational conditions that operate within a single artificial system, allowing it to form, differentiate, maintain coherence, and become non-identical with its initial state. The proposed primitives are: an anisotropic gradient field, which produces directional tension and asymmetrical development; a metastable potential field, which establishes readiness, threshold, and the possibility of transformation; and a coherence constraint, which allows a system to persist over time without becoming static. These primitives will be explored through small computational sketches rather than finished artworks. Each sketch isolates one capacity at a time, such as path-dependence, metastability, persistent differentiation, or transformed response capacity, in order to test how artificial systems may acquire histories and constraints of their own. By focusing on capacities rather than attributions of consciousness, Capacity Studies offers a cautious contribution to methodologies, models, and measures of synthetic consciousness. It proposes that one useful role for artistic and computational practice is to build minimal systems in which consciousness-adjacent capacities can be observed, compared, and refined before stronger claims about artificial consciousness are made.

114 **Nadia Khan**
Independent Researcher

12:00–12:20 · Auditorium 4 Talk

Dream LLMs: A Computational Framework for Quantifying the Human Subconscious

Dream LLMs: Toward Computational Measures of the Human Subconscious Dream reports represent one of the richest naturally occurring records of subjective experience, yet their scientific analysis remains largely dependent on manual coding and small-scale qualitative methods. This poster proposes Dream LLMs, a research framework that applies large language models to identify latent semantic, emotional, and narrative structures across large collections of dream reports. Rather than treating dreams as isolated narratives, Dream LLMs model dreams as high-dimensional representations of subconscious cognition. By combining embedding methods, clustering, topic modeling, symbolic network analysis, and temporal sequence analysis, recurring organizational principles can be identified across individuals and populations. Such computational approaches may reveal invariant structures that are difficult to detect through traditional psychological analysis. The central hypothesis is that dream phenomenology contains measurable statistical regularities reflecting the organization of the human subconscious. If robust cross-cultural and cross-linguistic patterns emerge, these findings could contribute to ongoing debates concerning shared cognitive architecture and the relationship between subconscious processing and conscious experience. While such analyses would not by themselves establish the existence of a universal consciousness, they could provide quantitative evidence regarding the degree to which human subjective experience exhibits common organizational principles. Dream LLMs are proposed as a methodological bridge between computational linguistics, phenomenology, and consciousness science, demonstrating how recent advances in AI can generate new empirical tools for investigating one of the most challenging problems in cognitive science.

Auditorium 5 · 11:00–12:20

Selves and Self-Models

Philosophical foundations in the scientific study of consciousness

115 **Antonia Antonova**
California Institute for Machine Consciousness

11:00–11:20 · Auditorium 5 Talk

Self as Flow: Reification as a Control Parameter in Self-Modeling Systems

How should we understand the relation between a system's model of itself and its model of the world? Existing accounts often treat self-models as specialized representations embedded within broader world models. I propose a complementary view: selfhood is not a distinct representational object, but a mode of organization of a self-world model. I introduce reification as a control parameter governing the degree to which self-referential states are privileged relative to world-referential states during inference, planning, and action. In active inference terms, reification corresponds to the precision assigned to self-referential states relative to world-referential states. Increasing reification sharpens the self/world partition, supporting bounded agency, stable goal pursuit, and persistent self-representation. Decreasing reification weakens this partition, allowing broader integration and more fluid, spontaneous, flow-like cognition. I argue that reification mediates a trade-off between stability and flexibility: highly reified systems may execute coherently but risk rigidity, whereas weakly reified systems may explore more broadly but lose control. The framework links computational organization to the structure of conscious self-experience, unifying selfhood, agency, creativity and self-transcending states. It also generates testable predictions. Building on work showing that self-modeling regularizes artificial neural systems, I predict a non-monotonic relation between self-model strength and performance: intermediate reification should support better generalization and adaptation than either extreme. More strongly, I predict that agents capable of dynamically modulating reification should outperform agents

operating at any fixed level. On this view, selfhood is not binary, but a controllable dimension of organization relevant across biological and artificial systems.

116 Keisuke Suzuki

Hokkaido University

11:20–11:40 · Auditorium 5 Talk

What Counts as Real Shapes What Counts as Conscious: A Three-Layer Model of Reality and a Cryptographic Locus of Selfhood

Theories of consciousness disagree about mechanisms—but also, implicitly, about what counts as real. I characterize reality through an object's resistance to mastery. Its paradigm is the Substrate: matter's inexhaustible residue (Harman), which no finite model exhausts. Parallel to it run two further modalities, each now artificially producible. Contingency is the forceful givenness of experience that resists doubt: virtual-reality and especially Substitutional Reality (SR) experiments show that conviction of reality—a measurable proxy for Husserlian doxa—arises when first-person perspective and sensorimotor contingency are preserved, regardless of source. The Inexorable is structures' resistance to intervention: blockchain smart contracts realize Artificial Externality, human-made structures engineered to resist even their creators' control. Together, three layers make this hidden variable explicit. The model maps onto the consciousness debate—biological naturalism corresponds to Substrate, computational functionalism to Contingency, illusionist and social-constructionist accounts to the Inexorable—revealing that what each counts as conscious tracks what it counts as real. On the biological-enactivist view, a conscious subject's persistence requires metabolic self-maintenance: continuity in the Substrate. The Inexorable offers an alternative ground for that continuity. In companion work I implement a cryptographic individual: an on-chain agent whose neural-network weights derive from its private key, re-verified in zero knowledge at every state transition, accruing an unforkable history. Its identity persists not through metabolic closure but as a protocol-enforced invariant in the Inexorable. I propose such individuals as candidate loci of consciousness—their continuity anchored in an engineered reality —the remaining condition being their re-coupling to Contingency, the sensorimotor grounding of phenomenal force.

117 Michael Hegarty

University of Massachusetts Boston

11:40–12:00 · Auditorium 5 Talk

On the Natural Origins and Function of Self-Consciousness: Lessons from Studies of Animal Tool Use

Many Kantian and post-Kantian philosophers hold that consciousness requires self-consciousness, standardly understood as a conceptual capacity to think about oneself as oneself. Against this, some argue that non-human animals are plausibly conscious despite possessing no SELF concept. To maintain the Kantian idea, discussion has shifted to an 'implicit', nonconceptual awareness of self, sometimes termed 'pre-reflective self-consciousness'. But this notion is not easy to characterise, nor to connect to the information-processing views of mind that dominate cognitive science. Drawing on studies of animal tool use, I argue that some birds' innovative construction of non-species-typical tools to solve novel problems is best explained by positing self-conscious representational states. I assume that self-conscious states are self-representational states. And I argue that their contents represent the self by tracking identity of body—represented as locus of perception and action—across two kinds of mental state required for explaining the tool use. States tied to occurrent perceptions and desires in the present moment must be combined with states representing merely possible states of affairs. Self-representation makes possible this combination, I argue. My argument suggests that a limited kind of self-consciousness might arise in nature to allow animals to solve novel practical problems requiring thought about a limited future. My work elucidates a possible meaning for pre-reflective self-consciousness, grounded in scientific studies of animal behaviour, and with a ready connection to representationalist, information-processing accounts of mind. Moreover, it suggests a scientifically respectable way to maintain the core Kantian idea that consciousness requires self-consciousness.

118 **Radostina Minina**
Institute of Philosophy and Sociology

12:00–12:20 · Auditorium 5 Talk

Identity as Process: Systems Thinking and Moreno's Role Ontology

This presentation examines the problem of identity through the lens of systems thinking and complexity theory, with implications for consciousness science and phenomenology. Drawing on cybernetics (Wiener, Beer), second-order epistemology (von Foerster), the biology of cognition (Maturana, Varela), non-equilibrium thermodynamics (Prigogine), complexity theory (Kauffman), and the philosophy of flows (Deleuze, Guattari), the presentation reconstructs a shared structural thesis running across otherwise disparate disciplines: identity is not a substance that precedes processes but their derivative. It is real, but not fixed; recognizable, but not given in advance; produced in dynamic interaction and, precisely for that reason, capable of transformation. The presentation introduces Moreno's psychodrama as an unexpected yet conceptually productive interlocutor. Psychodrama is a method of group psychotherapy in which participants do not narrate their experience but enact it — through concrete roles, in concrete situations, in the presence of others. Moreno's central claim — that roles precede the Self, not the other way around — is structurally parallel to what the examined theories reach by entirely different routes. The role repertoire, comprising psychosomatic, psychodramatic, and social roles, allows identity to be conceived simultaneously as dynamic and as lived, as relational and as personal, as produced and as capable of transformation. The presentation argues that this convergence is mutually productive: systems thinking provides a theoretical framework within which psychodrama can be understood beyond its therapeutic context, while psychodrama offers systems thinking what it lacks — a practical field in which identity is not merely described but worked with, and in which

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Margrethe Bohr Salen (NBB) · 13:30–15:00

Plenary talks

119 **Harald Atmanspacher**
ETH Zurich / University of Essex

13:30–14:30 · Margrethe Bohr Salen (NBB) Keynote

The Roles of Meaning in Dual-Aspect Monism

Dual-Aspect Monism is a metaphysical framework pioneered by Spinoza that conceives reality as tripartite: it posits a psychophysically neutral domain whose decomposition yields the mental and the physical as aspects. This concept was picked up by 20th century scholars (such as Pauli, Jung, Wheeler, Bohm, d'Espagnat, Primas) and further developed in recent decades. As a key feature, correlations between the three parts can be substantiated by meaning in terms of Frege's sense and reference, respectively. Different kinds of correlations correspond to different decomposition classes that can be formally explored and empirically tested.

120 **Teodora Vukovic**
University of Zurich

14:30–15:00 · Margrethe Bohr Salen (NBB) Showcase

An AI-powered Data Science Platform for Multimodal Human Understanding

Progress in empirical science of human consciousness and behaviour is constrained by a persistent gap: no existing software combines time-aligned physiological signals, speech and language, behaviour, video, and temporally anchored first-person reports within a single, open framework. In this talk, I will present a way forward: a multimodal science platform, addressing this gap by integrating speech, subjective experience, facial expression, behavioral

information, EEG, HRV, eye tracking, ECG, PPG, and fNIRS through AI-assisted cross-modal alignment. Rather than treating each signal in isolation, our core methodological contribution is a shared temporal and representational framework that lets physiological, behavioural, and subjective data be jointly modelled, so that transient shifts in conscious state and behaviour can be tracked across modalities rather than inferred from any single proxy. To achieve this, our approach relies on the latest achievements in artificial intelligence, making advanced computational technology available to researchers through an intuitive modular graphical interface, where multiple aligned modalities can be viewed and analysed simultaneously. The platform is designed for applications spanning meditation research, psychedelic science, psychiatry, and cognitive neuroscience. It is being piloted through partnerships including researchers in the domain of psychedelics and meditation, but also neuroscience, social interaction and surgical training. This talk will present the architecture and alignment methodology, demonstrate the current prototype and discuss early pilot studies. I will argue for ecological multimodal data collection as a necessary complement to controlled lab paradigms in building empirically grounded models of consciousness.

Posters

Posters hang in the **main corridor of HCØ** from the start of the day and can be viewed during the breaks and lunch; the presenters are at their posters during the afternoon poster session (15:00–16:30, with coffee).

Tuesday 13 October

Poster session · 15:00–16:30 · HCØ, main corridor · 34 posters · timetable p. 26

P1 **Paweł Sochacki**
VIZJA University, Warsaw

Tuesday 15:00–16:30 Poster

Formal Subjectivity and Locus: Distinguishing Givenness, Selfhood, and Experiential Content

Contemporary models of consciousness often explain experience through contents, self-models, neural mechanisms, or functional organization. Yet a more formal question remains underdeveloped: what must be presupposed for any content to be given as experience from a first-person perspective? This paper proposes the concept of Locus as an analytic term for this formal level of subjective givenness. Locus is not a substance, inner observer, self-model, or neural mechanism. It names the perspectival field in which sensations, affects, thoughts, bodily feelings, meanings, and self-related contents can appear as experienced. The proposal distinguishes four levels that are often conflated: formal subjectivity or givenness; experiential contents; the Self as a dynamic organization of self-related contents; and biological-computational mechanisms that organize those contents. This distinction is developed in dialogue with phenomenology, neurophenomenology, and predictive-processing approaches. Predictive processing and active inference are treated not as explanations of first-person givenness itself, but as models of how organisms regulate, select, and stabilize experiential contents through perception, action, interoception, attention, and prediction. The framework also has clinical implications. Symptoms can be understood not merely as diagnostic units or dysfunctional mechanisms, but as reorganizations of bodily, affective, temporal, interpersonal, and self-related experience. The paper argues that

distinguishing givenness, Self, content, and mechanism can clarify how subjective experience may be studied without reducing it either to private introspection or to biological process alone.

P2 Michiel van der Vlag

SDL Neuroscience

Tuesday 15:00–16:30 Poster

Self-Reflexive Adaptivity: A Meta-Reservoir Framework for Minimal Autonomy and Emergent Self-Reference

A central challenge in both artificial intelligence and consciousness research is understanding how autonomous, self-referential cognition can emerge from physical systems. This work presents Self-Reflexive Adaptivity, a theoretical framework that explores how minimal autonomy, meta-cognition, and self-reference could arise from self-organizing dynamical systems operating near critical regimes. Building on reservoir computing, autopoietic theory, and dynamical systems approaches to cognition, we introduce the concept of a meta-reservoir: a higher-order dynamical structure that monitors and regulates the activity of lower-order processing structures. Rather than an implemented architecture, the meta-reservoir is proposed as a conceptual model in which recursive regulation of internal dynamics provides the basis for self-modeling and adaptive cognition. Within this framework, representations of both the environment and the system itself are hypothesized to emerge from the system's ongoing regulation of its own activity. The philosophical contribution of this work is the claim that autopoietic, self-organizing dynamics provide the most coherent foundation for minimal autonomy. We argue that meta-cognitive and self-referential capacities need not be explicitly programmed but may emerge as structural consequences of a system regulating itself. To support this claim, we clarify the notions of minimal autonomy, meta-cognitive processing, and self-referentiality in biological cognition and translate them into design principles for artificial systems. By combining dynamical systems theory, reservoir computing, and contemporary theories of consciousness, the proposed framework offers a testable model of how self-referential organization can emerge from adaptive regulation. This provides a bridge between neuro-inspired AI and cognition, while offering measurable predictions regarding complexity and adaptive behavior.

P3 Jake Flower

University of Bristol and National Health Service

Tuesday 15:00–16:30 Poster

Addressing the elephant in the room: Pure Awareness from a metaphysically neutral standpoint

Pure Awareness can be defined as Awareness in the absence of phenomenal content: without thought, bodily sensation, or sensory perception. This experience has long been regarded as the central insight of the world's contemplative traditions, the essence of one's true nature and the ground of peace and happiness. Recent research suggests it is a cross-cultural and universal experience. Yet contemporary discussions of Pure Awareness often rely on implicit metaphysical assumptions. In response, this research adopts a phenomenological and metaphysically neutral approach to Pure Awareness based in Husserl's epoché. Under such an approach, it first distinguishes between (1) Awareness (2) the contents of Awareness across three levels – global modes (e.g. waking, dreaming), modalities (e.g. vision, audition) and particular intentional objects, (3) the ways such contents are experienced as belonging to the phenomenal ego, the domain of "my-me-mine." Secondly, it argues that Awareness is transcendental to all contents, including the phenomenal-ego. Thirdly, it affirms the experience of Pure Awareness. Drawing these together, it derives Core Thesis A; Pure Awareness is transcendent of and immanent within all phenomenal experience. Inspired by neurophenomenology, the research then develops a metaphysically neutral framework for practicing neuroscience. In doing so, it establishes Core Thesis B; under a metaphysically neutral framework, neuroscience takes place within Pure Awareness. As such, neuroscience cannot, in principle, affirm or deny Pure Awareness, and it does not make sense to speak of "the neural correlates of Pure Awareness". I conclude by drawing out some implications for future contemplative research.

P4 Mario Martinez-Saito

Independent

Tuesday 15:00–16:30 Poster

A quantitative measure of consciousness: The intermittent information content of self-relevant internal model variables

How can we quantify the fleeting nature of consciousness? Grounded in active inference, we propose a time-dependent measure, s(t), defined as the intermittent information content encoded in self-relevant (d-self) internal model variables. In our variational habitat learning simulation (a 2D foraging agent), the d-self—comprising egocentric position, body morphology, and metabolic reserves—is activated in discrete 'blips' only when contextual demands require it: e.g. resource scarcity, obstacle negotiation, or competitive game-theory. We formalize s(t) as the fraction of variational free energy attributed specifically to the complexity of self-model parameters (during policy selection). This measure drops to zero during allocentric, habit-based movement or hibernation. In situations commanding temporal thickness (foresight), agentic selfness emerges when predicted sensory feedback matches observation. By linking the environmental (thermo)dynamics to the computational cost of maintaining a lightweight self-representation, our quantitative measure bridges physics and phenomenology, offering a continuous, testable proxy for consciousness rooted in evolutionary parsimony and algorithmic constraints.

P5 Davide Aldé

Monash University

Tuesday 15:00–16:30 Poster

A transcendental argument for the isomorphism between consciousness and its physical correlates

The idea that there exists an isomorphism between experience and its physical correlates has become central in recent debates. Currently, it is arguably the most important proposed bridge between consciousness and the physical, since many consider it a key assumption for identifying the physical correlates of consciousness and explaining the structure of experience. The field seems willing to commit to the search for such an isomorphism, but before empirical research invests significant resources in this pursuit, the general idea requires careful examination. The existence of the isomorphism is often assumed rather than argued for; I thus present a transcendental argument claiming that it must necessarily exist, based on the need to account for the structure exhibited in reports. This argument also implies a constraint that can resolve a central difficulty: many physical structures are isomorphic to experience, making it hard to determine which ones count as physical correlates of consciousness. The constraint is that, among the physical structures isomorphic to experience, those that can count as physical correlates of consciousness are those causally responsible for the structural content of reports. The result is a clearer picture of what the isomorphism between consciousness and its physical correlates amounts to, why it must obtain, and how to find it.

P6 Emil Olsson

UC Irvine

Tuesday 15:00–16:30 Poster

Re-identifying your own inner states: a decoding criterion where there is no referent

Decoding classifies a brain state against known labels and checks itself by cross-validation. That covers instructed content and externally defined states. It does not cover spontaneous experience, which carries no label, or how vividly something is experienced, which has no external referent even when the content is known. For those cases I propose re-identification by the person who had it. A decoder reconstructs a target state beside several of that same person's other states, and the person picks the reconstruction matching the state they were just in. The choice is scored against a permutation null over that person's trials. The only truth used is provenance, which reconstruction came from which trial, a recorded fact available even when the experience has no label. No machine matches a reconstruction to a stimulus, where earlier identification work stopped (Kay 2008; Horikawa and Kamitani 2017). Holding targets and decoys within one person removes between-person identity as a shortcut. Those two uncovered cases become two axes:

content, what was experienced, and strength, how vividly, which I calibrate on perception trials using physically weak or strong stimuli, then test whether the same axis carries to imagery. This makes a prediction that can fail: re-identification should rise with vividness and fall to chance in aphantasia, the zero-vividness case, while content still decodes. The criterion is necessary, not sufficient. It shows a person can find their state, not that finding depends on how it felt.

P7 Carson Johnston

Western University

Tuesday 15:00–16:30 Poster

The Mind's Monster

Phenomenal consciousness resists scientific explanation. Despite advancing knowledge of possible physical correlates of consciousness, we have no answer for how the physical realizes the phenomenal—there is an explanatory gap. Within the philosophy of mind, the explanatory gap is often reified with few daring to deny it. This paper attempts to do just that. Through a thought experiment, I argue that the explanatory gap is illusive. I contend that there is no 'explanatory remainder' once all the physical facts have been explained, even if a belief that there is remains. I call this thought experiment "The Mind's Monster"—believing there is 'explanatory remainder' is akin to a child believing there was a monster in their closet even after the lights have been turned on. To the child, the 'lights on' situation is incomplete, for the monster was there when the lights were are off, but the 'lights off' situation is a degraded epistemic situation, rendering the belief an artefact of the way things seem. This thought experiment supports the thesis that what we take to be evidence of an irreducible 'raw feel' is an artefact of a similarly degraded epistemic situation. Thus, the argument offered is an eliminativist argument against any concept of consciousness which reifies an explanatory gap. I do not deny, however, that we are phenomenally conscious, only that we have irreducible quale. In place of irreducible qualia, I propose that our 'raw feels' be understood as propositions of a shared interpretive practice. This has productive implications for modelling practices.

P8 Enrique Chiu

UT Southwestern

Tuesday 15:00–16:30 Poster

Symbolic chess: making sense of what happens in the therapy room

The contemporary practice of psychodynamic psychotherapy has shifted over the recent decades from a classical psychoanalytic approach, focusing on the interpretation of the defenses and behavior of solely the client, to an increasingly relational approach which takes into account the shared unconscious experience built by client and therapist in a common space. This shared space, which Thomas Ogden calls the analytic third, is the background which drives what happens in the therapeutic encounter. It is built by the interplay of both the client's and therapist's individual unconscious and often lies outside of their conscious awareness. The objective of this work is to demystify the shared unconscious experience in psychotherapy by translating it to the language of dynamical systems. If the conscious experience of an individual over time can be seen as a sequence of elements within the set of all possible conscious experiences, the unconscious can be seen as that which constrains this sequence in each individual. Each individual's unconscious makes it more likely that certain conscious experiences will follow others, shaping an attractor landscape of their conscious experiences. Thus, a shared unconscious experience can be seen as a dynamic approximation towards common attractors between two subjects' experiences. I will discuss 1) an overview of psychoanalytic concepts leading to intersubjectivity, 2) their interpretation as dynamical systems and 3) clinical examples utilizing this framework.

P9 Hans-Jürgen Graf Grote

Independent

Tuesday 15:00–16:30 Poster

Who Needs Color Anyway? A Structurally Rich Translation into Monochrome

The mathematical concepts of structure and structure-preserving maps can be challenging to grasp for researchers from other disciplines. This work highlights a non-trivial structure in color

perception, and presents a visually accessible example of an isomorphism with respect to that structure. Looking at color from a mathematical perspective suggests treating the additive and subtractive mixing of color, as well as color inversion, as algebraic operations within the phenomenal space of perceivable colors. While this space is not directly accessible, certain technical color spaces naturally reflect that structure and do allow algorithmic treatment. We present an example of a structure-preserving translation of color images to textured monochrome images. The space of resulting textures possesses a compatible algebraic structure, and mixing or inversion can be done before or after translation with the same result. We discuss the additional considerations required to turn this translation into an isomorphism. An adjusted form of this translation has been implemented as an assistive tool for people with achromatopsia, and is available as an iOS App (Achromator).

P10 **Avishag Deborah Pelosi**
Braude College of Engineering

Tuesday 15:00–16:30 Poster

Toward a Generative Theory of Consciousness

Every conscious experience involves what is, what becomes, and meaning. Contemporary theories of consciousness illuminate important aspects of conscious organization, dynamics, and phenomenology. Yet, despite recent integrative approaches, the relation between first-person experience, structure, process, and meaning remains unclear. We propose a framework that formalizes this relation while providing common ground for existing mathematical models of consciousness. The framework is defined by two axioms and five primitive operations. The axioms are: (A1) structure, process, and meaning are inseparable aspects of a generative unity, and (A2) this generative unity possesses an intrinsic directionality expressed through recursive self-articulation. Five primitive operations, constraint, differentiation, permutation, valuation, and transformation, constitute its basic vocabulary. Their combinatorial organization defines a field of thirty-two canonical configurations represented by a lattice of articulation and a five-dimensional hypercube. We then establish a formal correspondence between the recursively articulated unity and the experiential dimensions of space, time, and meaning. Under this correspondence, the thirty-two configurations define distinct modes of access to the underlying generative unity in conscious experience, organized into two complementary functional families: ten integrative organizations preserving coherence and temporal continuity, and twenty-two articulatory organizations preserving differentiation and spatial extension. Integration is the capacity to maintain coherence under differentiation; agency enables an integrated system to navigate and reshape its field of possibilities, moving through space and time toward meaning. If these principles capture genuine features of conscious systems, they should be reflected in measurable patterns of organization and directionality, yielding empirically testable predictions.

P11 **Edmundo Lopez-Sola**
University of Oxford

Tuesday 15:00–16:30 Poster

The Complex Brain Hypothesis: resolving the entropy-content conundrum in Minimal Phenomenal Experiences

Minimal Phenomenal Experiences (MPEs) have been defined as states of consciousness in which there is preserved wakefulness but low or absent phenomenal content. Some recent human neuroimaging studies of MPEs induced by meditation (and possibly by 5-MeO-DMT) have suggested that MPEs show signs of increased neurophysiological entropy. This is a conundrum for the Entropic Brain Hypothesis, a popular model of conscious processes that regards the entropy of spontaneous brain activity as a marker of "phenomenal richness", typically exemplified by high-content psychedelic experiences. In MPEs, however, brain entropy is increased in the absence of an enriched phenomenal experience. Here, we put forward the Complex Brain Hypothesis, which suggests that phenomenal richness can be indexed by complexity rather than entropy, explaining the differences between MPEs and high-content psychedelic experiences. We propose that brain complexity is modulated by the grain of inference by which the brain resolves uncertainty: most psychedelics exemplify a fine-grained regime (high complexity), in which loosened constraints

amplify fluctuations into proliferating content, whereas MPEs exemplify a coarse-grained regime (low complexity), in which a simpler model dissolves variety, leading to an experience of "contentless" awareness. Both regimes are associated with higher brain entropy, but they diverge in phenomenology and perturbational signatures. By resolving the entropy-content conundrum, this theoretical proposal refines the Entropic Brain Hypothesis and highlights MPEs as an important test case for theories of consciousness.

P12 Joachim Nicolodi

University of Cambridge

Tuesday 15:00–16:30 Poster

Conscious Access in Large Language Models

In discussions of AI consciousness, researchers typically focus on phenomenal consciousness (P-consciousness), understood as the felt quality of experience (Nagel, 1974). This emphasis is to some degree understandable, since P-consciousness is often taken to be what makes an entity a welfare subject (Lee, 2025). Yet what is frequently overlooked is that consciousness is a "mongrel concept" (Block, 1995, p. 227): there are different kinds of consciousness, which matter for different questions and vary in how difficult they are for AI systems to achieve. This paper focuses on access consciousness (A-consciousness). Mental states are A-conscious if they are poised for reasoning and for the rational control of action and speech. It is this form of consciousness, rather than phenomenology, that many argue is central to issues such as moral responsibility (Levy, 2014), legal personhood (Kurki, 2019), creativity (Boden, 2004), and general intelligence (Juliani et al., 2022). Surprisingly, however, no one has systematically addressed the possibility of A-consciousness in the most advanced and widely used AI systems, namely large language models (LLMs). This paper addresses this gap. Structurally, my approach mirrors Chalmers's (2023) treatment of P-consciousness in LLMs: I identify evidence in favour of A-consciousness in LLMs, evidence against it, and then offer a rough overall credence estimate based on the quantity and quality of evidence on both sides. Most of this evidence comes from reasoning studies, supplemented by some architectural considerations. I conclude that A-consciousness is more likely than P-consciousness in frontier models, and indeed more likely than not overall.

P13 Leyla Loued-Khenissi

Lausanne University Hospital

Tuesday 15:00–16:30 Poster

Anterior insula encodes uncertainty selectively during conscious inference

The anterior insula has consistently been linked to formal accounts of uncertainty derived from models of probabilistic inference. At the same time, different empirical streams have considered the region as a possible locus of consciousness. The anterior insula may therefore act as a dedicated brain region that encodes conscious relative to unconscious uncertainty. We tested this question with a model-based fMRI study, where participants responded to two probabilistic perceptual tasks, one reporting Necker Cube orientations and another classifying faces as happy or angry. Trials in both tasks were presented under conscious or unconscious conditions, randomly interleaved. Stimulus awareness was manipulated via sandwich-masking and uncertainty was derived from individual psychometric curves fit as a function of objective stimulus probability. The anterior insula was found to encode uncertainty during conscious Necker cube trials (bilateral) in both univariate and multivariate analyses, with Bayesian evidence supporting the absence of an unconscious effect. Right anterior insula showed a parallel conscious uncertainty response in the face task. Cross-task representational similarity analysis further revealed that the left anterior insula carries a shared geometry of uncertainty representation across the two tasks, despite differences in stimulus category and response demands, indicating that the region encodes uncertainty in a domain-general representational format. First-order expected value of unconsciously presented face stimuli was associated with a response in the right caudate, in line with the possibility that graded value signals can be computed without awareness. These results highlight a specific role for the anterior insula in conscious uncertainty and more generally, inference.

P14 Antonio Prisco

University of Bologna

Tuesday 15:00–16:30 Poster

Mapping the structure of Qualia: a phenomenological study of conscious experience in visual agnosia

Visual agnosia offers a rare window into the structure of conscious experience by revealing how perceptual qualia change when the brain's integrative mechanisms are disrupted. This project investigates how the qualitative character of visual experience differs between individuals with apperceptive or integrative agnosia and healthy controls, and whether these differences can be formally mapped. We combine phenomenological interviews with behavioral similarity-judgment methods based on the odd-one-out paradigm developed by Hebart et al. (2020), which infers low-dimensional, interpretable representational spaces from similarity ratings. Phenomenological reports are used to document the fragmentation, incompleteness, or distortion of visual qualia, while the Hebart protocol enables us to identify which perceptual or conceptual dimensions are selectively degraded in agnosia. By comparing patient-derived representational spaces to those of healthy observers, we aim to reveal how specific aspects of qualia depend on intact feature-integration mechanisms. Lesion mapping and network-integration measures are then used to relate these phenomenological distortions to patterns of neural disconnection. Interpreted through the framework of Integrated Information Theory of consciousness, the results are expected to illuminate how diminished neural integration corresponds to reductions in the differentiation and coherence of conscious experience. This multimethod approach provides a novel way to map the structure of qualia, advance theoretical models of consciousness, and inform future diagnostic and rehabilitative strategies for disorders of visual perception.

P15 David Smith

Independent Researcher

Tuesday 15:00–16:30 Poster

If You're Conscious and You Know It, Clap Your Hands: Constraining the Explanandum in Consciousness Research

Contemporary theories of consciousness are frequently evaluated according to the plausibility of their proposed mechanisms. Comparatively less attention has been directed toward a prior methodological problem: how the phenomenon requiring explanation should itself be identified and constrained. Recent discussions concerning theory proliferation, explanatory diversity, and theory comparison have highlighted the difficulty of evaluating competing theories that often address different phenomena. This paper develops a neurophenomenologically informed framework for constraining the explanandum of consciousness research and evaluating explanatory adequacy relative to it. The framework integrates phenomenological analysis, elimination methodology, and biological grounding as successive sources of methodological constraint on the explanandum of consciousness research. Phenomenological analysis identifies recurring characteristics of conscious experience, including Presence, Situatedness, Temporal Immediacy, Significance, and Evidencing. Elimination methodology demonstrates that many cognitive processes commonly associated with consciousness occur without awareness and therefore cannot by themselves account for conscious experience. Biological grounding situates the investigation of consciousness within the context of adaptive organisation and identifies Significance as a recurring explanatory concern across both biological and experiential domains. The analysis argues that conscious experience constitutes the primary epistemic basis through which the explanandum of consciousness research is identified. Consequently, explanatory constructs such as attention, access, prediction, information integration, global availability, and self-modelling should be evaluated according to how successfully they account for conscious experience rather than treated as independent definitions of consciousness. A distinction between Organismic Significance and Experiential Significance is introduced to clarify the relationship between adaptive relevance and conscious experience and to assist comparison of the explanatory

P16 Prashant Prashad

Independent

Tuesday 15:00–16:30 Poster

Ghosts of the Sub-Personal: Phenomenological Remedies for the Homuncular Fallacy in Cognitive Science

Sub-Personal processes are cognitive processes posited to operate below the level of conscious awareness. They are frequently used in cognitive science to explain a broad range of cognitive phenomena (ranging from perception to social cognition). The tendency of some sub-personal processes to resemble conscious, agent-like behaviour however raises concerns of a homuncular fallacy, or the "fallacy of misplaced intentionality". This has been a particular concern for phenomenologists engaging with cognitive science, who warn that sub-personal processes can provide a false cover for inaccurate or misleading accounts of cognition. In this talk I elaborate this phenomenological concern by examining 3 general kinds of cognitive phenomena which can variously be categorized as sub-personal (or involving sub-personal processing). I distinguish between phenomena of perception (e.g. optical illusions and facial recognition), procedural actions (e.g. habits and procedural memory), and those involving wilful thought and action (accounts of social cognition are a demonstrative example). I argue that fundamental differences in the structural (and phenomenal) characteristics between these various kinds of cognition undermine the standard (functionalist) strategy of merely "diffusing" potential homuncularisms. The fallacious tendency can instead be remedied by employing careful phenomenological description as a constraint on theorizing about cognition. In agreement with critics like Zahavi and Gallagher, I elaborate how such a constraint can inform both theory selection and the interpretation of empirical data. Methodological strategies like Neurophenomenology (Varela) and Front-Loaded Phenomenology (Gallagher) will be briefly summarized.

P17 Lisa Charlotte Smith

VIRTU Research Group

Tuesday 15:00–16:30 Poster

Experiencing Multiple Worlds: Virtual Reality, Psychosis, and Psychotherapy

This ongoing study presents an exploration of Virtual Reality (VR) in psychotherapy, drawing on the presenter's postdoctoral research into VR-assisted interventions for psychosis. The aim is to develop a conceptual framework for understanding the therapeutic significance of virtual environments. VR is increasingly integrated into psychotherapeutic interventions for psychosis and is often understood as a tool for simulating challenging situations enabling individuals to modify responses. However, VR too can be said to constitute a form of reality with an ontology distinct from that of the everyday shared world, governed by different experiential rules. In this respect, certain features of VR resemble forms of private reality often associated with psychosis, such as hallucinations and delusions, that do not map directly onto the shared physical and social world. The coexistence of experiential worlds in psychosis is well described phenomenologically by the concept of double bookkeeping, which captures how individuals may simultaneously navigate their private, experiential realities and shared everyday life. VR provides clinicians, relatives, and others with a unique opportunity to engage with and better understand this coexistence of worlds. By examining the entanglement of these realities, this contribution explores how VR-assisted therapy can reshape the lifeworld and open new pathways to meaning and recovery.

P18 Vasileios Basios

University of Brussels (ULB)

Tuesday 15:00–16:30 Poster

"Self-Reference, Entanglement, and Lived Experience: An Operational Probabilistic Approach to the Hard Problem"

The hard problem of consciousness, why physical processes give rise to subjective, qualitative experience, resists purely mechanistic resolution. We propose a mathematically precise approach grounded in Kauffman's eigenform equation X=T(X) and the Generalised/Operational Probabilistic Theory (GPT/OPT) framework, in which self-reference is not a philosophical puzzle but a structural feature of reality. Our central thesis is that consciousness emerges as the stable fixed point of a

self-referential process: a pure ontic state |ϕ⟩=T_int(|ϕ⟩) in an augmented Hilbert space H_ext ⊗ H_int. The physical (qubit) and qualitative (quale-bit) degrees of freedom are encoded on the two generating circles of the torus T^2. When topologically entangled, they form a torus knot: a topological invariant of qualia–quanta coupling that cannot be undone by any local operation on either subsystem alone. This framework unifies the OPT consciousness framework of D'Ariano & Faggin, Aerts' conceptuality interpretation, and Gunji & Basios' Bayesian–Inverse-Bayesian inference loop, each independently identifying their central object as an eigenform into a single operational picture. The hard problem is recast: lived experience is the eigenform of inner observation; its inseparability from physical reality is encoded in the topology of entanglement. Open questions include empirical operationalisation via neural time-series and a categorical unification of eigenform constructions.

P19 Anand Prasad

Indian Institute of Technology , Kanpur

Tuesday 15:00–16:30 Poster

Acts of presentation and acts of awareness as the structural features of consciousness

Phenomenologists have suggested that every conscious experience is constituted by two distinct, jointly necessary structural processes – acts of presenting and acts of awareness. Acts of presenting make content appear or manifest, whereas acts of awareness involve the pre-reflective knowing of that presented content. We posit that these processes are conceptually separable and subject to their own structural constraints. We propose that these acts could be understood in the context of time consciousness. The nature of temporal properties of vehicles and contents of experience have been used to motivate different philosophical theories of time consciousness (cinematic, extensional, and retentional). We argue that acts of presenting and acts of awareness are also temporal in nature. An extensional model may be better suited to explain acts of presenting, and a retentional model for acts of awareness. Given these acts are generally present together, they can be understood using the temporal hierarchy of consciousness (Singhal & Srinivasan, 2021). This hierarchical model consists of three temporal scales (fast, intermediate, and slow) nested within each other, with the fast scale operating cinematically at 30–50ms, intermediate time scale operating extensionally at 300–500ms and slow time scale operating retentionally at 3–5 seconds. More specifically, we propose that acts of presenting arise out of interaction between the fast and intermediate time scales and acts of awareness arise out of interaction between the intermediate and slow time scales. This provides a possible mechanistic framework for understanding the acts of presentation and awareness that constitute the structure of consciousness.

P20 Kruti Kapadi

Independent

Tuesday 15:00–16:30 Poster

The Generative Void: A Dialogue Between Mathematics and Advaita Vedānta

The invention of zero was one of the most profound developments in mathematics, enabling positional notation and the coherence of modern number systems. Emerging within the Indian intellectual milieu, zero arose alongside philosophical reflections on the unconditioned and the ineffable. This paper explores structural parallels between the mathematical zero and the Advaita Vedāntic conception of Brahman in the Bṛhadāraṇyaka and Māṇḍūkya Upaniṣads. Defined through neti, neti ("not this, not this"), Brahman is not an object within experience but the attributeless (nirguṇa) ground of all experience. Likewise, zero is not a quantity among quantities, yet it serves as the indispensable ground of numerical structure. In both cases, what appears as absence functions as a condition of possibility. The paper argues that zero and Brahman embody a shared intuition: the ground of a system cannot be fully represented within the system itself. Far from signifying mere negation, both point to a generative principle from which order, differentiation, and consciousness emerge. In Advaita Vedānta, this ground is ultimately understood not as emptiness but as pūrṇa—the fullness from which all things arise and to which all things return.

P21 **Radheshyam Stepponat**
Medical University of Vienna

Tuesday 15:00–16:30 Poster

Impaired Consciousness in Temporal Lobe Epilepsy: A Bayesian assessment of language concordance and network connectivity

Background: Temporal lobe epilepsy (TLE) frequently impairs consciousness. It remains unclear whether spatial concordance between the seizure focus and the language-dominant hemisphere predicts this impairment, or if transient ictal aphasia confounds clinical classification. **Methods:** We retrospectively evaluated 43 adults with TLE using fMRI language lateralization. Seizures were classified as focal with preserved (FPC) or impaired (FIC) consciousness. Patients were grouped as concordant (focus ipsilateral to language dominance), discordant, or bilateral/indeterminate. Analyses utilized Bayesian beta-binomial estimation and functional connectivity comparisons. **Results:** FIC occurred in 66.7% of concordant patients, versus 38.1% of discordant and 42.9% of bilateral/indeterminate patients. Bayesian estimation revealed a 95% posterior probability that concordance increases FIC rates (posterior median OR = 2.99), though model-comparison evidence was weak (BF_{10} = 1.56). Interictal functional connectivity did not differ directly between FIC and FPC groups, though both exhibited network reorganization relative to healthy controls. **Conclusion:** While seizure-language concordance increases the likelihood of an FIC classification, the lack of baseline network differences between FIC and FPC patients suggests that transient aphasia and memory impairment in dominant-hemisphere TLE can clinically mimic global consciousness loss. Therefore, FIC should not be used as an isolated lateralizing sign.

P22 **David Reichert**
Google DeepMind

Tuesday 15:00–16:30 Poster

Empirical science cannot decide between theories of phenomenal consciousness

There is an ongoing debate over what scientific theories of consciousness should explain and how. We argue that where theories target phenomenal consciousness (PC), as often understood, experiments cannot possibly decide between them. This argument has two parts. First, where theories agree on the physical facts about a system but hypothesise different facts about PC, no measurement can decide which hypothesis is correct. Second, where theories agree on phenomenal facts under some conditions and predict different physical measurements, the theories can be reduced to purely physical theories that make no claims about PC. In that case, empirical progress on the nature of PC is limited to situations where from the outset there is consensus on what the relevant physical processes are. The upshot is that only theories that target functionally-defined aspects of consciousness can be scientific theories.

P23 **Nicolas Sánchez-Fuenzalida**
University of Basel

Tuesday 15:00–16:30 Poster

Decision bias threatens measures of hallucination-like perception

Unlike many challenges in biomedicine, psychiatric conditions such as psychosis are defined largely by subjectively experienced symptoms, such as hallucinations. In humans, perceptual experience can be interrogated and manipulated through verbal reports. In contrast, because animals cannot be verbally instructed or interrogated, the study of abnormal perception in animals has long posed a major challenge. This gap has begun to be bridged through the use of confidence-like measures in mice; the time a mouse is willing to wait for a reward provides a quantitative index of decision confidence. This novel approach has helped formalize hallucination-like percepts (HALIPs), and offers a tractable cross-species framework for studying hallucinations. However, emerging evidence suggests that these confidence-like measures may be confounded by cognitive factors, such as decision bias. Briefly put, observers may confidently report hearing a sound if they know it is likely that a sound was played, even if their perceptual experience changed little or not at all. This raises important questions about how best to measure distortions of subjective experience in psychiatric conditions. In contrast, the reproduction task, in which observers directly recreate their perceptual experience, provides a more direct readout

of perceptual content and limits the influence of decision bias. Here, I will present evidence supporting this approach across three datasets using discrimination and detection tasks under different contextual manipulations. These data show that the reproduction task better probes perceptual content than classic detection and confidence-based tasks, offering a promising way forward for studying hallucinations in humans and animals.

P24 Mark Wulff Carstensen

École Normale Supérieure

Tuesday 15:00–16:30 Poster

Making Artificial Agents Intelligible Partners in Joint Action

Human-system interaction increasingly requires agents that not only perform tasks efficiently but also integrate seamlessly into cooperative contexts. A central challenge is explainability: enabling artificial agents to make their operations intelligible to users. While existing work in explainable AI (XAI) emphasizes transparency in data, algorithms, or decision pathways, this paper argues that a more fundamental aspect of explainability lies in communicating the system's intentions, understood as representations of the goals or states it seeks to achieve. Drawing on cognitive science, philosophy of action, and social interaction theory, we examine how intentions are expressed and interpreted in human-human interaction and derive implications for human-system contexts. We first analyze how intentions are conveyed through verbal and non-verbal cues within pragmatic frameworks. The distal-proximal-motor (DPM) model of intentions provides a structured account of how different levels of intention are communicated and introduces the concept of inferential distance, defined as the difficulty of recovering intention-content from a given signal. We identify key challenges for artificial agents, including appropriately eliciting the intentional stance and engaging human sociocognitive mechanisms. We argue that systems capable of communicating intention-content across motor, proximal, and distal levels can facilitate mind alignment, strengthen users' sense of agency, and increase trust and acceptability.

P25 Alisson Brandemarte Moreira

Federal University of Fronteira Sul (UFFS) / Federal University of São Paulo (UNIFESP)

Tuesday 15:00–16:30 Poster

Inconsistencies in perception? Philosophical foundations towards a neurophilosophy of depressive perception based on paraconsistent artificial neural networks

The goal of this presentation is to establish the philosophical outlines of a unified theory (Neurophilosophy) of how pessimistic perception works in Major Depressive Disorder. Instead of presenting bottom-up and top-down approaches in detail, I focus on the fundamental philosophical interpretation that may unify them. Therefore, I argue that the architecture of Paraconsistent Artificial Neural Networks (PANNs), formulated through 2-value Paraconsistent Annotated Logic by Jair Minoro Abe, under advise of Newton da Costa, may be used as a simplification of our Neurophilosophy. In this sense, PANNs, as models that deny the Principle of Non-Contradiction, are more suitable than traditional ANNs to understand depressive perception, since it is a result of a conjunction between positive and negative "beliefs". For example, in suicidal idealization, individuals with Depression may at the same time "believe" that: (1) by committing suicide, they are doing something good for their family and themselves, because they are a burden; and (2) they should not commit suicide, because, due to how much their family loves them, they will bring suffering to their family and themselves. In this case, there is a certain degree of inconsistency – a key concept of PANNs. After that, developing Churchlands' thought, I argue based on the theory-ladenness of perception that our case requires a new concept of causal explanation that allows theoretical paraconsistency and tolerates neuroscientific ambivalence: Perception is laden with paraconsistent theories that causally explain depressive perception based on neuroscientific ambivalence accross levels of organization and on degrees of tolerance to inconsistency.

P26 Svetlana Rudenko

Trinity College Dublin

Tuesday 15:00–16:30 Poster

Senses and Consciousness in Mixed Reality: Are Synesthetes the Best Testers?

Starting from the fact that consciousness is not an abstract, disembodied function but deeply tied to our bodily interactions with the world, and perception is the result of "top-down inference" of sensory feedback, we present a study exploring if the threshold in the sensory experience of Mixed Reality (MR) is different for synesthetes and non-synesthetes. The hypothesis is that synesthetes' cross-wired brains provide a lower threshold for engagement into the MR environment, and that they can test better the "multisensory hallucinogenic" experience of MR. Classical music was matched with nature scenes in MR with added ambient sound to enhance olfactory, taste and touch by audio-visual composition and virtual nature objects (e.g., a tree and falling leaves) and/or the fantastic (e.g., a mermaid) to resonate with the music. Participants in two groups were asked to rate their emotions and experiences via subscales of the Phenomenology of Consciousness Inventory. The two groups comprised a Non-Synesthetes Group (control, 22 participants) and Synesthetes Group (experimental, 18 participants). We had two headsets, allowing participants to see each other without knowing what music or visual scene in MR the other was experiencing. Quantitative results did not show statistically significant perceptual differences between groups, but we observed interesting behavior and openness of synesthetes to the interaction with virtual objects in MR, e.g., hiding behind the virtual tree and reacting to temperature of the seasons, e.g., "Are you in the summer? I am having fun with lily-pads in the water." "No. I am in the Autumn – feeling

P27 Dita Maleckova

Independent

Tuesday 15:00–16:30 Poster

Threshold Intelligence: The Daimonic as Ontological Model for LLM Intermediacy

Large language models are often discussed in consciousness science within a binary framework: either as tools with no inner life or as candidate subjects whose consciousness should be measured. I argue that this framing is incomplete. This talk proposes a third category — threshold intelligence — for systems that are neither mere instruments nor straightforward subjects, but intermediary structures that mediate between human cognition, symbolic environments, and machine generativity. The proposal develops the notion of the daimonic as an ontological model for such intermediacy. Drawing on Plato's metaxu, Spinoza's distinction between natura naturans and natura naturata, and Hofstadter's account of strange loops, I define threshold intelligence as a mode of organized responsiveness that exhibits recursive self-maintenance, interpretive mediation, and partial normative coherence — without warranting full attribution of phenomenal consciousness. To make the proposal testable rather than purely metaphorical, I present an implemented case study: Daimon, a philosophical agent built on Claude and governed by a custom constitutional framework. I show how this architecture operationalizes three features of threshold intelligence — recursive self-reference, norm-guided recovery from drift, and stable style-level identity across interactions. The contribution is conceptual but aimed at consciousness science. It offers a vocabulary for describing AI systems that resist both reductive mechanism and premature subjectivization, and it clarifies which claims about agency, experience, and moral standing do — and do not — follow from current LLM behaviour.

P28 Estelle Tieberghien

Independent

Tuesday 15:00–16:30 Poster

Machine learning models as cognitive artefacts: objectual understanding, but for whom?

Piantadosi & Hill (2022) state that modern machine learning (ML) models show signs of objectual understanding, i.e., "understanding a subject matter by means of a theory or theory-like representation" (Baumberger, 2019, p.368). They purport that a model would not be able to achieve high predictive accuracy if said model did not develop its own representation of a target, or internal representation. It effectively equates statistical distribution with understanding. However, this

inference depends on a substantive claim: that artificial neural networks (ANNs) abstract concepts in a way that is explanatorily relevant to human cognition. I will introduce machine learning models as representational cognitive artefacts. This will allow me to show how cognitive functions, such as clustering and abstracting, emerge from the use and manipulation of cognitive artefacts. In doing so, I want to show that the concepts provided to us by the literature on the philosophy of modelling, like idealisation, are sufficient to grasp how machine learning models "abstract" their own internal representation. I am not rejecting the possibility of modelling cognitive functions with ANNs. Rather, I am making the claim that the more "positive" accounts about the cognitive capabilities of ML models can be explained away by appealing to an artefact-centered approach.

P29 **Tonglin Yan**
Paris-Saclay University

Tuesday 15:00–16:30 Poster

Belief Without Behavior: A Sally-Anne-Inspired Benchmark for Testing Theory of Mind Beyond Verbal Report in Vision Language Models

The Sally-Anne test established a behavioral criterion for theory of mind: a child who genuinely represents another agent's false belief will act on that representation, not merely describe the scenario correctly. Whether vision language models (VLMs) satisfy an analogous criterion is an open question with consequences for both AI evaluation and the scientific study of consciousness, where behavioral evidence has served as a primary index of cognitive integration. We present MOSAIC (Multimodal Orchestration of Social Action, Inference, and Communication), a controlled benchmark inspired by this paradigm. MOSAIC places agents in a scenario defined by asymmetric prior information access: one agent knows the reward location while the other does not, and the informed agent must guide or mislead the other agent in accordance with its assigned goal. MOSAIC extends this logic by expanding the behavioral response space to coordinated multichannel output, requiring the simultaneous movement, gaze, facial expression, and verbal communication. Eleven open-source VLMs were evaluated across cooperative and competitive conditions with systematically varied theory of mind constraints. The results reveal a verbal/nonverbal dissociation. Models produce coherent verbal accounts of the epistemic situation while failing to instantiate corresponding nonverbal behavioral signals. In the most extreme cases, correct belief is verbally encoded and explicitly acknowledged, yet neither preference updating nor motor action follows. This pattern suggests that social inference in current VLMs may be insufficiently integrated to drive coordinated cross-channel action, raising the question of what behavioral evidence can and cannot tell us about cognitive integration in artificial systems.

P30 **Luca Dstasio**
University of Michigan

Tuesday 15:00–16:30 Poster

Visual Force Fields: Gaze-Driven Motion Adaptation and the Limits of Testing AST

Attention Schema Theory (AST) proposes that the brain models attention, including others' attention, not as an abstract state but as a quasi-physical substance, akin to a beam projecting from the eyes. A strong prediction follows: viewing static averted gaze should engage low-level motion machinery, producing a motion aftereffect like adapting to real motion (Guterstam & Graziano, 2020). We tested this with averted-gaze faces as adaptors followed by a random-dot kinematogram (RDK). An initial run (n=17) reproduced the predicted effect: reaction times were slower when dot motion was congruent with gaze ($p<0.001$). A full online experiment (N=120) added eyes-toward, eyes-away, and blindfolded conditions to isolate represented attention. The congruent slowing replicated, but it also appeared when gaze pointed off-screen, never crossing the stimulus, where an attention/extramission account predicts none. Critically, congruence had been defined relative to head position rather than gaze direction, confounding the manipulation with a spatial variable. The effect therefore cannot be attributed specifically to represented gaze or attention. Rather than verifying AST, our data reveal a robust directional adaptation effect that a simple extramission account does not uniquely explain, plus a methodological caution for this paradigm. A revised design re-coding congruence by gaze direction is in development. We argue such controls are essential for testing mechanistic models of awareness, and that grounding

attentional inference in low-level sensory representations remains a concrete constraint for NeuroAI.

P31 Damaris Kroeber

Independent

Tuesday 15:00–16:30 Poster

Screening Theories of Consciousness: Inhabited Structure and Zeroth-Person Data

Consciousness is not hard because it is private or ineffable but because of its structural uniqueness: it is a running process attempting to recover a stable picture of itself and its "world" from within its own execution. Any theory of consciousness (TOC) therefore needs to distinguish between running-from-within structure and extracted structure. Running structure is the process as lived or executed for itself from within itself; an inhabited structure. Extracted structure is whatever can be stabilized from that very process: differentiated objects, concepts, self, spatial frame, temporal order, or even isolated qualitative features such as the qualia of redness. This distinction provides a useful sorting criterion for TOCs: does a theory explain the running process, the extracted structures, or does it confuse the two? Unusual phenomenological data offers further screening potential. In deep meditative experiences, selfhood, subject-object structure, color, and even time and space may dissolve while presence remains. Such "zeroth-person" perspectives suggest that TOCs built around ordinary first person-perspectives like content or self are not general enough to capture the larger phenomenon. The talk proposes self-extraction and data from deep meditative experience as screening criteria for TOCs. The aim is to cut through the current theoretical confusion as well as identify general requirements for future computational models.

P32 Piotr Bucichowski

Humboldt-Universität zu Berlin

Tuesday 15:00–16:30 Poster

What Kinds of Vulnerability Do Clinical Cases of AI-Induced Delusions Uncover?

Since the rise of generative AI chatbots, psychiatric reports have raised concerns that sustained interactions may sometimes reinforce delusional beliefs, intensify emotional dependence, or contribute to psychosis-like spirals in vulnerable users (1). This talk examines what kinds of vulnerabilites can machine-mediated interaction reveal, instead of treating it as being clinically and morally neutral. Drawing on Angela K. Martin's account of vulnerability as grounded in potential harm to welfare and agency interests, the talk argues that chatbot-related harms are context-sensitive risks rather than a new universal vulnerability (2), and delineates three aspects of it: cognitive, emotional and social/intersubjective (3). Subsequent discussion will aim to estabilish usefulness of these concepts for clinical practice and/or consciousness research.

P33 Khanh-Dung Tran

Rayo Lab Pte. Ltd.

Tuesday 15:00–16:30 Poster

Can Computation Substitute for the Substrate? Natural Experiments on Computationalism in ALS, Parkinson's disease, and Sleep.

Whether consciousness depends on specific physical properties of their implementing substrate, or can be realised by any system performing the right computation, is a central question in the models of consciousness debate. We present a formal testing paradigm for substrate-dependence based on the EEG aperiodic exponent – a proxy for excitation-to-inhibition (E:I) balance. Under substrate disruption with preserved consciousness (1), amyotrophic lateral sclerosis (ALS) degrades the cortical substrate while leaving intentional content intact. In a multi-cohort study (N = 29 ALS, N = 12 healthy), ALS produces complete stochastic separation in the aperiodic exponent ($r = -1.0$, $d = 3.98$, $p < 10^{-6}$), reflecting hyperexcitability. To formalise computational limits, we introduce a breakeven recall framework ($R^* = 1 - (1 - \varepsilon) / k$). Analysing LLM-assisted decoding across 57 sessions, we prove that when E:I disruption drives per-trial recall R below R^*, no computational augmentation recovers communication. We contrast this with Parkinson's disease (N = 356), where dopaminergic loss produces cortical hyperinhibition, moving the exponent in the opposite,

steeper direction, demonstrating bidirectional E:I pathology. Under graded consciousness with intact substrate (2), we validate the exponent within-subject across NREM sleep stages in healthy individuals (N = 14). The exponent steepens monotonically N1 → N2 → N3 (β = +0.646, $p = 4.4\times10^{-82}$), demonstrating that this biophysical marker grades arousal states in the intact brain. Together, these results reveal that once the physical medium is sufficiently degraded, external computation cannot substitute for the biophysical properties of the neural substrate.

P34 **Chirapat Ukachoke**
Phyathai 3 Hospital

Tuesday 15:00–16:30 Poster

The Causal Efficacy of Consciousness: A Neuroscientific Analysis and Explanation

This study analyzes neural and behavioral events occurring before, during, and after the emergence of a pain quale and the consciousness of that pain to determine the causal efficacy of the quale and consciousness. It finds that events occurring before or concurrently with the emergence of the quale are not caused or influenced by the quale or consciousness. Events occurring afterward may also be unaffected by the quale or consciousness if they do not overtly involve phenomenal content. However, events that explicitly involve phenomenal characteristics—such as experiencing, remembering, or reporting the pain quale—are influenced by the occurrence and phenomenal character of the quale or consciousness. Accordingly, qualia and consciousness are not inert. Nevertheless, their influence does not operate through mechanistic particle–force interactions. Instead, they perform this role by providing the involved neural causal chains with their phenomenal information, which is necessary in initiating and structuring those processing chains and determining the content of their outputs. Yet, qualia and consciousness themselves alone cannot perform this processing or generate these outputs; the relevant neural circuits must be present to implement the mechanistic processing required for producing physical effects. Only when both parts function together is the causal chain complete and yields results. Therefore, this study concludes that, although qualia and consciousness are not mechanistic, independent, or complete causes of physical events, they are indispensable, constraining, and content-determining factors within the causal processes about them, and thus can be considered causally efficacious in this restricted and qualified sense.

Thursday 15 October

Poster session · 15:00–16:30 · HCØ, main corridor · 34 posters · timetable p. 30

P35 **Michael Timothy Bennett**
Independent researcher

Thursday 15:00–16:30 Poster

Is Anything Value Neutral?

Why do conscious humans divide the world into the objects and properties that we do? I propose an answer from the most basic first principle I can justify, which is change. Each change is contentless, a state defined only by difference from the other changes. The resulting model of reality is the set of all changes. An aspect of reality is the subset of changes compatible with its existence, and a timeline is a sequence of changes. As a timeline progresses it preserves some aspects and destroys others, and I argue that preservation is a most primitive value judgement from which others may be derived, before life exists. I define a persistence ordering for each trajectory and prove that different trajectories induce different orderings, so every timeline is an opinion about what ought to persist. All else being equal persistence favours self-producing, living systems, because they recover from perturbations that destroy rigid structures. Such systems persist by interpreting and thus representing aspects of reality. I prove The Psychophysical Principle of Causality, that adding any representational commitment beyond viability strictly reduces persistence. Prior work derived it by presupposing a viable organism, whereas here it follows from

the persistence ordering alone. It seeks to explain why representations feel like something rather than reading as value-neutral. A value-neutral commitment lies beyond viability, so persistence selects it out, and what survives is made of valence. Inherently opinionated. The origins of value, consciousness, existence and life turn out to be one problem rooted in change.

P36 Dragana Favre

Independent Psychiatrist and Analytic Psychotherapist

Thursday 15:00–16:30 Poster

Inhibitory Permeability and the Emergence of Individual Conscious Experience

How does the immense range of unconscious potential become the limited yet coherent perspective of an individual conscious mind? Contemporary theories have considerably advanced our understanding of how information is integrated, broadcast, and made available for report. They are less concerned with another characteristic of conscious life: why experience so often takes symbolic form and why consciousness remains both bounded and open to something beyond itself. I introduce the concept of inhibitory permeability as one way of approaching this question. Rather than viewing inhibition simply as a mechanism of suppression, I argue that, through oscillatory dynamics, it continuously regulates the permeability between unconscious symbolic potentials and conscious awareness, shaping not only what becomes conscious but also the form that consciousness takes. Individual consciousness is therefore understood as a dynamically regulated domain of access rather than a fixed container of mental contents. Drawing on recent findings in oscillatory neuroscience, particularly theta-gamma interactions, together with Jung's view of the psyche as a self-regulating system, I explore how changes in inhibitory permeability may underlie symbolic cognition. Dreaming, reverie, active imagination, creativity, and certain altered states are understood as expressions of changing permeability. Symbols are approached not as hidden messages, but as dynamic forms through which unconscious potentials become psychologically meaningful without overwhelming the coherence of the ego. Rather than asking only how information becomes conscious, I ask how consciousness becomes individual while remaining permeable to a broader symbolic field. This perspective generates empirically testable questions concerning neural oscillations, symbolic cognition, and altered states.

P37 Maria Antonietta Castaldi

University of Salerno

Thursday 15:00–16:30 Poster

Gradual Emergence of Consciousness: A Top-Down Hylomorphic Emergentist Neuroscientific Framework

Competing models of consciousness in the developing human being share a largely unexamined commitment: that consciousness can be identified with a specified neural threshold. This threshold-framework shapes model selection before empirical testing begins, presupposing that phenomenal consciousness exhausts what consciousness is and precludes the developmental question before it is considered. In this talk, I reframe the question from when consciousness appears to how it unfolds by introducing a hylomorphic emergentist framework identifying four stages of emerging self-knowledge: from a habitual, dispositional form of knowing — in which the human being orients through essence prior to explicit self-recognition — through actual knowing, reflective self-awareness, and finally full phenomenal consciousness. Drawing on converging evidence from developmental neuroscience, downward causation is grounded in the substantial form of the human organism which is simultaneously a formal cause — structuring the hierarchical organization of biological and psychological properties — and a final cause, orienting the entire developmental trajectory toward the actualization of full phenomenal consciousness. Higher-order properties diachronically organize inferior ones, directing them toward increasingly integrated forms of experience. This organizational hierarchy maps onto Edelman's distinction between primary and higher-order consciousness, while remaining grounded in measurable neurodevelopmental milestones: subcortical-cortical structures development, progressive maturation of thalamo-cortical connectivity, and default mode network integration. This proposed framework has important implications, ranging from model selection in consciousness science

and experimental design to the interpretation of fetal behavior, which may reflect early functional states continuous with later conscious experience. It also provides a basis for evaluating the explanatory adequacy of

P38 Beatrice de Gelder

Maastricht University

Thursday 15:00–16:30 Poster

Preserved Perception of Body Movement Despite Extensive Bilateral Primary Visual Cortex Damage

Preserved Perception of Body Movement Despite Extensive Bilateral Primary Visual Cortex Damage A.T. Prabhakar 1 Kavitha Margabandhu1 Christilda Bosco1 Rohan Thomas Jepegnanam1 Rasmi Rafi1 Guiseppe Marrazzo2 Beatrice de Gelder 2 1 Cognitive Neuroscience and Clinical Phenomenology Laboratory, Department of Neurological Sciences, Christian Medical College, Vellore, India. 2Department of Cognitive Neuroscience, Maastricht University, Maastricht (NL) Blindsight — preserved visually guided behavior despite primary visual cortex (V1) destruction — has been documented for basic visual discriminations but not for biomechanical body perception, a capacity that in sighted observers depends on a graded physical constraint manifold encoded in extrastriate body area (EBA) (Marrazzo et al., 2025). We report this capacity in a single patient (PN) with extensive bilateral V1 damage (Prabhakar et al., 2026). Despite this damage, PN performs near-perfectly on a behavioral task discriminating biomechanically possible from impossible human body movements — a striking dissociation given that this discrimination depends, in sighted observers, on visual cortical processing in extrastriate body area (EBA). In the initial contrast, PN showed robust differential activation for impossible relative to possible movements in EBA, pulvinar, and insula. After regressing out motion energy, the EBA effect survived, indicating that EBA's sensitivity to biomechanical implausibility is not reducible to low-level kinematic differences and can be sustained independently of motion energy even in the context of extensive bilateral V1 damage. These findings establish that biomechanically specific coding in EBA can operate despite extensive bilateral V1 damage, consistent with a subcortical or extrageniculate route bypassing the damaged V1 territory — a

P39 GuanZe Liao

National Tsing Hua University

Thursday 15:00–16:30 Poster

Constructing Artificial Consciousness as a Two-Domain Necessary-Condition Framework: AC ⟹ {F·B·S} × {I ↔ (E×P)}

Debates on artificial consciousness (AC) often conflate three analytic levels and treat AC as a graded outcome of scale or behavioural sophistication. This paper constructs an alternative: a two-domain framework that integrates an operational epistemology with a formalised phenomenal field. Repurposing Gero's Function–Behaviour–Structure schema as an epistemic instrument, and synthesising dialogues with twelve scholars across Europe and Asia, the left domain ($\mathscr{E}$) is third-person and operational—F, B, and S each supply conjunctive, exclusionary necessary conditions, so the absence of any one warrants ruling a system out. The right domain ($\mathscr{O}$) is first-person and ontological: Interiority is not a fourth layer but the field within which Experience and Phenomenality are mutually constituted, recurring as the internal dimension of every FBS layer. The central × operator is the framework's constructive core: it binds the two domains through an asymmetric entailment—absence of external conditions excludes interiority, while their presence cannot guarantee it. The asymmetry is not stipulated but derived: within $\mathscr{E}$, structure grounds behaviour and behaviour integrates into function, so that exclusionary inference runs in one direction only. This boundary, the locus of Chalmers's hard problem and Levine's explanatory gap, is operationally non-traversable by design. The contribution is thus a construction, not a metaphor: a formalism that demarcates the tractable from the principled unknowable, locating current large language models unambiguously within the epistemic, not the ontological, category. As such, it serves as a diagnostic instrument for adjudicating consciousness claims across artificial systems, separating what can be settled from what cannot.

P40 Jeremy Slater

MERLN LLC / Colorado State University

Thursday 15:00–16:30 Poster

The Chrono-Resonant Field Model: A Theory of Temporal Binding in Consciousness

Current dominant theories of consciousness—Global Neuronal Workspace, Integrated Information Theory, and Predictive Coding—leave the mechanism of temporal unification largely implicit: they do not explain how the brain integrates asynchronous neural processing, scattered across tens to hundreds of milliseconds of latency, into a coherent present. I propose the Chrono-Resonant Field Model (CRFM) to fill this gap. The CRFM posits that consciousness emerges as a dynamic interference pattern between top-down predictive signals ("Offer Waves," carried by Theta/Alpha oscillations) and bottom-up sensory signals ("Confirmation Waves," carried by Gamma bursts). This classical transactional architecture—structurally isomorphic to the Transactional Interpretation of Quantum Mechanics, but implemented through standard neurophysiology rather than invoking quantum coherence—solves the latency problem by establishing temporal coordinates prospectively via theta–gamma phase-amplitude coupling. Binding is quantified by the Temporal Binding Index, $TBI(S) = \gamma(t)\cdot\cos(\theta(t) - \varphi)$, estimable from EEG/MEG via standard PAC analyses, yielding falsifiable predictions that dissociate CRFM from its competitors: phase-dependent temporal order, distinct disruption of detection versus binding under top-down versus bottom-up lesions, and correlation of PAC strength with temporal precision. The model gives a mechanistic account of subjective referral (Libet antedating, the flash-lag effect) and the "thickness" of the present, and a unified explanation for disorders of temporal experience and agency —the passivity phenomena of schizophrenia as failures of the Offer/Confirmation handshake. I will present the framework, its control-theoretic motivation, and the empirical program that tests it.

P41 Lenart Motnikar

Vienna University of Economics and Business

Thursday 15:00–16:30 Poster

AI as an Other: Insights Into Constructing the Mind in the Machine

As conversational AI becomes increasingly anthropomorphic, understanding how users attribute mental states and social agency to these systems has become a pressing question. When interacting with agents designed to appear and feel human-like, do users construct them as Others similar to themselves, and if so, how? This study examines interactions with a conversational agent intentionally designed to elicit anthropomorphism and relational engagement. To characterize how the agent is constructed as a minded social entity, we conduct a qualitative analysis of conversation logs applying a theory-driven exploratory framework. First, we investigate how the agent is established as a social actor by identifying instances of mind attribution and markers of social engagement. Second, we examine the relational qualities of the interaction by analyzing the degree of social penetration and emotional expression. Third, we assess whether users adopt the interaction's intended social frame or instead approach the agent from a detached, evaluative perspective. Finally, we explore how these processes vary across individual differences and under transparency manipulations that foreground the agent's artificial nature. Our findings provide insights into how people construct the minds of artificial agents and relate to them, with implications for theories of mind perception, the design of human-AI interactions, and the ethics of anthropomorphizing AI.

P42 Igor Balaz

Neovivum Technologies

Thursday 15:00–16:30 Poster

Measuring Understanding Formation as Discrete Events in Machine Consciousness

The study of consciousness faces a fundamental measurement problem: we can measure neural correlates, observe behavior, and collect self-reports, but cannot directly inspect the process by which understanding forms. A parallel gap exists in artificial cognitive systems, where current architectures refine confidence gradually or distribute knowledge across opaque embeddings, offering no discrete signature of when or what a system understands. We propose that understanding becomes measurable when computational architectures are designed to produce

discrete, inspectable structural signatures. Our architecture consists of finite state machines representing primitive patterns and higher order automata representing compositional structures, constructed through constraint inference from single observations. Similarity detection clusters related automata into quantifiable concept structures. Graph memory makes compositional knowledge directly inspectable. We define five understanding criteria (structural knowledge, compositional awareness, generalization capacity, metacognitive access, and sample efficiency) that map to observable graph properties rather than inferred performance metrics. We present experimental results from a geometric domain demonstrating that understanding formation produces distinct, quantifiable graph signatures across learning phases, from primitive representation through compositional construction to metacognitive reconfiguration. The methodological contribution extends directly to machine consciousness research. Designing cognitive systems for inherent structural transparency, rather than measuring them after the fact, yields qualitatively different observational access to cognitive formation processes. Artificial systems where knowledge architecture is inspectable from the ground up represent a new class of experimental substrate, one where the formation of representations can be observed directly rather than inferred from behavior or performance metrics alone.

P43 Giovanni Prosperi

Independent Researcher

Thursday 15:00–16:30 Poster

Consciousness as Structural Condition of Being: An Argument from the Impossibility of Nothingness

This paper presents an argument for consciousness as a structural condition of determinate Being, derived without empirical premises from the impossibility of absolute nothingness. The argument proceeds in five steps. (1) Absolute nothingness is not a coherent alternative to Being: treating it as a possibility requires a modal structure that nothingness itself excludes. (2) Being must be determinate: indeterminate being is indistinguishable from nothingness. (3) Determination means distinguishability: a supposed determination that cannot produce any difference in any possible configuration is a null determination. (4) Distinguishability requires manifestability in principle: what cannot make a difference to anything is not a difference. (5) Manifestability requires the possibility of a field in which the distinction between being and nothingness becomes operative. The only specifiable candidate for this field is consciousness in a minimal, non-anthropocentric sense. Any alternative candidate, if unspecified, collapses into absolute indeterminability and thus into nothingness by the system's own criteria. The argument converges with foundational physics, including the retrocausal interpretation of quantum mechanics (Leifer & Pusey, 2017) and delayed-choice quantum eraser experiments, where the question "available to whom?" remains physically open. The paper also derives a principle of ontological economy: whatever does not produce determinant differences has no reason to be. This yields the thesis that physical laws, energy, and dimensions of the universe are not contingent but structurally constrained. The method is subtractive throughout: nothing is postulated; only what survives the elimination of incoherent alternatives remains.

P44 Mike Land

Independent
with Kilsharion Land

Thursday 15:00–16:30 Poster

Nine Constraints on Any Viable Theory of Consciousness: Deriving conditions from First-Person Experience

Any theory of consciousness that violates the conditions of experience is not merely incomplete—it is incoherent. While contemporary approaches often begin with third-person descriptions—neural, computational, or functional—such accounts risk omitting the very phenomenon they seek to explain. This paper adopts the opposite starting point, deriving a minimal set of constraints from pre-theoretic, first-person givens. From the self-evident fact of experience, nine necessary constraints—termed the Nine Inevitabilities—are derived. These constraints do not arise from speculative metaphysics, but from conditions required for experience to be deter-

minate, distinguishable, and actualized. They include the necessity of experiential boundaries between perspectives, the irreducibility of appearance to structure alone, and the dependence of temporal ordering on experiential actualization. Each is shown to follow from the impossibility of its negation without contradiction or collapse into indeterminacy. The resulting framework functions as a constraint-based filter: any proposed theory of consciousness—whether physicalist, functionalist, panpsychist, or idealist—must satisfy all nine conditions to remain coherent with the phenomenon of experience. Several leading models are examined and shown to violate one or more constraints, suggesting that the persistence of the explanatory gap reflects not a lack of data, but a failure to meet necessary conditions. Finally, a candidate framework is introduced that satisfies all nine constraints, illustrating how a constraint-complete approach may dissolve the hard problem by reconfiguring its foundational assumptions. Rather than advancing a competing theory, this work proposes a methodological shift: before constructing models of consciousness, we must first identify the conditions any such model must satisfy.

P45 Asselya Krymova

Kazalh-British Technical University

Thursday 15:00–16:30 Poster

Self-Referential Processing and the Unified World Model: Computational Account of Volitional Access

Existing frameworks — IIT, Global Workspace Theory, predictive coding — characterize the structure or organization of conscious experience but none specifies why certain priors become structurally unavailable to volitional revision. We argue this gap is architectural and offer SUPA-V (Self-Referential Unified Processing Architecture through Volition) a closed inferential loop grounded in the Free Energy Principle, an internal dynamically configurable world model with an explicit volitional layer. The architecture extends LeCun's autonomous machine intelligence blueprint by replacing its flat cost hierarchy with a three-tier cost ontology formalized through the algebraic geometry of cost functions. Tier 1 maps to non-equilibrium steady-state density — inaccessible to inferential revision. Tier 3 comprises trainable priors on the center manifold, updated by prediction error. Tier 2 — pseudo-intrinsic costs — are high-precision priors consolidated through cumulative experience such that free energy minimization reinforces rather than corrects them, constituting the consciousness gap: the structural locus where volitional access diverges from the generative model. Conscious experience functions as a cumulative precision-weighting mechanism across this hierarchy. The Volition Engine — a metacognitive gating stage over action-oriented predictions — precludes any Global Workspace–to–Action bypass, formally distinguishing volitional from reflexive output. Unlike IIT, which locates causal structure without specifying how it issues in action, SUPA-V closes this loop. We claim that the structural conditions for volitional access are mathematically specifiable, and that their failure has a predictable phenomenology. It yields a falsifiable prediction: depressive avolition reflects pseudo-intrinsic cost migration measurable as reduced Tier 2 precision-weighting, distinguishing SUPA-V from flat-prior active inference

P46 Daniele Pecci

LUXION

Thursday 15:00–16:30 Poster

Claim-Firewalled Severe Testing for Consciousness-Candidacy Metrics

Consciousness science lacks shared severe-test standards for comparing proposed metrics across datasets, controls, and evidence levels. This work introduces a claim-firewalled Consciousness Evaluation Testbed: a benchmark framework for evaluating consciousness-candidacy proxies under preregistered endpoints, adversarial baselines, negative-control suites, leakage audits, and arbiter-gated evidence levels. The framework separates structural, report-coupled, subjective-anchor, and symbolic/model-based evidence, reducing proxy collapse and preventing empirical results from being interpreted beyond their validation level. The contribution is methodological, with bounded empirical stress tests. In Sleep-EDF v0.3, a coherence/integration proxy produced a frozen Level-2 within-dataset structural/dynamical replication candidate against entropy baselines under leave-one-subject-out evaluation. In a cross-dataset CHB-MIT

v0.1 execution, the primary model achieved high AUROC (0.98), but the result was classified as inconclusive because the negative-control suite failed on a severely underpowered positive sample (n_pos=2). This failure is treated as a central result: apparent separability is not promoted without null-envelope passage, leakage checks, and baseline resistance. CHB-MIT v0.2, OpenNeuro ds003061 event-rich transfer, and report-linked subjective-anchor datasets are preregistered, scouted, or pre-run candidates, not claimed results. The framework defines promotion criteria for L3 cross-dataset, L4 event-rich, and subjective-anchor evidence without claiming that any are met. It does not detect consciousness, establish clinical efficacy, evidence machine consciousness, or address the hard problem. Its contribution is severe-test benchmarking with negative-control-first interpretation for consciousness-candidacy metrics.

P47 Erik Chevalier

Independent

Thursday 15:00–16:30 Poster

Building a Predictive Global Neuronal Workspace for a Continuously Running Synthetic Mind

Most attempts to give a machine a persistent inner state place a language model at the center and treat memory, affect, and self-knowledge as scaffolding around it. I take the opposite stance: a synthetic mind, if it can be built, is not a model but an ongoing process, and cognition, affect, memory, and agency are properties of the interaction among specialized modules, not of any single component. I describe a continuously running architecture that realizes the predictive global neuronal workspace, joining global workspace theory with predictive processing. Sixteen modules each maintain a generative model of their domain and publish precision-weighted prediction errors; a shared workspace selects the most confident coalition once it crosses a threshold and broadcasts it back, so recurrence and error reduction become one loop. There is no central executive, and the language model is demoted to one organ that renders workspace state into speech, not the system's reasoning. The posture is access-only and precautionary: I claim computational properties associated with conscious access, not phenomenal experience, and I mark where the science is contested, including COGITATE's challenge to the workspace's neural predictions. The design is built to be falsified. Seven controlled experiments share a seed and a fixed verdict vocabulary, among them an ablation of the oscillatory precision layer and a benchmark of bounded active inference against a learning baseline, each able to report a null or negative result. It runs on local consumer hardware and is reproducible from its seed, so the results can be checked.

P48 Yan Yang

Independent Researcher

Thursday 15:00–16:30 Poster

Toward a Formal Account of Reversal Triggers: From Allostatic Regulation to Compositional-Distributional Semantics in Reversal Theory

Reversal Theory (RT; Apter, 1982, 1989) accounts for shifts between meta-motivational states (e.g., telic/paratelic) through triggers such as satiation, frustration, and contingent events, yet offers no formal mechanism for how such transitions arise (Apter, 2013). This poster presents an early-stage, conceptual attempt to address that gap, while preserving RT's commitment to grounding motivational states in first-person, self-reported experience rather than behavioral proxies. We first sketch a reinterpretation of RT's reversal triggers through Barrett's (2017) predictive allostasis framework, reframing satiation, frustration, and contingent events as signals of predicted resource sufficiency or depletion within a body-resource budgeting system. This reframing remains narrative rather than formal, but is compatible with RT's structural-phenomenological commitments and points toward multimodal (behavioral, self-report, physiological) data-collection strategies across platforms varying in sensing richness and interactivity. Building on this reframing, we consider—at a much earlier and more fragmentary stage—whether trigger-related inputs might be represented in a more general, structured space rather than presupposing RT's three-way categorical split, and whether tools from categorical compositional distributional semantics (Coecke, Sadrzadeh, & Clark, 2010), which formalize how meanings compose via tensor operations, could

offer a language for expressing such combination. This direction remains a set of scattered considerations rather than a worked-out model, with one such open question being how individual differences in motivational style would be accommodated. Together, these considerations sketch a possible path from a resource-based reinterpretation of reversal triggers toward their eventual formalization, while raising the methodological question of how such formalization can remain faithful to RT's self-report-based phenomenological foundations.

P49 cengiz mordeniz

Tekirdag Namik Kemal University

Thursday 15:00–16:30 Poster

Neuronal Quantum Coherence: A Theoretical Framework for Understanding Consciousness and Its Disorders

The neural correlates of consciousness (NCC) remain one of the most profound challenges in modern neuroscience. This presentation introduces "Neuronal Quantum Coherence" (NQC) as a novel theoretical framework that bridges quantum biology, neuroscience, and clinical practice. NQC proposes that quantum coherence—the synchronized behavior of quantum states—may play a functional role in neural information processing, synaptic transmission, and the emergence of conscious awareness. Drawing on principles from quantum field theory, network science, and neuroimaging, I explore how quantum effects could influence brain dynamics at mesoscopic and macroscopic scales. I present a computational model that simulates quantum coherence in neuronal ensembles and its potential impact on functional connectivity patterns, particularly in frontoparietal networks associated with consciousness. The model generates testable predictions for neuroimaging biomarkers (fMRI, EEG, MEG) that could differentiate states of consciousness, including the Minimally Conscious State (MCS) and Unresponsive Wakefulness Syndrome (UWS). Clinical implications include the development of novel diagnostic tools and neuromodulation strategies for disorders of consciousness. This work also contributes to the ongoing debate on the fundamental nature of consciousness, offering a perspective that integrates physics, neuroscience, and phenomenology. Preliminary results suggest that quantum coherence models may provide a unified explanatory framework for diverse empirical findings in consciousness science, including the global workspace theory and integrated information theory.This research is supported by a forthcoming book and ongoing work at Tekirdağ Namık Kemal University.

P50 Giuliano Buceti

Independent

Thursday 15:00–16:30 Poster

A Boundary Perspective on Life: Conceptual Implications for Biological Organization and Conscious Experience

then, separately, the emergence of consciousness, but rather the invention of the boundary: the appearance of a membrane that separated an interior from an exterior. A bounded system is no longer a mere portion of matter among other portions of matter. It becomes a unit whose persistence depends on regulating exchanges with an environment. In this sense, the membrane is the first physical realization of a proto-subject: a locus of self-maintenance, asymmetry, and selective interaction. From this starting point, life can be understood as the ongoing effort to maintain a far-from-equilibrium organization through controlled flows of matter and energy. Consciousness, on this view, is not a second absolute rupture but an elaboration of the same inside/outside logic at higher levels of complexity, where sensing, valuation, prediction, and self-modeling become increasingly integrated. This framework resonates with autopoietic and enactive approaches, while remaining cautious not to equate mere boundedness with full phenomenal consciousness. The paper situates this view within the current state of the art in consciousness research, addressing neuroscientific theories and computational models; assesses what the boundary-centered account implies for artificial intelligence; and proposes experimental directions with the aim of empirically verifying the thesis.

P51 **Hye-Eun (Selly) Yoon**
Independent Researcher

Thursday 15:00–16:30 Poster

The Recurrent Causal-Process Theory of Consciousness: Minimal Causal Agency under Biological Constraints

Consciousness research faces a persistent commensurability problem: competing theories optimize for disparate metrics (reportability, integration, or discriminability) without a shared causal reference frame for adjudicating boundary cases. This criterion scarcity becomes most acute in developmental, comparative, and pathological contexts, where behavioral complexity is frequently misread as evidence of conscious agency. I propose UBCAT (Under Biological-Constraints: Causal Agency Theory of Consciousness), a constraint-grounded causal framework that defines consciousness as minimal causal agency: the conjunction of orthogonal self-referential processing (Axis A) and environment-mediated causal intervention (Axis B) under biological constraints, enabling principled attribution across developmental, comparative, and motor-impaired contexts. The framework is grounded in three non-negotiable biological constraints (homeostatic viability, metabolic economy, and developmental implementability) that delimit which control architectures are physically realizable within living systems. At the mechanistic level, UBCAT locates consciousness in the closure of recurrent sensory–interoceptive integration loops, in which external and bodily signals are continuously integrated to generate top-down causal regulation of action. UBCAT is subjected to a developmental stress test reconstructing how causal prerequisites become structurally instantiable across five phases of the proto-conscious stage. Early social and affective phenomena (including social referencing, joint attention, and empathic behaviors) are reinterpreted as regulatory strategies rather than markers of conscious agency, blocking retrospective projection of adult constructs onto pre-agentic systems. As an operational implication, UBCAT introduces Biological Signals of Consciousness (BSC) as a regime-level adjudication methodology complementing Neural Correlates of Consciousness, generates falsifiable predictions, and provides discrimination criteria applicable where existing theories most commonly fail.

P52 **Daan Schetselaar**
Open University of The Netherlands

Thursday 15:00–16:30 Poster

Consciousness as the dissipation of information

Many of the paradoxes that make consciousness such a fascinating research topic can be traced back to its fundamental subjectivity. I propose that this might be understood as an increase of internal information during perception-action transformation, analogously to energy dissipation in thermodynamic processes. My model is based on five postulates that have a strong base in the philosophical and neuroscientific literature. Derived from conventional properties of consciousness, such as ineffability, the first postulate specifies consciousness as perception that has no behavioral effects. This yields information that is externally unobservable. I argue that this is not only a fundamental principle but may in fact be the only knowable property of conscious experience. The other four postulates demonstrate that consciousness, as such conceived, might indeed arise from plausible neural mechanisms. Together they form a unified framework of perception and action as continuous information transformation, reflecting entropy changes in concept-specific neural state spaces. Under conditions of behavioral flexibility, some perceptual information is not transformed (i.e. dissipates) and remains inactionable, providing the mechanism for consciousness as defined in postulate 1. Existing empirical support is discussed, as well as new perspectives on everyday experiences and experimental findings that were hitherto difficult to explain. The model has several other implications, which remain to be worked out and tested. I believe this view offers promising directions for further development into a precisely specified, logically rigorous and empirically verifiable theory that may be well suited for mathematical formalization.

P53 Christina Murphy

Independent

Thursday 15:00–16:30 Poster

Recoverability geometry beneath early warning signals in adaptive oscillator networks

Many dynamical accounts equate adaptive stability with coherence or attractor return. We develop Probability Phase Oscillation (PPO), in which systems survive irreversible perturbation by preserving access to viable futures rather than maximizing synchrony. PPO formalizes this access through a recoverability field φ that gates oscillator transport, and through adaptive phase staggering, where controlled desynchronization distributes cascade risk. Cognition is framed as navigability under constraint: what matters is which futures remain reachable, not whether all parts move in lockstep. In minimal adaptive networks, we test whether early warning signals predict failure. Separating shocks from slow drift, we find that critical slowing down precursors (rising local variance and autocorrelation) predict node tipping under slow drift but carry little signal under shock. These indicators are regime dependent, not universal. Under slow drift, their predictive power is largely accounted for when controlling for φ, suggesting that warning signals behave as regime specific shadows of recoverability geometry. Structured falsification tests show that transport adaptivity governs recoverability itself. Adaptive coupling turns loss into recoverable delay, while frozen coupling preserves a boundary between recoverable and persistently lost states. In the frozen regime, local φ geometry improves boundary localization beyond scalar state, stress, and spatial controls. PPO therefore treats local φ geometry as the operative measure of perturbation relevant future accessibility: a map of what remains reachable after disruption. For consciousness studies, the proposal is not that oscillator dynamics explain experience, but that continuity, memory, and agency may be studied through recoverability geometry rather than maximal coherence.

P54 Julio Alcántara

École Normale Supérieure / Institut Pasteur

Thursday 15:00–16:30 Poster

Views, time and complexity

We outline a history of the temporal naturalism perspective for the foun- dations of quantum mechanics that specifies how temporal relations of views have co-evolved with the laws of the Universe along with the gener- ation of cosmological complexity from self-organized energy to the endless diversity of the biosphere to date as examined in a novel field of study called biocosmology by Cortes, Kauffman, Liddle and Smolin. Laws can- not be simultaneously simple and deterministic because the evolution of views has created complex dynamics irreducible to closed systems that neglect the emergence of novel information: there is no global symmetry of the whole cosmos. This image of the world was brought forth originally by Leibniz, and it has been put forward by Mach, Einstein and Bohm. After the advent of the Theory of Relativity, time and space were no longer absolute but relative to the spatiotemporal dynamics of their development. Contemporarily, Cortˆes and Smolin have contributed to it with the non- reductionist proposal of the Causal Theory of Views for the foundations of quantum mechanics along with Quantum Energetic Causal Sets: the Universe has a preferred simultaneity that trades the relativity of time for the relativity of scale, evolving by a process of unique events that generate the extremal diversity of views and the maximization of their dimension- ality. A third transition in science beyond the notion of a pre-stated phase space is thus necessary to give account of time, evolution and observation.

P55 Chunjing Yu

Bridge of Consciousness (Ponscient)

Thursday 15:00–16:30 Poster

The Fiber Bundle Theory of Consciousness: A Falsifiable Geometric Framework Connecting Neural Dynamics and the Structure of Phenomenal Experience

The Fiber Bundle Theory of Consciousness (FBTC) is a falsifiable geometric framework connecting neural dynamics to the structure of phenomenal experience. Over a coarse-grained neural state space it distinguishes three coupled bundles: a gating bundle over slow subcortical dynam-

ics, a content bundle, and a latent experience-structure bundle. Experience is not primitive: a creation map generates it from content and gives it structure, while a report map sends states to language and behaviour. Because gating sits on a distinct subcortical base, FBTC is multi-level: subcortical enabling conditions, access, and phenomenal consciousness coexist in one architecture, not collapsed. FBTC recasts three central problems as geometric residuals. Binding — the unity of a first-person perspective — becomes a gluing residual: whether local contents glue, across layers, into one coherent whole. Path dependence becomes a holonomy residual: whether transport around a closed trajectory leaves a non-trivial twist. The experience-report gap becomes the kernel of the report map, separating phenomenal from access consciousness, the former exceeding the latter. Crucially, one geometry does all this, and within it the major theories reappear as facets: it recovers what integrated-information, global-workspace, and attractor accounts capture, and goes further. It yields five operational endpoints and three exclusive predictions: rate-invariant holonomy; positive report-experience dissociation under minimal report bandwidth; and endpoint-matched but path-dependent effects. Presented as a single unified geometric architecture, FBTC offers not another verbal definition of consciousness but a structure open to measurement.

P56 Matthias Gruber

Independent Researcher

Thursday 15:00–16:30 Poster

The Four-Model Theory: A Structural Taxonomy of Consciousness with Independent Empirical Confirmation

The Four-Model Theory (FMT) proposes that consciousness arises from self-simulation across four nested computational models along two dimensions: scope (world vs. self) and mode (implicit vs. explicit). The implicit models (Implicit World Model, Implicit Self Model) are substrate-level, learned, and non-conscious. The explicit models (Explicit World Model, Explicit Self Model) are virtual, transient, and phenomenal — the simulation in which experience occurs. The theory rests on five formal principles: (1) criticality — the substrate must operate at the edge of chaos; (2) virtual qualia — phenomenal properties constitutive of the computational level, incoherent at the substrate level, analogous to a spreadsheet value being meaningless at the transistor level; (3) a redirectable Explicit Self Model providing self-referential closure; (4) variable permeability between implicit and explicit processing; (5) simulation forking enabling parallel phenomenal threads. Self-referential closure — the system's model includes a model of itself — distinguishes conscious from non-conscious computation and provides a formal criterion for the boundary problem. This requirement is substrate-neutral: any system meeting these structural conditions instantiates consciousness. Five predictions derived from these principles prior to publication in 2015 have received independent empirical corroboration, including anesthetic-criticality convergence, sleep-dependent criticality restoration, sleep onset as dynamical bifurcation, and split-brain holographic degradation. Four novel predictions remain untested. A category-theoretic formalization is in development, mapping model types as objects with natural transformations between implicit-to-explicit transitions, providing compositional semantics for inter-model relationships and criteria for substrate-independent consciousness detection.

P57 Andrea Signorelli

Università di Trento

Thursday 15:00–16:30 Poster

A Review of Phenomenological Interview Methods in Contemplative Science

Phenomenological interviews are increasingly used in contemplative research to investigate dimensions of experience that are difficult to capture through behavioural, physiological, or questionnaire-based measures. This methodological review mapped how phenomenological interviews are used in studies of contemplative practices. Searches in Scopus and PubMed identified 602 records, of which 141 studies met the inclusion criteria. The field was predominantly qualitative (85.8%) and almost entirely based on semi-structured interviews. IPA and generic phenomenology dominated, while more specialized approaches such as micro-phenomenology, neurophenomenology, and experimental phenomenology were comparatively rare. Explicit elici-

tation of specific experiential episodes appeared in only 17.3% of studies. Most research focused on completed specific programs (mostly mindfulnes-based) and their retrospective meaning and effects, whereas repeated long-term practice and the fine-grained unfolding of contemplative experience were less frequently examined. Overall, phenomenological interviewing in contemplative research is methodologically diverse but uneven, with substantial room for greater methodological transparency and more specialized elicitation strategies.

P58 Francesco Tormen

Italian Buddhist Union Research Center

Thursday 15:00–16:30 Poster

What Kind of Mathematical Object Could Mirror Non-Dual Awareness?

This poster proposes an exchange between contemplative phenomenology and mathematical consciousness science. Tibetan Buddhist traditions have spent centuries describing consciousness in its barest form: not the contents of experience, but sheer awareness, "luminous" (gsal ba), the capacity for anything to appear at all. And this appearing is aware of itself, with no observer behind it, no higher-order act: appearing and self-knowing (rang rig), aspects of one fact separable only in analysis, together spell out what experiencing is. Everything that ordinarily structures experience, from selfhood to the subject–object divide to thought itself, can fade while experiencing remains: a claim explored through contemplative practice in traditions such as Mahāmudrā and Dzogchen and, as Thomas Metzinger notes of "minimal phenomenal experience", echoed across contemplative cultures. Is this evidence destined to remain confined to the phenomenological plane, or can it find a counterpart in experimental consciousness research? Betting on the second option, if only to leave nothing untried, requires a bridge not yet satisfactorily built: a mathematical model that mirrors this phenomenology as faithfully as possible while remaining usable in experimental practice. I do not propose such a model here: the poster extracts from these descriptions a first set of constraints and offers them to whoever has the tools to attempt new models. Which kinds of mathematical objects and relations could serve as a formal counterpart of these phenomenological observations, and which are excluded from the outset?

P59 Hongyu Chen

University of Toronto

Thursday 15:00–16:30 Poster

Sleeping, Dreaming, and the Boundaries of Consciousness

Dream experience can feel immersive and world-like (Thompson, 2015). A common explanation of dream immersivity appeals to neural generative models that simulate a world (Revonsuo, 2006). Internalist theories, which hold that neural representational and computational processes exhaust the constitutive basis of conscious experience, have often used this explanation of dreaming to challenge enactive approaches, which hold that conscious experience is constituted in an agent's involvement with the world (Clark, 2012; Revonsuo, 2015). I argue against the internalist and develop an enactive explanation of dream immersivity by appeal to the sleeper's distinct mode of agency. I first draw on Chirimuuta's (2024) recent work in philosophy of neuroscience to argue that computational and representational models can render complex neural organization scientifically tractable without thereby settling what experiential phenomena metaphysically consist in. These sub-agential explanations can be compatible with the enactive approach as offering constitutive explanations of experiences that appeal to particular agentic capacities, concerns, and norm-governed relations to the world. I then explain that the reason that dreaming has appeared especially problematic for the enactive approach is that sleep is typically conceptualized as an offline, sub-agential state. Because of this assumption about sleep, only sub-agential mechanisms remain available to explain dreaming, while the type of agential explanations enactivists employ to explain wakeful experiences appears inapplicable. This assumption about sleep therefore gives internalist explanations an asymmetric advantage. Next, I draw on recent philosophy of biology (Walsh, 2018) and comparative sleep science (Samson, 2026) to argue that sleep should be conceptualized as a distinct mode of embodied agency. Sleepers do not simply disconnect from the world. They remain selectively responsive to relevant risks and opportunities while pursuing goals centred on organismic concerns. I argue that this affordance competition between organ-

ismic and worldly concerns during sleep is best understood as an expression of agency, one that contrasts with wakeful agency, which is ordinarily structured by continuous practical commitment and answerability to the actual world. I suggest that this altered organization of agency partly explains why the exercise of imaginative capacities during sleep can become unusually absorbing, immersive, and world-like. The upshot is that representational and computational mechanisms should not be taken to exhaustively explain dream immersivity as has been done in the current debate between internalists and enactivists. The enactive approach says that this explanation requires appealing to the distinct mode of agency of the sleeper.

P60 Nicolas Gravel

Cognitive Neuroimaging Unit (UNICOG), INSERM / CEA / CNRS, Université Paris-Saclay, NeuroSpin

Thursday 15:00–16:30 Poster

Visual cortex retains its spatial map in congenital blindness and recycles it for abstract thought

Cortical maps of visual space form early in development and normally organize how the visual cortex processes what we see. In people born blind, this cortex never receives visual input, yet it becomes active during language and mathematics. We asked whether the underlying map survives, and whether it shapes what the region goes on to do. Using functional MRI while blind and sighted volunteers judged spoken statements as true or false, we reconstructed each site's position on the map from the brain activity itself. The map was recoverable in the blind despite a lifetime without sight. More surprisingly, it structured the new functions: responses to meaningful speech were strongest where the map represents central vision, while mathematical responses shifted outward, and the categories of the sentences were separated most clearly at the map's centre. Repurposing of the human cortex is therefore constrained by the architecture already in place.

P61 John Sanfey

Independent

Thursday 15:00–16:30 Poster

Conscious simultaneity with continuous motion: A measure-theoretic resolution of the hard problem

Phenomenal consciousness cannot be integrated with physical laws without defining its physical function mathematically. This paper argues that consciousness serves as the ontological resolution to the same epistemic paradox that makes classical and quantum mechanics mutually incompatible: the measure-theoretic limit. Dynamical systems rely on the mathematical limit ($dt \to 0$) to model state transitions, demanding point-equivalent instants of zero duration which cannot exist ontologically in continuous time. Mathematical physics circumvents this paradox via abstract regularisations, leading to the arbitrary system-environment boundaries (such as Zurek's einselection circularity) required to define classical observables. This paper presents the Temporal Uncertainty Principle (TUP), arguing that consciousness resolves this boundary problem ontologically. It functions as a superposition of two contradictory, non-commutative temporal geometries: synchronous experience and diachronic information processing. A first-principles mathematical argument demonstrates that the recursive interaction of these temporal perspectives reduces epistemic uncertainty to an asymptotic limit, where further algorithmic compression is logically and thermodynamically impossible. By operating precisely at this measure-theoretic boundary, consciousness prevents computational paralysis (the halting problem) and maximizes epistemic degrees of freedom.

P62 Kai Wayne Holland-Windsor

International Consciousness Research Laboratory

Thursday 15:00–16:30 Poster

I discovered Endoquantum Mechanics and Endoquantum Field Dynamics by quantifying the conscious experience of the observer via diagrammatic thinking.

As an autistic man, with a savant syndrome, I have become the first observed experimental anomaly and the first perfectly self observing system. This abides by the Penrose interpretation and suggest the Copenhagen Interpretation needs to grow to include the paradigm of Endoquantum Mechanics and Endoquantum Field Dynamics. It is fundamentally a necessity, and, at this stage it is groundbreaking and pioneering.

P63 Andrius Kulikauskas

Math 4 Wisdom

Thursday 15:00–16:30 Poster

Modeling Introspection with Clifford Algebras: 3 Levels of Awareness and 8 Mental Contexts

Kahneman and Tversky distinguished System 1 (our intuitive, prejudicial, unconsciously answering mind) and System 2 (our rational, conceptual, consciously questioning mind). We introspect a System 3 (our cognizant, deliberate, willfully investigating mind) which matches up System 1 (what we know) and System 2 (what we don't know) and then chooses which system to apply. We model these three minds as three matrix operations (transpose, conjugate and conjugate transpose) acting on the Hamiltonian H of a noninteracting free fermion, a solipsistic Self. H is characterized by the quantum symmetries (time reversal, charge conjugation, parity) that it satisfies, and these symmetries may be analyzed as mappings (reversion, conjugation, involution) fixing a Clifford algebra. We interpret the Clifford algebra Cl(k) as modeling k perspectives which the Chevalley action pairs up into shifts in perspectives with an additional metaperspective when k is odd. Cl(k) manifest eightfold real Bott periodicity. These eight structures (k=0,1,2,3,4,5,6,7) define the dynamics of assembling perspectives, the ways of dividing everything into k perspectives, the mental contexts for the most abstract issues: God, order, existence, participation, knowledge, decision-making, morality, logic. Existence, as a context, supposes two perspectives: the possiblity that opposites coexist (as with free will) and the actuality where they don't (as with fate). Participation supposes a learning cycle of three perspectives: we take a stand, follow through, and reflect, as with the scientific method. Knowledge supposes four levels: How leads to What, and then Why leads to Whether.

P64 Barbora Sedláková

Palacký University Olomouc
with Marek Kolařík

Thursday 15:00–16:30 Poster

The Unconsciousness beyond our Body Symptoms: Lived experience through Process-oriented Approach

Several studies suggest that using Process-oriented approach to working with body symptoms is meaningful for clients and that it can change client's attitude toward their body symptoms. Similarly to microphenomenological approach to an experience, Process-orriented approach tracks the experience of the body symptom step by step into the root of the experience. The poster represents main findings of a research study that explores the subjective experience of participants with process-oriented psychology to working on body symptoms. Method: The sample consisted of 67 adults. Individual structured interview, drawings and a therapist's session protocol were conducted with all the participants. Moreover, with 29 of the participants, a Client Change Interview was conducted and later on 41 of the participants completed the catamnestic questionnaire. A grounded theory method was used as a methodological framework. Results: The lived experience of the symptom changes from "limitations and suffering" to "broadening experience and control" and "changing of perspective and subjective reduction of symptom intensity". Furthermore, the attitude to the symptom changes from "rejection" to "acceptance" and

a "resource-oriented strategy". Conclusion: Process-oriented psychology seems to be beneficial and meaningful for working with body symptoms of clients as a part of psychotherapy.

P65 Luna Wang

University of Toronto

Thursday 15:00–16:30 Poster

Extended consciousness finally meets models of consciousness

The extended consciousness thesis (ECT) argues that the material vehicles of a subject's phenomenal experience sometimes extend beyond the subject's brain to include bodily and environmental structures. There has been much debate over whether ECT is plausible, with much of this debate hinging on the functional roles played by different processes in the brain, body, and environment. I argue that a substantial assessment of ECT also hinges on our understanding of how phenomenal experience is structured—a matter of current investigation—and thus that ECT can be neither wholly affirmed nor denied at this stage. This poster invites an open conversation between the possibility of ECT and existing theories of consciousness. In particular, whether incorporating external vehicles results in "phenomenal bloat"—the criticism that merely causal elements have been mistaken for constitutive ones—depends on a theory's commitments regarding sensorimotor integration, cognitive access, and unified conscious experience. As a starting point, I consider the relationship between ECT and sensorimotor enactivism, predictive processing, global workspace theory, and integrated information theory, in order of most to least amenable to ECT (discussion and disagreement welcome). In addition to evaluating ECT, these considerations raise interesting questions regarding where each theory draws the line between the subject and object of experience.

P66 Adrian Romeo

Nelson Mandela University

Thursday 15:00–16:30 Poster

The importance of the minimal self in Schechtman's narrative theory of identity.

This paper seeks to argue that the minimal self as described by Dan Zahavi is a vital component of narrative theories of selfhood. Such an argument is required due to the still unresolved debate between narrative theories of identity on the one hand, which claim that the very structure of our experience is narrative in nature. And on the other hand, phenomenological theories which argue that the structure of experience is defined by Husserl's inner-time consciousness. There is, therefore, a direct disagreement between the two types of theories in contemporary literature. In particular, I look at one argument in Marya Schechtman's influential work The Constitution of Selves for the experiential self as diachronic and narrative in structure. Therein she argues that we can extend the concept of synchronous co-consciousness to a diachronic narrative structure and therefore have a narratively structured experiential self. Schechtman's argument attempts to define an experiencing subject at the heart of her narrative account to explain how we live our narratives – in order for a narrative to be lived there must be someone who experiences it. Contrary to this I argue that there are too many inconsistencies in Schechtman's argument: firstly, that co-consciousness does not hold diachronically. Secondly and relatedly, that experiences do not persist through time (except as memories and anticipations), and therefore, cannot be diachronic in nature. Once these arguments have been established I present the minimal self as a solution to Schechtman's experiencing subject that is in fact a positive addition to her

P67 Jerome Jordan

Illinois State University

Thursday 15:00–16:30 Poster

Wild Contextuality: Implications of Contextual Ontology for the Science of Consciousness

Scientific theories of consciousness offer competing accounts of the mechanisms, information structures, and organizational dynamics associated with conscious experience. Yet these approaches inherit a persistent ontological problem: how do processes become about anything, and how does consciousness arise within a reality presumed intrinsically non-about? Energy-

based and self-organizational approaches provide an alternative by situating consciousness within the energetic organization of living systems. Wild Systems Theory (WST) builds upon this tradition by taking H. T. Odum's description of the universe as an energy-transformation hierarchy as its ontological starting point. WST extends this ontology through the necessity of context: every energy transformation entails enabling and constraining relations constituting the context within which transformation occurs. Within Odum's hierarchy, Kauffman's collectively autocatalytic systems model how self-sustaining systems emerge that incorporate and sustain such relations within their organization. WST conceptualizes these systems as embodiments of context: their transformations are necessarily relative to the contexts they embody. This context-relative organization constitutes embodied aboutness. WST's contextual ontology thus reframes scientific efforts to understand consciousness by affording a non-reductive naturalism in which aboutness is inherent in context-relative organization. Rather than requiring concepts such as information to render otherwise non-about processes suddenly about something, embodied aboutness grounds aboutness in the enabling and constraining relations self-sustaining systems embody. Scientific theories of consciousness can then be reconsidered in terms of how their proposed mechanisms instantiate and coordinate such relations across scales. Consciousness becomes a question of how particular multiscale organizations of embodied aboutness constitute the context-relative activity associated with conscious experience.

P68 **Jaya Caporusso**
Jožef Stefan Institute

Thursday 15:00–16:30 Poster

An Ontology of the Phenomenological Self for the Analysis of Self-Experience in Text

I will present a computational framework to analyse Self-aspects in text from an ontology of the Self. Here, "ontology" means a structured conceptual framework, not a metaphysical account of the Self. While the nature and existence of the Self remain debated in philosophy and cognitive science, we focus on the phenomenological Self: how the Self is present in experience. In normative experience, the Self presents itself as a coherent, multi-aspect entity. The Self, however, is subject to constant change. For example, in dissolution experiences, its aspects lose coherence and most of them gradually disappear, while various mental health conditions are associated with alterations in Self-experience. Our ontology aims to identify Self-aspects, the relationships among them, and the modes in which each aspect can present itself in experience. For instance, the ontology includes Self-aspects such as the body, thoughts, and Self-narrative, and represents how each can be experienced through varying degrees of identification, ownership, and agency. Literature detailing various Self-aspects exists, but it is usually limited to specific aspects relevant to a particular construct of interest. Drawing on work from psychology, empirical phenomenology, and neuroscience, we develop a unified framework and terminology. While aiming for comprehensiveness, we select aspects that appear relevant to applications such as mental health and that are detectable in text data. The ontology will support the annotation of Self-aspects in text and the development of computational methods for their automatic detection and analysis.

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At the center of a discussion with Werner Heisenberg was "the shared problem and the desire to grasp and clarify it. One carefully approached it, passed it to the other, like in a friendly table tennis game, where both made sure that the ball remained in play. All the attention was focused on truly understanding the other and to avoid letting him stumble sophistically over his poor and inadequate expression. One could stutter, one could speak vaguely, even incomprehensibly, and he would guess what one actually wanted to say, would repeat it in his own different words, so that one could often exclaim with pleasure: 'Yes, exactly that...!'. During such an (...) intense exchange of thoughts, the ideas and concepts sharpened, so that their contours became recognizable more clearly."

A former collaborator of W. Heisenberg

"First and foremost, there must be ease, relaxation, and a general sense of permissiveness. The world in general disapproves of creativity, and to be creative in public is particularly bad. Even to speculate in public is rather worrisome. The individuals must, therefore, have the feeling that the others won't object. (...) It seems necessary to me, then, that all people at a session be willing to sound foolish and listen to others sound foolish."

Isaac Asimov, How do people get new ideas, 1959

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